

Chapter 20

Analog-to-Digital Converter (ADC)



The analog-to-digital converter (ADC) module described in this chapter is a Type 4 ADC. See the [C2000 Real-Time Control Peripheral Reference Guide](#) for a list of all devices with modules of the same type, to determine the differences between the types, and for a list of device-specific differences within a type.

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20.1 Introduction

The ADC module is a successive approximation (SAR) style ADC with selectable resolution of either 16 bits or 12 bits. The ADC is composed of a core and a wrapper. The core is composed of the analog circuits which include the channel select MUX, the sample-and-hold (S/H) circuit, the successive approximation circuits, voltage reference circuits, and other analog support circuits. The wrapper is composed of the digital circuits that configure and control the ADC. These circuits include the logic for programmable conversions, result registers, interfaces to analog circuits, interfaces to the peripheral buses, post-processing circuits, and interfaces to other on-chip modules.

Each ADC module consists of a single sample-and-hold (S/H) circuit. The ADC module is designed to be duplicated multiple times on the same chip, allowing simultaneous sampling or independent operation of multiple ADCs. The ADC wrapper is start-of-conversion (SOC) based (see [Section 20.3](#)).

20.1.1 ADC Related Collateral

Foundational Materials

- [ADC Input Circuit Evaluation for C2000 MCUs \(TINA-TI\) Application Report](#)
- [C2000 Academy - ADC](#)
- [PSpice for TI design and simulation tool](#)
- [Real-Time Control Reference Guide](#)
 - Refer to the ADC section
- [TI Precision Labs - ADCs](#)
- [TI Precision Labs: Driving the reference input on a SAR ADC \(Video\)](#)
- [TI Precision Labs: Introduction to analog-to-digital converters \(ADCs\) \(Video\)](#)
- [TI Precision Labs: SAR ADC input driver design \(Video\)](#)
- [TI e2e: Connecting VDDA to VREFHI](#)
- [TI e2e: Topologies for ADC Input Protection](#)
- [TI e2e: Why does the ADC Input Voltage drop with sampling?](#)
 - Sampling a high impedance voltage divider with ADC
- [Understanding Data Converters Application Report](#)

Getting Started Materials

- [ADC-PWM Synchronization Using ADC Interrupt](#)
 - NOTE: This is a non-TI (third party) site.
- [Analog-to-Digital Converter \(ADC\) Training for C2000 MCUs \(Video\)](#)
- [Hardware Design Guide for F2800x C2000 Real-Time MCU Series](#)

Expert Materials

- [ADC Oversampling Application Report](#)
- [Analog Engineer's Calculator](#)
- [Analog Engineer's Pocket Reference](#)
- [Charge-Sharing Driving Circuits for C2000 ADCs \(using PSpice-FOR-TI\) Application Report](#)
- [Charge-Sharing Driving Circuits for C2000 ADCs \(using TINA-TI\) Application Report](#)
- [Debugging an integrated ADC in a microcontroller using an oscilloscope](#)
- [Methods for Mitigating ADC Memory Cross-Talk Application Report](#)
- [TI Precision Labs: ADC AC specifications \(Video\)](#)
- [TI Precision Labs: ADC Error sources \(Video\)](#)
- [TI Precision Labs: ADC Noise \(Video\)](#)
- [TI Precision Labs: Analog-to-digital converter \(ADC\) drive topologies \(Video\)](#)
- [TI Precision Labs: Electrical overstress on data converters \(Video\)](#)
- [TI Precision Labs: High-speed ADC fundamentals \(Video\)](#)
- [TI Precision Labs: SAR & Delta-Sigma: Understanding the Difference \(Video\)](#)
- [TI e2e: ADC Bandwidth Clarification](#)

- [TI e2e: ADC Calibration and Total Unadjusted Error](#)
- [TI e2e: ADC Reference Driver Options](#)
- [TI e2e: ADC Resolution with Oversampling](#)
- [TI e2e: ADC configuration for interleaved mode](#)
- [TI e2e: Simultaneous Sampling with Single ADC](#)

20.1.2 Features

Each ADC has the following features:

- Selectable resolution of 12 bits or 16 bits
- Ratiometric external reference set by VREFHI and VREFLO pins
- Differential signal conversions (16-bit mode only)
- Single-ended signal conversions
- Input multiplexer with up to 16 channels (single-ended) or 8 channels (differential)
- 16 configurable SOC's
- 16 individually addressable result registers
- Multiple trigger sources
 - S/W - software immediate start
 - All ePWMs - ADCSOC A or B
 - GPIO XINT2
 - CPU Timers 0/1/2 (from each C28x core present)
 - ADCINT1/2
- Four flexible PIE interrupts
- Configurable interrupt placement
- Burst mode
- Four post-processing blocks, each with:
 - Saturating offset calibration
 - Error from set-point calculation
 - High, low, and zero-crossing compare, with interrupt and ePWM trip capability
 - Trigger-to-sample delay capture

Note

Not every channel is pinned out from all ADCs. Check the device data sheet to determine which channels are available.

20.1.3 Block Diagram

Figure 20-1 shows the block diagram for the ADC core and ADC wrapper.

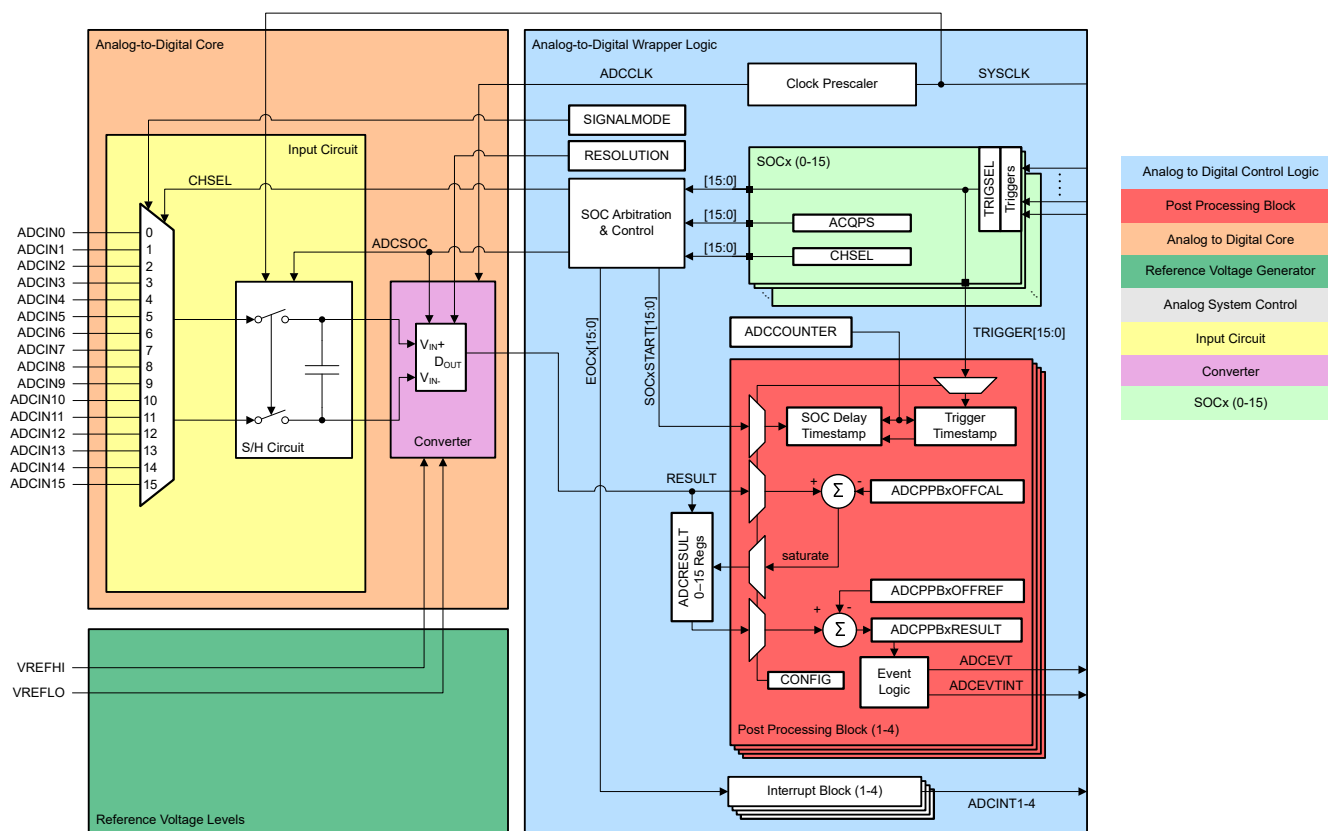


Figure 20-1. ADC Module Block Diagram

Note

The ADC block diagram reflects the number of ADC channels internally configurable on the device. The actual number of available external ADC inputs varies depending on part and package.

20.2 ADC Configurability

Some ADC configurations are individually controlled by the SOC's, while others are globally controlled per ADC module. [Table 20-1](#) summarizes the basic ADC options and the level of configurability. The subsequent sections discuss these configurations.

Table 20-1. ADC Options and Configuration Levels

Options	Configurability
Clock	Per module ⁽¹⁾
Resolution	Per module ⁽¹⁾
Signal mode	Per module
Reference voltage source	Not configurable (external reference only)
Trigger source	Per SOC ⁽¹⁾
Converted channel	Per SOC
Acquisition window duration	Per SOC ⁽¹⁾
EOC location	Per module
Burst Mode	Per module ⁽¹⁾

(1) Writing these values differently to different ADC modules can cause the ADCs to operate asynchronously. See [Section 20.13.1](#) for guidance on when the ADCs are operating synchronously or asynchronously.

20.2.1 Clock Configuration

The base ADC clock is provided directly by the system clock (SYSCLK). SYSCLK is used to generate the ADC acquisition window. The register ADCCTL2 has a PRESCALE field that determines the ADCCLK. ADCCLK is used to clock the converter, and is only active during the conversion phase. At all other times, including during the sample-and-hold window, the ADCCLK signal is gated off.

In 16-bit mode, the core requires approximately 29.5 ADCCLK cycles to process a voltage into a conversion result, while in 12-bit mode, this process requires approximately 10.5 ADCCLK cycles. The choice of resolution also determines the necessary duration of the acquisition window, see [Section 20.13.2](#).

Note

To determine an appropriate value for ADCCTL2.PRESCALE, see the device data sheet to determine the maximum SYSCLK and ADCCLK frequency.

20.2.2 Resolution

The resolution of the ADC determines how finely the analog range is quantized into digital values. Each ADC module supports a configurable resolution of 16 bits or 12 bits.

The resolution must be configured by using either the `AdcSetMode()` or `ADC_setMode()` functions, depending on the header files used, provided in C2000ware in `F2838x_Adc.c`. These functions make sure that the correct trim is loaded into the ADC trim registers, and must be called at least once after a device reset. Do not configure the resolution by directly writing to the ADCCTL2 register.

The resolution can be changed at any time when the ADC is idle (no active or pending SOC's). No wait time is necessary after changing the resolution before conversions can be initiated. If SOC's are active or pending when the resolution is changed, those SOC's can produce incorrect conversion results.

20.2.3 Voltage Reference

20.2.3.1 External Reference Mode

Each ADC has a VREFHI input and a VREFLO input. In external reference mode, these pins are used as a ratiometric reference to determine the ADC conversion input range.

See [Section 20.13.6](#) for information on how to supply the reference voltage.

Note

- On devices with no external VREFLO pin, VREFLO is internally connected to the device analog ground, VSSA.
 - See the device data sheet to determine the allowable voltage range for VREFHI and VREFLO.
 - The external reference mode requires an external capacitor on the VREFHI pin. See the device data sheet for the specific value required.
-

20.2.4 Signal Mode

The ADC supports two signal modes: single-ended and differential.

In single-ended mode, the input voltage to the converter is sampled through a single pin (ADCINx), referenced to VREFLO.

In differential signaling mode, the input voltage to the converter is sampled through a pair of input pins, one of which is the positive input (ADCINxP) and the other is the negative input (ADCINxN). The actual input voltage is the difference between the two (ADCINxP – ADCINxN).

Note

- In differential signal mode, VREFLO must be connected to VSSA.
- In differential signal mode, the common mode voltage is $V_{CM} = (ADCINxP + ADCINxN)/2$

The data sheet for a particular device places some requirements on how close this voltage needs to be to: $(VREFHI + VREFLO)/2$

Note: The above condition is not met by connecting the negative input to VSSA or VREFLO.

- Differential signaling mode is advantageous because noise encountered on both inputs is largely canceled. The effect can be maximized by routing the positive and negative traces for the same differential input as close together as possible and keeping them symmetrical with respect to the signal reference.
-

The signal mode must be configured by using either the ADC_setMode() or AdcSetMode() functions, depending on the header files used, provided in C2000ware in F2838x_Adc.c. These functions make sure that the correct trim is loaded into the ADC trim registers. These functions must be called at least once after a device reset. The signal mode must not be configured by writing to the ADCCTL2 register directly.

20.2.5 Expected Conversion Results

Based on a given analog input voltage, the expected digital conversion is given in [Table 20-2](#) and [Table 20-3](#). Fractional values are truncated.

Table 20-2. Analog to 12-bit Digital Formulas

	Analog Input	Digital Result
Single-Ended	when $ADCIN_y \leq VREFLO$	$ADCRESULT_x = 0$
	when $VREFLO < ADCIN_y < VREFHI$	$ADCRESULT_x = 4096 \left(\frac{ADCIN_y - VREFLO}{VREFHI - VREFLO} \right)$
	when $ADCIN_y \geq VREFHI$	$ADCRESULT_x = 4095$
Differential	when $ADCIN_yP - ADCIN_yN \leq -(VREFHI - VREFLO)$	$ADCRESULT_x = 0$
	when $-(VREFHI - VREFLO) < ADCIN_yP - ADCIN_yN \leq (VREFHI - VREFLO)$	$ADCRESULT_x = 4096 \left(\frac{ADCIN_yP - ADCIN_yN + VREFHI - VREFLO}{2(VREFHI - VREFLO)} \right)$
	when $ADCIN_yP - ADCIN_yN \geq (VREFHI - VREFLO)$	$ADCRESULT_x = 4095$

Table 20-3. Analog to 16-bit Digital Formulas

	Analog Input	Digital Result
Single-Ended	when $ADCIN_y \leq VREFLO$	$ADCRESULT_x = 0$
	when $VREFLO < ADCIN_y < VREFHI$	$ADCRESULT_x = 65536 \left(\frac{ADCIN_y - VREFLO}{VREFHI - VREFLO} \right)$
	when $ADCIN_y \geq VREFHI$	$ADCRESULT_x = 65535$
Differential	when $ADCIN_yP - ADCIN_yN \leq -(VREFHI - VREFLO)$	$ADCRESULT_x = 0$
	when $-(VREFHI - VREFLO) < ADCIN_yP - ADCIN_yN \leq (VREFHI - VREFLO)$	$ADCRESULT_x = 65536 \left(\frac{ADCIN_yP - ADCIN_yN + VREFHI - VREFLO}{2(VREFHI - VREFLO)} \right)$
	when $ADCIN_yP - ADCIN_yN \geq (VREFHI - VREFLO)$	$ADCRESULT_x = 65535$

20.2.6 Interpreting Conversion Results

Based on a given ADC conversion result, the corresponding analog input is given in [Table 20-4](#) and [Table 20-5](#). This corresponds to the center of the possible range of analog voltages that can produce this conversion result.

Table 20-4. 12-Bit Digital-to-Analog Formulas

	Digital Value	Analog Equivalent
Single-Ended	when ADCRESULTy = 0	$ADCIN_x \leq VREFLO$ (2)
	when $0 < ADCRESULTy < 4095$	$ADCIN_x = (VREFHI - VREFLO) \left(\frac{ADCRESULTy}{4096} \right) + VREFLO$ (3)
	when ADCRESULTy = 4095	$ADCIN_x \geq VREFHI$ (4)
Differential	when ADCRESULTy = 0	$ADCIN_{xP} - ADCIN_{xN} \leq (VREFHI - VREFLO)$ (5)
	when $0 < ADCRESULTy < 4095$	$ADCIN_{xP} - ADCIN_{xN} = (VREFHI - VREFLO) \left(\frac{ADCRESULTy}{2048} - 1 \right)$ (6)
	when ADCRESULTy = 4095	$ADCIN_{xP} - ADCIN_{xN} \geq (VREFHI - VREFLO)$ (7)

Table 20-5. 16-Bit Digital-to-Analog Formulas

	Digital Value	Analog Equivalent
Single-Ended	when ADCRESULTy = 0	$ADCIN_x \leq VREFLO$ (8)
	when $0 < ADCRESULTy < 65535$	$ADCIN_x = (VREFHI - VREFLO) \left(\frac{ADCRESULTy}{65536} \right) + VREFLO$ (9)
	when ADCRESULTy = 65535	$ADCIN_x \geq VREFHI$ (10)
Differential	when ADCRESULTy = 0	$ADCIN_{xP} - ADCIN_{xN} \leq (VREFHI - VREFLO)$ (11)
	when $0 < ADCRESULTy < 65535$	$ADCIN_{xP} - ADCIN_{xN} = (VREFHI - VREFLO) \left(\frac{ADCRESULTy}{32768} - 1 \right)$ (12)
	when ADCRESULTy = 65535	$ADCIN_{xP} - ADCIN_{xN} \geq (VREFHI - VREFLO)$ (13)

20.3 SOC Principle of Operation

The ADC triggering and conversion sequencing is accomplished through configurable start-of-conversions (SOCs). Each SOC is a configuration set defining the single conversion of a single channel. In that set, there are three configurations: the trigger source that starts the conversion, the channel to convert, and the acquisition (sample) window duration. Upon receiving the trigger configured for a SOC, the wrapper makes sure that the specified channel is captured using the specified acquisition window duration.

Multiple SOC's can be configured for the same trigger, channel, and acquisition window as desired. Configuring multiple SOC's to use the same trigger allows the trigger to generate a sequence of conversions. Configuring multiple SOC's to use the same trigger and channel allows for oversampling.

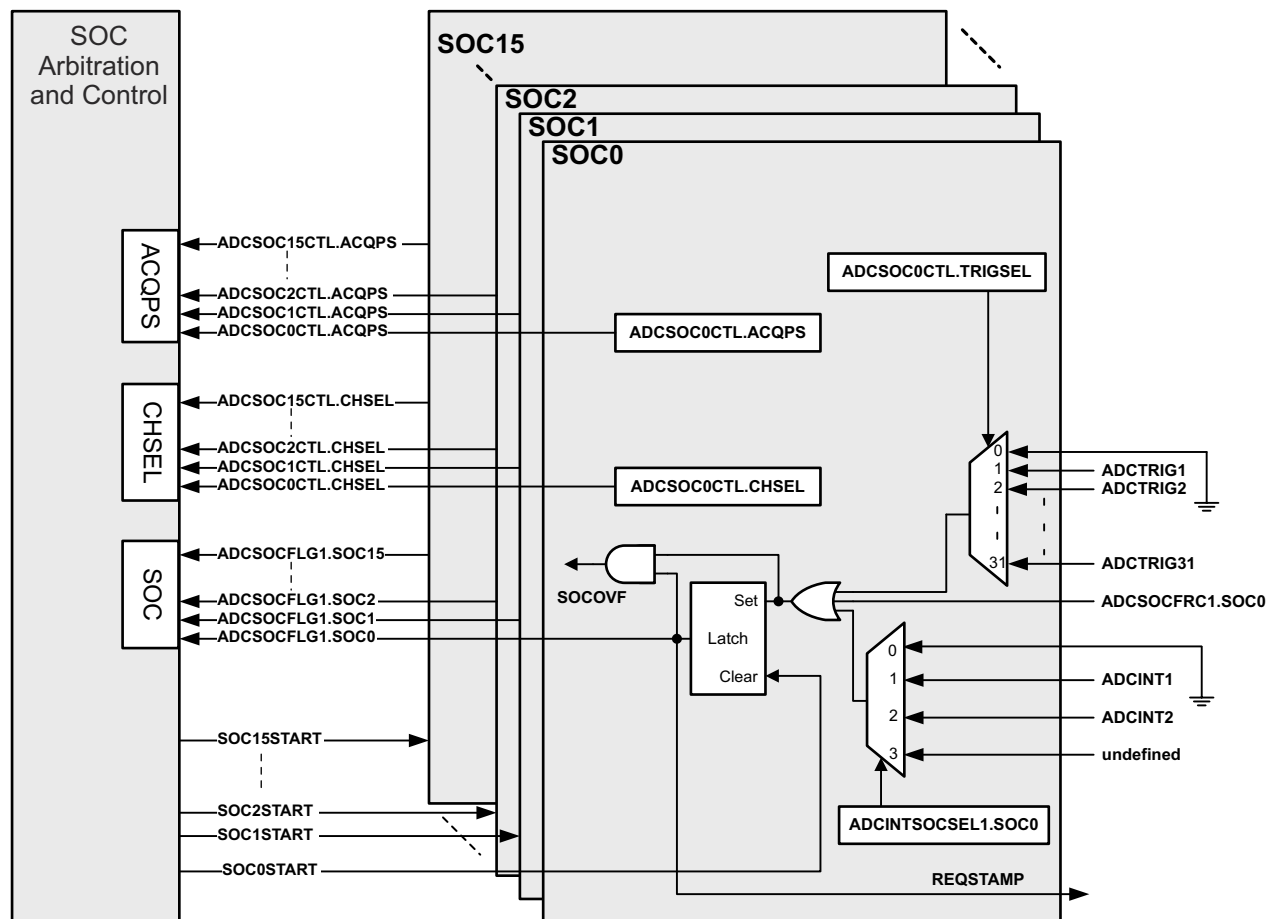


Figure 20-2. SOC Block Diagram

20.3.1 SOC Configuration

Each SOC has a configuration register, ADCSOCxCTL. Within this register, SOCx can be configured for trigger source, channel to convert, and acquisition (sample) window duration.

20.3.2 Trigger Operation

Each SOC can be configured to start on one of many input triggers. The primary trigger select for SOCx is in the ADCSOCxCTL.TRIGSEL register, which can select between:

- Disabled (software only)
- CPU Timers 0/1/2 (from each C28x core present)
- GPIO: Input X-Bar INPUT5
- ADCSOCA or ADCSOCB from each ePWM module

In addition, each SOC can also be triggered when the ADCINT1 flag or ADCINT2 flag is set. This is achieved by configuring the ADCINTSOCSEL1 register (for SOC0 to SOC7) or the ADCINTSOCSEL2 register (for SOC8 to SOC15). This is useful for creating continuous conversions.

20.3.3 ADC Acquisition (Sample and Hold) Window

External signal sources vary in the ability to drive an analog signal quickly and effectively. To achieve rated resolution, the signal source needs to charge the sampling capacitor in the ADC core to within 0.5 LSBs of the signal voltage. The acquisition window is the amount of time the sampling capacitor is allowed to charge and is configurable for SOCx by the ADCSOCxCTL.ACQPS register.

ACQPS is a 9-bit register that can be set to a value between 0 and 511, resulting in an acquisition window duration of:

Acquisition window = (ACQPS + 1) * (System Clock (SYSCLK) cycle time)

- The acquisition window duration is based on the System Clock (SYSCLK), not the ADC clock (ADCCLK).
- The selected acquisition window duration must be at least as long as one ADCCLK cycle.
- The data sheet specifies a minimum acquisition window duration (in nanoseconds). The user is responsible for selecting an acquisition window duration that meets this requirement.

20.3.4 ADC Input Models

For single-ended operation, the ADC input characteristics for values in the single-ended input model (see [Figure 20-3](#)) can be found in the device data sheet.

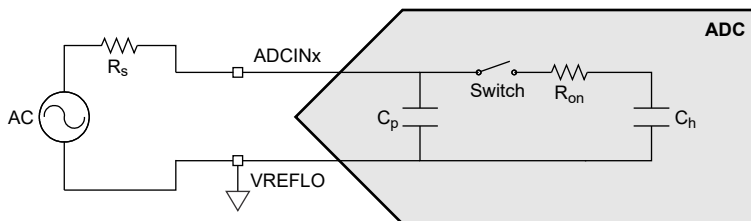


Figure 20-3. Single-Ended Input Model

For differential operation, the ADC input characteristics for values in the differential input model (see [Figure 20-4](#)) can be found in the device data sheet.

These input models must be used along with actual signal source impedance to determine the acquisition window duration. See [Section 20.13.2](#) for more information.

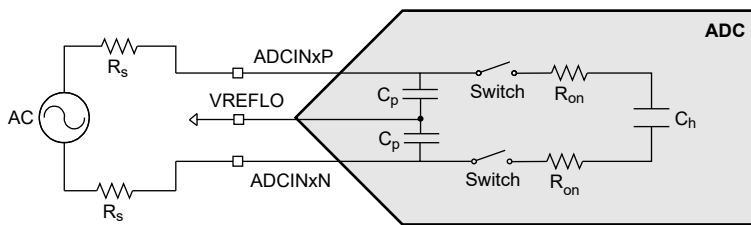


Figure 20-4. Differential Input Model

20.3.5 Channel Selection

Each SOC can be configured to convert any of the ADC channels. This behavior is selected for SOCx by the ADCSOCxCTL.CHSEL register. Depending on the signal mode, the selection is different. For single-ended signal mode, the value in CHSEL selects a single pin as the input. For differential signal mode, the value in CHSEL selects an even-odd pin pair to be the positive and negative inputs. This is summarized in [Table 20-6](#).

Note

Regardless of configured resolution and signal mode, channel 13 on ADC-A (temperature sensor) and channel 12 on all ADCs are sampled in 12-bit single-ended mode.

Table 20-6. Channel Selection of Input Pins

Input Mode	CHSEL	Input	
Single-Ended	0	ADCIN0	
	1	ADCIN1	
	2	ADCIN2	
	3	ADCIN3	
	4	ADCIN4	
	5	ADCIN5	
	6	ADCIN6	
	7	ADCIN7	
	8	ADCIN8	
	9	ADCIN9	
	10	ADCIN10	
	11	ADCIN11	
	12	ADCIN12	
	13	ADCIN13	
	14	ADCIN14	
	15	ADCIN15	
	CHSEL	Positive Input	Negative Input
Differential	0 or 1	ADCIN0	ADCIN1
	2 or 3	ADCIN2	ADCIN3
	4 or 5	ADCIN4	ADCIN5
	6 or 7	ADCIN6	ADCIN7
	8 or 9	ADCIN8	ADCIN9
	10 or 11	ADCIN10	ADCIN11
	12 or 13	ADCIN12	ADCIN13
	14 or 15	ADCIN14	ADCIN15

20.4 SOC Configuration Examples

The following sections provide some specific examples of how to configure the SOC5s to produce some conversions.

20.4.1 Single Conversion from ePWM Trigger

To configure ADC5 to perform a single conversion on channel ADCINA1 when the ePWM timer reaches the period match, a few things are necessary. First, ePWM3 must be configured to generate an SOCA or SOCB signal (in this statement, SOC refers to a signal in the ePWM module). See the *Enhanced Pulse Width Modulator Module (ePWM)* chapter on how to do this. Assume that SOCB was chosen.

SOC5 is chosen arbitrarily. Any of the 16 SOC5s can be used.

Assuming a 100ns sample window is desired with a SYSCLK frequency of 200MHz, then the acquisition window duration must be $100\text{ns}/5\text{ns} = 20$ SYSCLK cycles. The ACQPS field must be set to $20 - 1 = 19$.

```
AdcRegs.ADCSOC5CTL.bit.CHSEL = 1;      //SOC5 converts ADCINA1
AdcRegs.ADCSOC5CTL.bit.ACQPS = 19;     //SOC5 uses a sample duration of 20 SYSCLK cycles
AdcRegs.ADCSOC5CTL.bit.TRIGSEL = 10;   //SOC5 begins conversion on ePWM3 SOCB
```

As configured, when ePWM3 matches the period and generates the SOCB signal, the ADC begins sampling channel ADCINA1 (SOC5) immediately if the ADC is idle. If the ADC is busy, ADCINA1 begins sampling when SOC5 gains priority (see [Section 20.5](#)). The ADC control logic samples ADCINA1 with the specified acquisition window width of 100ns. Immediately after the acquisition is complete, the ADC begins converting the sampled voltage to a digital value. When the ADC conversion is complete, the results are available in the ADCRESULT5 register (see [Section 20.12](#) for exact sample, conversion, and result latch timings).

20.4.2 Oversampled Conversion from ePWM Trigger

To configure the ADC to oversample ADCINA1 4 times, we use the same configurations as the previous example, but apply them to SOC5, SOC6, SOC7, and SOC8.

```
AdcRegs.ADCSOC5CTL.bit.CHSEL = 1;      //SOC5 converts ADCINA1
AdcRegs.ADCSOC5CTL.bit.ACQPS = 19;     //SOC5 uses a sample duration of 20 SYSCLK cycles
AdcRegs.ADCSOC5CTL.bit.TRIGSEL = 10;   //SOC5 begins conversion on ePWM3 SOCB
AdcRegs.ADCSOC6CTL.bit.CHSEL = 1;      //SOC6 converts ADCINA1
AdcRegs.ADCSOC6CTL.bit.ACQPS = 19;     //SOC6 uses a sample duration of 20 SYSCLK cycles
AdcRegs.ADCSOC6CTL.bit.TRIGSEL = 10;   //SOC6 begins conversion on ePWM3 SOCB
AdcRegs.ADCSOC7CTL.bit.CHSEL = 1;      //SOC7 converts ADCINA1
AdcRegs.ADCSOC7CTL.bit.ACQPS = 19;     //SOC7 uses a sample duration of 20 SYSCLK cycles
AdcRegs.ADCSOC7CTL.bit.TRIGSEL = 10;   //SOC7 begins conversion on ePWM3 SOCB
AdcRegs.ADCSOC8CTL.bit.CHSEL = 1;      //SOC8 converts ADCINA1
AdcRegs.ADCSOC8CTL.bit.ACQPS = 19;     //SOC8 uses a sample duration of 20 SYSCLK cycles
AdcRegs.ADCSOC8CTL.bit.TRIGSEL = 10;   //SOC8 begins conversion on ePWM3 SOCB
```

As configured, when ePWM3 matches the period and generates the SOCB signal, the ADC begins sampling channel ADCINA1 (SOC5) immediately if the ADC is idle. If the ADC is busy, ADCINA1 begins sampling when SOC5 gains priority (see [Section 20.5](#)). Once the conversion is complete for SOC5, SOC6 begins converting ADCINA1 and the results for SOC5 are placed in the ADCRESULT5 register. All four conversions eventually are completed sequentially, with the results in ADCRESULT5, ADCRESULT6, ADCRESULT7, and ADCRESULT8 for SOC5, SOC6, SOC7, and SOC8, respectively.

Note

It is possible, but unlikely, that the ADC can begin converting SOC6, SOC7, or SOC8 before SOC5 depending on the position of the round-robin pointer when the ePWM trigger is received. See [Section 20.5](#) to understand how the next SOC to be converted is chosen.

20.4.3 Multiple Conversions from CPU Timer Trigger

This example shows how to sample multiple signals with different acquisition window requirements. CPU1 Timer 2 is used to generate the trigger. To see how to configure the CPU timer, see the *System Control and Interrupts* chapter.

A good first step when designing a sampling scheme with many signals is to list out the signals and the required acquisition window. From this, calculate the necessary number of SYSCLK cycles for each signal, then the ACQPS register setting. This is shown in [Table 20-7](#), where a SYCLK of 200MHz is assumed (5ns cycle time).

Table 20-7. Example Requirements for Multiple Signal Sampling

Signal Name	Acquisition Window Requirement	Acquisition Window (SYSCLK Cycles)	ACQPS Register Value
Signal 1	>120ns	120ns/5ns = 24	24 – 1 = 23
Signal 2	>444ns	444ns/5ns = 89 (round up)	89 – 1 = 88
Signal 3	>110ns	110ns/5ns = 22	22 – 1 = 21
Signal 4	>291ns	291ns/5ns = 59 (round up)	59 – 1 = 58

Next decide which ADC pins to connect to each signal. This is highly dependent on the application board layout. Once the pins are selected, determining the value of CHSEL is straightforward (see [Table 20-8](#)).

Table 20-8. Example Connections for Multiple Signal Sampling

Signal Name	ADC Pin	CHSEL Register Value
Signal 1	ADCINA5	5
Signal 2	ADCINA0	0
Signal 3	ADCINA3	3
Signal 4	ADCINA2	2

With the information tabulated, generate the SOC configurations:

```
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 5;           //SOC0 converts ADCINA5
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 23;          //SOC0 uses a sample duration of 24 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 3;         //SOC0 begins conversion on CPU1 Timer 2
AdcaRegs.ADCSOC1CTL.bit.CHSEL = 0;           //SOC1 converts ADCINA0
AdcaRegs.ADCSOC1CTL.bit.ACQPS = 88;          //SOC1 uses a sample duration of 89 SYSCLK cycles
AdcaRegs.ADCSOC1CTL.bit.TRIGSEL = 3;         //SOC1 begins conversion on CPU1 Timer 2
AdcaRegs.ADCSOC2CTL.bit.CHSEL = 3;           //SOC2 converts ADCINA3
AdcaRegs.ADCSOC2CTL.bit.ACQPS = 21;          //SOC2 uses a sample duration of 22 SYSCLK cycles
AdcaRegs.ADCSOC2CTL.bit.TRIGSEL = 3;         //SOC2 begins conversion on CPU1 Timer 2
AdcaRegs.ADCSOC3CTL.bit.CHSEL = 2;           //SOC3 converts ADCINA2
AdcaRegs.ADCSOC3CTL.bit.ACQPS = 58;          //SOC3 uses a sample duration of 59 SYSCLK cycles
AdcaRegs.ADCSOC3CTL.bit.TRIGSEL = 3;         //SOC3 begins conversion on CPU1 Timer 2
```

As configured, when CPU1 Timer 2 generates an event, SOC0, SOC1, SOC2, and SOC3 eventually is sampled and converted, in that order. The conversion results for ACINA5 (Signal 1) are in ADCRESULT0. Similarly, The results for ADCINA0 (Signal 2), ADCINA3 (Signal 3), and ADCINA2 (Signal 4) are in ADCRESULT1, ADCRESULT2, and ADCRESULT3, respectively.

Note

There is a possibility, but unlikely, that the ADC can begin converting SOC1, SOC2, or SOC3 before SOC0 depending on the position of the round-robin pointer when the CPU Timer trigger is received. See [Section 20.5](#) to understand how the next SOC to be converted is chosen.

20.4.4 Software Triggering of SOC's

At any point, whether or not the SOC's have been configured to accept a specific trigger, a software trigger can set the SOC's to be converted. This is accomplished by writing bits in the ADCSOCFRC1 register.

Software triggering of the previous example without waiting for the CPU1 Timer 2 to generate the trigger can be accomplished by the statement:

```
AdcaRegs.ADCSOCFRC1.all = 0x000F;           //set SOC flags for SOC0 to SOC3
```

20.5 ADC Conversion Priority

When multiple SOC flags are set at the same time, one of two forms of priority determines the converted order. The default priority method is round-robin. In this scheme, no SOC has an inherent higher priority than another. Priority depends on the round-robin pointer (RRPOINTER). The RRPOINTER reflected in the ADCSOCPRIORITYCTL register points to the last SOC converted. The highest priority SOC is given to the next value greater than the RRPOINTER value, wrapping around back to SOC0 after SOC15. At reset the value is 16 since 0 indicates a conversion has already occurred. When RRPOINTER equals 16 the highest priority is given to SOC0. The RRPOINTER is reset when the ADC module is reset or when the reset value is written to the SOCPRCTL register. The ADC module is reset by writing and clearing the SOFTPRES bit corresponding to the ADC instance.

An example of the round-robin priority method is given in [Figure 20-5](#).

The SOC PRIORITY field in the ADCSOCPRIORITYCTL register can be used to assign high priority from a single to all of the SOC's. When configured as high priority, an SOC interrupts the round-robin wheel after any current conversion completes and inserts in as the next conversion. After the conversion completes, the round-robin wheel continues where the conversion was interrupted. If two high priority SOC's are triggered at the same time, the SOC with the lower number takes precedence.

High priority mode is assigned first to SOC0, then in increasing numerical order. The value written in the SOC PRIORITY field defines the first SOC that is not high priority. In other words, if a value of 4 is written into SOC PRIORITY, then SOC0, SOC1, SOC2, and SOC3 are defined as high priority, with SOC0 the highest.

An example using high priority SOC's is given in [Figure 20-6](#).

- A** After reset, SOC0 is highest priority SOC ;
SOC7 receives trigger;
SOC7 configured channel is converted immediately .
- B** RRPOINTER changes to point to SOC 7;
SOC8 is now highest priority SOC .
- C** SOC2 & SOC12 triggers rcvd. simultaneously ;
SOC12 is first on round robin wheel ;
SOC12 configured channel is converted while
SOC2 stays pending .
- D** RRPOINTER changes to point to SOC 12;
SOC2 configured channel is now converted .
- E** RRPOINTER changes to point to SOC 2;
SOC3 is now highest priority SOC .

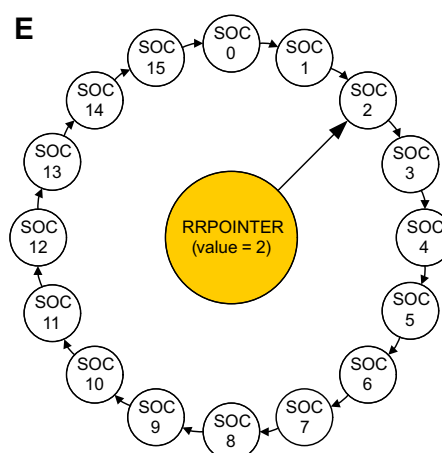
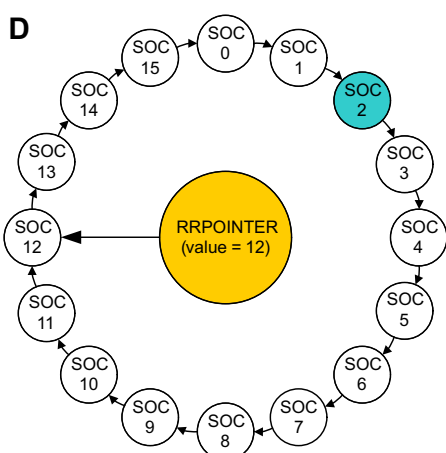
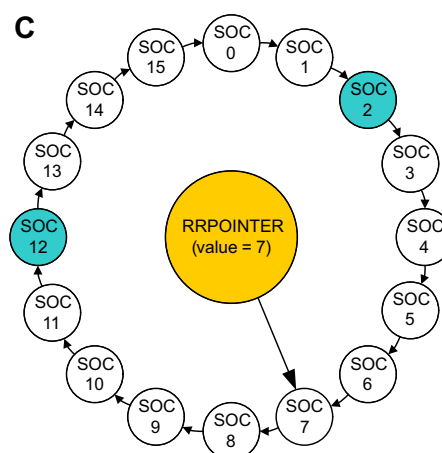
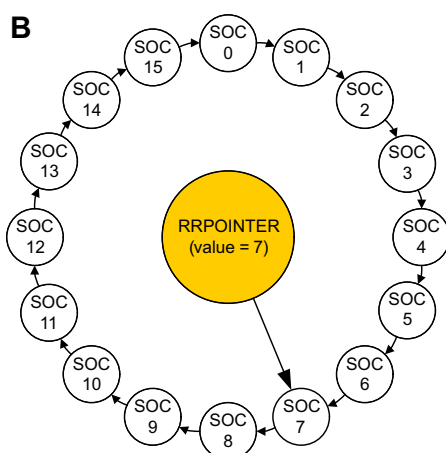
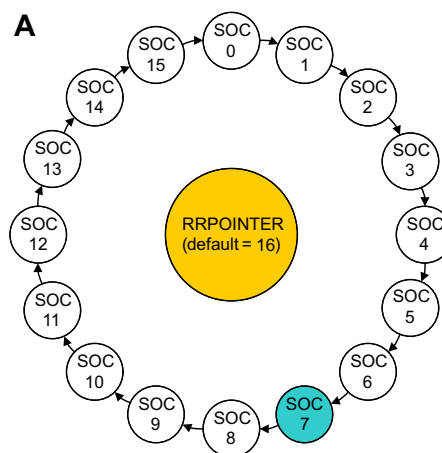


Figure 20-5. Round Robin Priority Example

Example when $SOC_{PRIORITY} = 4$

- A** After reset, SOC4 is 1st on round robin wheel ;
SOC7 receives trigger ;
SOC7 configured channel is converted immediately .
- B** RRPOINTER changes to point to SOC 7 ;
SOC8 is now 1st on round robin wheel .
- C** SOC2 & SOC12 triggers rcvd. simultaneously ;
SOC2 interrupts round robin wheel and SOC 2 configured channel is converted while SOC 12 stays pending .
- D** RRPOINTER stays pointing to 7 ;
SOC12 configured channel is now converted .
- E** RRPOINTER changes to point to SOC 12 ;
SOC13 is now 1st on round robin wheel .

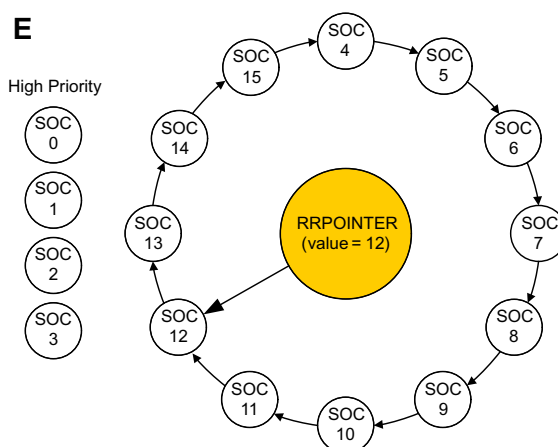
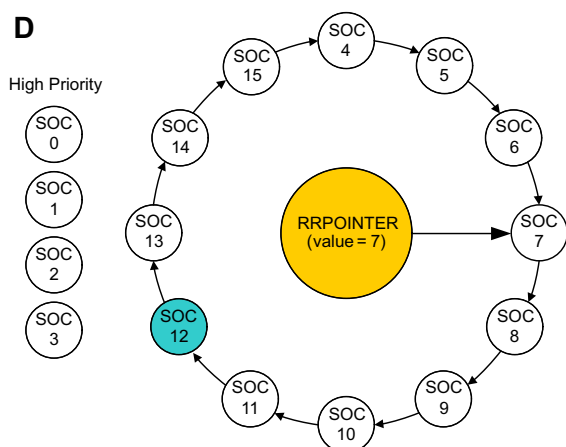
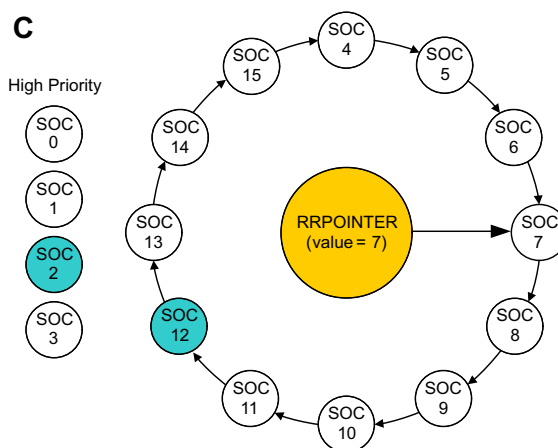
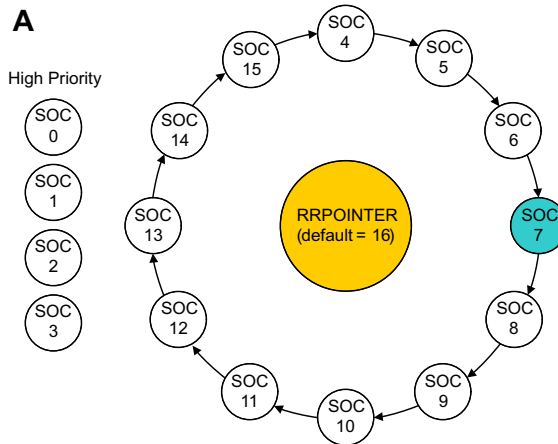
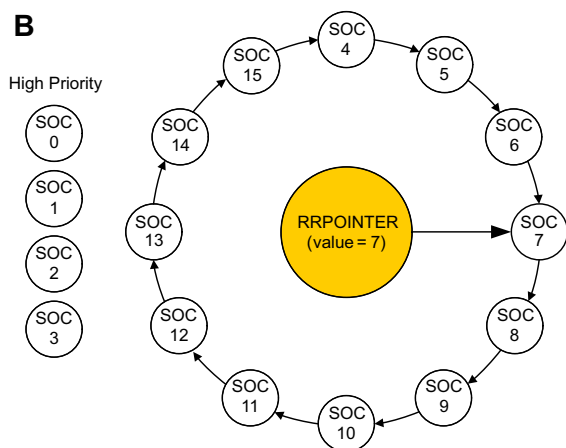


Figure 20-6. High Priority Example

20.6 Burst Mode

Burst mode allows a single trigger to walk through the round-robin SOC's one or more at a time. Setting the bit BURSTEN in the ADCBURSTCTL register configures the ADC wrapper for burst mode. This causes the TRIGSEL field to be ignored, but only for SOC's that are configured for round-robin operation (not high priority). Instead of the TRIGSEL field, all round-robin SOC's are triggered based on the BURSTTRIG field in the ADCBURSTCTL register. Upon reception of the burst trigger, the ADC wrapper does not set all round-robin SOC's to be converted, but only (ADCBURSTCTL.BURSTSIZE + 1) SOC's. The first SOC to be set is the SOC with the highest priority based on the round-robin pointer, and subsequent SOC's are set until BURSTSIZE SOC's have been set.

Note

When configuring the ADC for burst mode, the user is responsible for ensuring that each burst of conversions is allowed to complete before the next burst trigger is received. The value of (ADCBURSTCTL.BURSTSIZE + 1) must be less than or equal to the number of SOC's configured for round-robin priority.

For example, if SOC PRIORITY = 12, that is, SOC12, SOC13, SOC14, and SOC15 are in round-robin, ADCBURSTCTL.BURSTSIZE setting must be ≤ 3 for burst mode to operate correctly.

20.6.1 Burst Mode Example

Burst mode can be used to sample a different set of signals on every other trigger. In the following example, ADCIN7 and ADCIN5 are converted on the first trigger from CPU1 Timer 2 and every other trigger thereafter. ADCIN2 and ACIN3 are converted on the second trigger from CPU1 Timer 2 and every other trigger thereafter. All signals are converted with 20 SYSCLK cycle wide acquisition windows, but different durations can be configured for each SOC as desired.

```
AdcaRegs.BURSTCTL.BURSTEN = 1;           //Enable ADC burst mode
AdcaRegs.BURSTCTL.BURSTTRIG = 3;          //CPU1 Timer 2 triggers burst of conversions
AdcaRegs.BURSTCTL.BURSTSIZE = 1;          //conversion bursts are 1 + 1 = 2 conversions long
AdcaRegs.SOCPRCTL.bit.SOC PRIORITY = 12; //SOC0 to SOC11 are high priority
AdcaRegs.ADCSOC12CTL.bit.CHSEL = 7;        //SOC12 converts ADCINA7
AdcaRegs.ADCSOC12CTL.bit.ACQPS = 19;       //SOC12 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC13CTL.bit.CHSEL = 5;        //SOC13 converts ADCINA5
AdcaRegs.ADCSOC13CTL.bit.ACQPS = 19;       //SOC13 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC14CTL.bit.CHSEL = 2;        //SOC14 converts ADCINA2
AdcaRegs.ADCSOC14CTL.bit.ACQPS = 19;       //SOC14 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC15CTL.bit.CHSEL = 3;        //SOC15 converts ADCINA3
AdcaRegs.ADCSOC15CTL.bit.ACQPS = 19;       //SOC15 uses sample duration of 20 SYSCLK cycles
```

When the first CPU1 Timer 2 trigger is received, SOC12 and SOC13 are converted immediately if the ADC is idle. If the ADC is busy, SOC12 and SOC13 are converted once the SOC's gain priority. The results for SOC12 and SOC13 are in ADCRESULT12 and ADCRESULT13, respectively. After SOC13 completes, the round-robin pointer gives the highest priority to SOC14. Because of this, when the next CPU1 Timer 2 trigger is received, SOC14 and SOC15 is set as pending and eventually converted. The results for SOC14 and SOC15 are in ADCRESULT14 and ADCRESULT15, respectively. Subsequent triggers continue to toggle between converting SOC12 and SOC13, and converting SOC14 and SOC15.

While the above example toggles between two sets of conversions, three or more different sets of conversions can be achieved using a similar approach.

20.6.2 Burst Mode Priority Example

An example of priority resolution using burst mode and high-priority SOC is presented in [Figure 20-7](#).

Example when SOC PRIORITY = 4, BURSTEN = 1, and BURSTSIZE = 1

- A** After reset, SOC4 is 1st on round robin wheel; BURSTTRIG trigger is received; SOC4 & SOC5 are set and configured channels converted immediately.
- B** RRPOINTER changes to point to SOC5; SOC6 is now 1st on round robin wheel.
- C** BURSTTRIG & SOC1 triggers rcvd. simultaneously; SOC1, SOC6, and SOC7 are set; SOC1 interrupts round robin wheel and SOC1 configured channel is converted while SOC6 and SOC7 stay pending.
- D** RRPOINTER stays pointing to 5; SOC6/SOC7 configured channels are now converted.
- E** RRPOINTER changes to point to SOC7; SOC8 is now 1st on round robin wheel, waiting for BURSTTRIG.

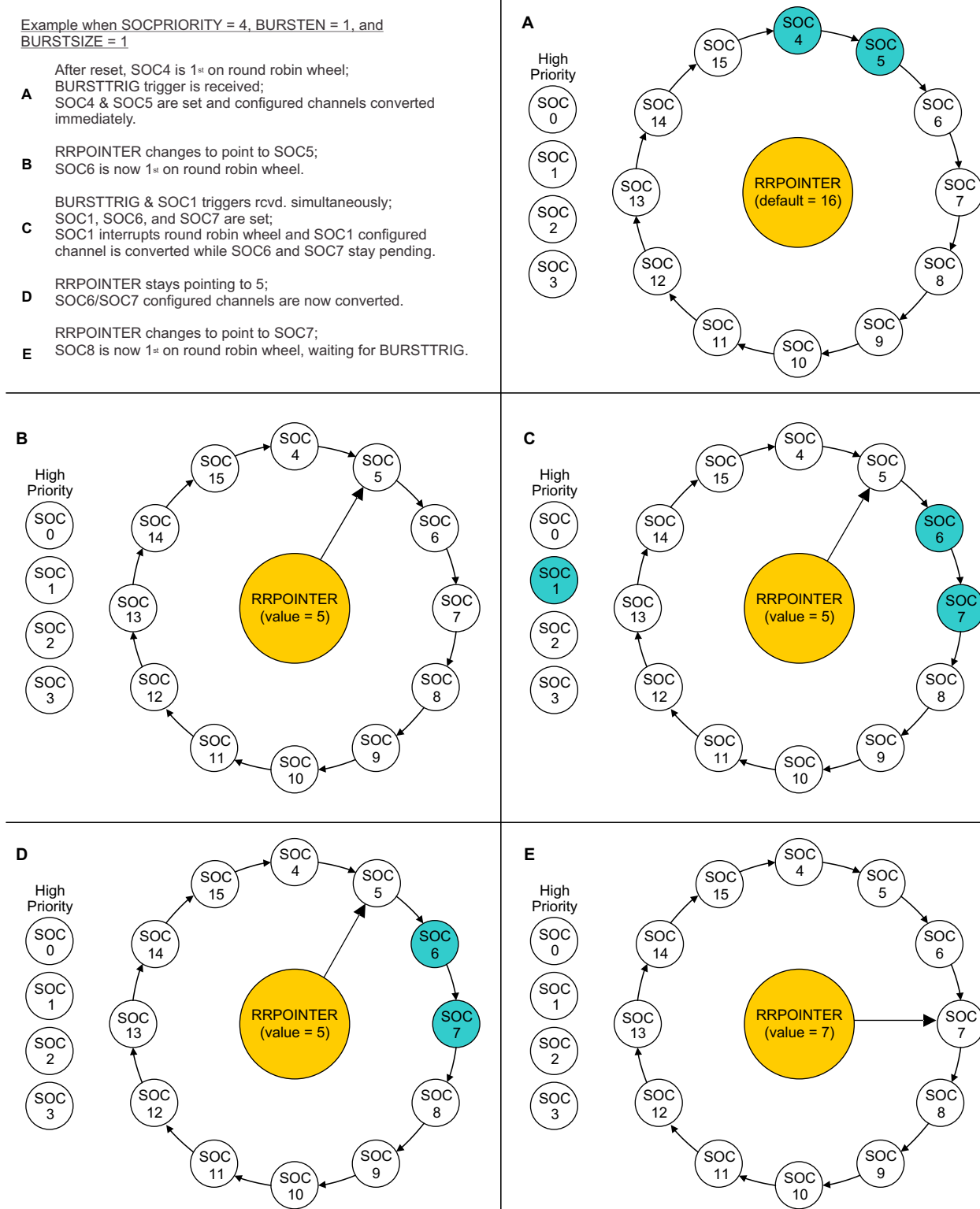


Figure 20-7. Burst Priority Example

20.7 EOC and Interrupt Operation

Each SOC has a corresponding end-of-conversion (EOC) signal. This EOC signal can be used to trigger an ADC interrupt. The ADC can be configured to generate the EOC pulse at either the end of the acquisition window or at the end of the voltage conversion. This is configured using the bit `INTPULSEPOS` in the `ADCCTL1` register. See [Section 20.12](#) for exact EOC pulse location.

Each ADC module has 4 configurable ADC interrupts. These interrupts can be triggered by any of the 16 EOC signals. The flag bit for each `ADCINT` can be read directly to determine if the associated SOC is complete or the interrupt can be passed on to the PIE.

Note

The `ADCCTL1.ADCBSY` bit being clear does not indicate that all conversions in a set of SOC's have completed, only that the ADC is ready to process the next conversion. To determine if a sequence of SOC's is complete, link an `ADCINT` flag to the last SOC in the sequence and monitor that `ADCINT` flag.

Figure 20-8 shows a block diagram of the ADC interrupt structure.

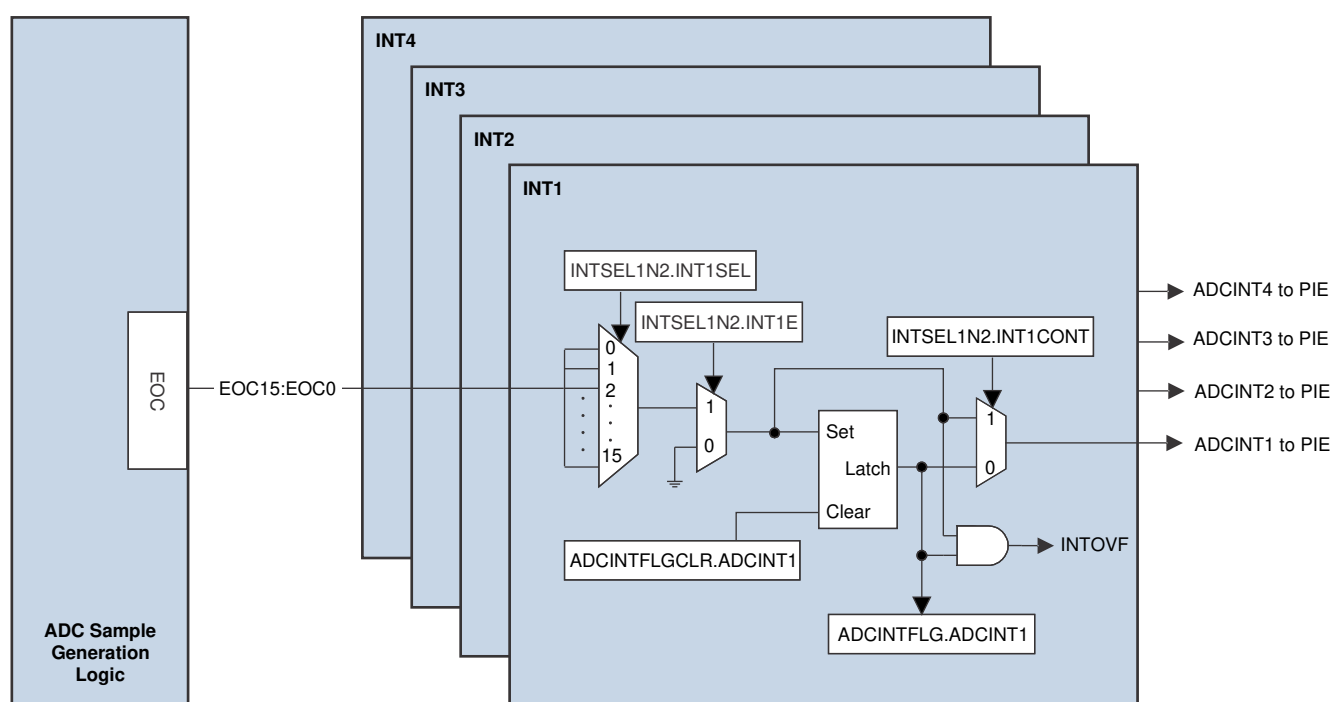


Figure 20-8. ADC EOC Interrupts

20.7.1 Interrupt Overflow

If the EOC signal sets a flag in the ADCINTFLG register, but that flag is already set, an interrupt overflow occurs. By default, overflow interrupts are not passed on to the PIE module. When an overflow occurs on a given flag in the ADCINTFLG register, the corresponding flag in the ADCINTOVF register is set. This overflow flag is only used to detect that an overflow has occurred; the flag does not block further interrupts from propagating to the PIE module.

When an ADC interrupt overflow occurs, the application must check the appropriate ADCINTOVF flag inside the ISR or in the background loop and take appropriate action when an overflow is detected. The following code snippets demonstrate how to check the ADCINTOVF flag inside the ISR after attempting to clear the ADCINT flag.

```
// Clear the interrupt flag
AdcaRegs.ADCINTFLGCLR.bit.ADCINT1 = 1;    //clear INT1 flag for ADC-A

// check if an overflow has occurred
if(1 == AdcaRegs.ADCINTOVF.bit.ADCINT1)    //ADCINT overflow occurred
{
    AdcaRegs.ADCINTOVFCLR.bit.ADCINT1 = 1    //Clear overflow flag
    AdcaRegs.ADCINTFLGCLR.bit.ADCINT1 = 1    //Re-clear ADCINT flag
}
```

```
//
// Clear the interrupt flag
//
ADC_clearInterruptStatus(ADCA_BASE, ADC_INT_NUMBER1);

//
// Check if an overflow has occurred
//
if(true == ADC_getInterruptOverflowStatus(ADCA_BASE, ADC_INT_NUMBER1))
{
    ADC_clearInterruptOverflowStatus(ADCA_BASE, ADC_INT_NUMBER1);
    ADC_clearInterruptStatus(ADCA_BASE, ADC_INT_NUMBER1);
}
```

20.7.2 Continue to Interrupt Mode

The INTxCONT bits in the ADCINTSEL1N2 and ADCINTSEL3N4 registers configure how interrupts are handled when an ADCINTFLG has not yet been cleared from a prior interrupt. This mode is disabled by default and additional overlapping interrupts are not issued to the PIE. By activating this mode, ADC interrupts always reach the PIE. If interrupts occur while ADCINTFLG is set, the ADCINTOVF register remains set regardless of the configuration of the INTxCONT bits.

20.7.3 Early Interrupt Configuration Mode

Enabling early interrupt mode can allow the application to enter the ADC interrupt service routine before the ADC results are ready. This allows the application to do any necessary pre-work so that the application can act on the ADC results immediately when the ADC results become available. If the timing of the early interrupt is too early, then the application needs to waste time until the updated ADC results become available. To prevent this situation, the time the ADC interrupt is entered in early interrupt mode is configurable by way of the DELAY field in the ADCINTCYCLE register.

- To use the configurable interrupt time, the ADC must be in early interrupt mode. To achieve this, clear the bit INTPULSEPOS to 0 in ADCCTL1.
- The DELAY value in the ADCINTCYCLE register sets the number of additional SYSCLK cycles after the falling edge of the SOC pulse before the ADCINT flag is set.
- If the value of DELAY goes beyond EOC, the ADC interrupt is generated along with EOC.
- Writing values to DELAY when INTPULSEPOS is set to 1 does not have any effect on the interrupt generation.

20.8 Post-Processing Blocks

Each ADC module contains four post-processing blocks (PPB). These blocks can be associated with any of the 16 RESULT registers using the ADCPPBxCONFIG.CONFIG bit field. The post-processing blocks have the ability to:

- Remove an offset associated with the ADCIN channel
- Subtract out a reference value
- Flag a zero-crossing point, with the option to trip a PWM and generate an interrupt
- Flag a high or low compare limit, with the option to trip a PWM and generate an interrupt
- Record the delay between the associated SOC trigger and when sampling actually begins

Figure 20-9 presents the structure of each PPB. Subsequent sections explain the use of each submodule.

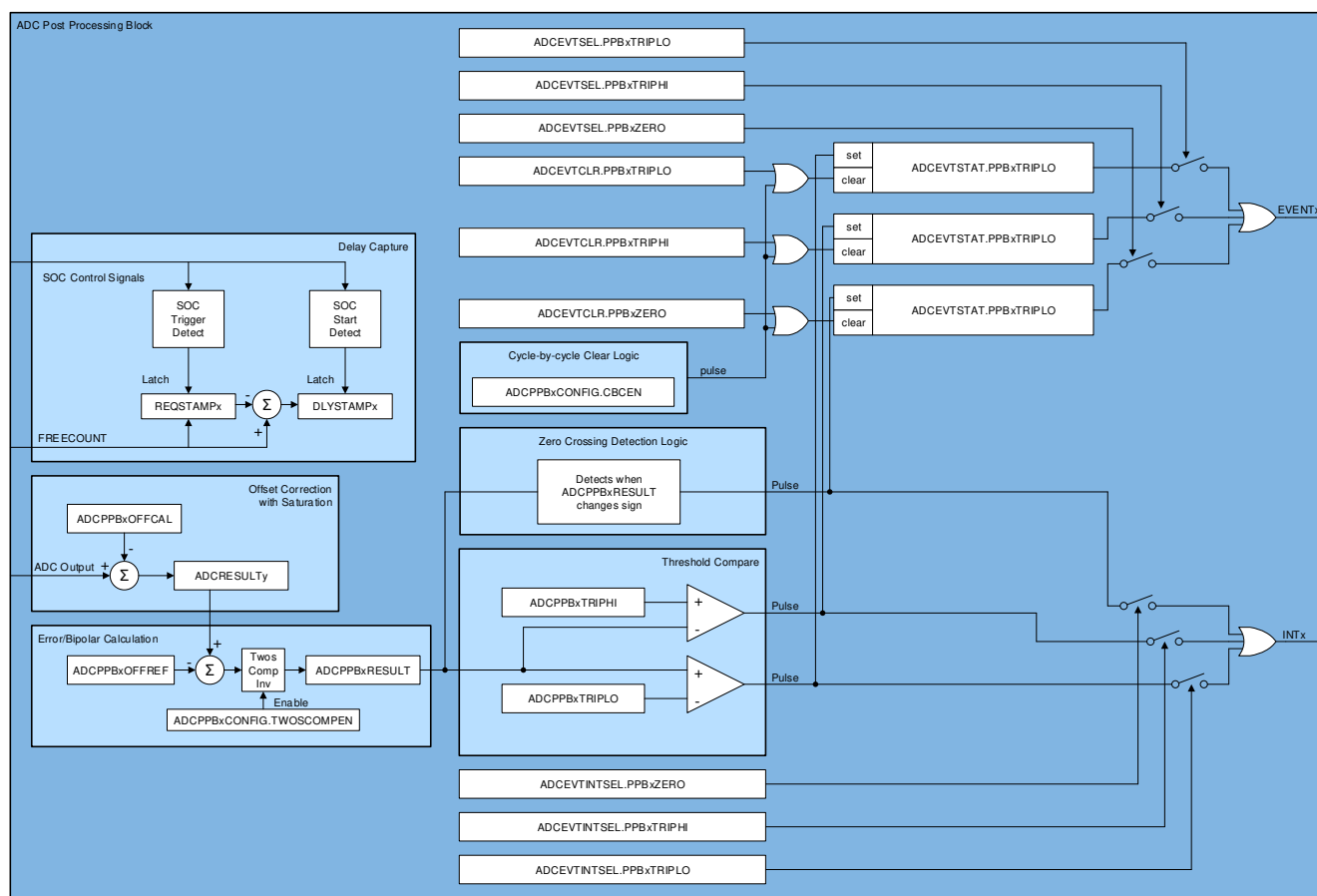


Figure 20-9. ADC PPB Block Diagram

20.8.1 PPB Offset Correction

In many applications, external sensors and signal sources produce an offset. A global trimming of the ADC offset is not enough to compensate for these offsets, which vary from channel to channel. The post-processing block can remove these offsets with zero overhead, saving numerous cycles in tight control loops.

Offset correction is accomplished by first pointing the ADCPPBxCONFIG.CONFIG to the desired SOC, then writing an offset correction value to the ADCPPBxOFFCAL.OFFCAL register. The post-processing block automatically adds or subtracts the value in the OFFCAL register from the raw conversion result and stores the value in the ADCRESULT register. This addition/subtraction saturates at 0 on the low end and either 4095 or 65535 on the high end for 12-bit or 16-bit mode, respectively.

Note

- Writing a 0 to the OFFCAL register effectively disables the offset correction feature, passing the raw result unchanged to the ADCRESULT register.
 - To point multiple PPBs to the same SOC is possible. In this case, the OFFCAL value that is actually applied comes from the PPB with the highest number.
 - In particular, care needs to be taken when using the PPB on SOC0, as all PPBs point to this SOC by default. This can cause unintentional overwriting of offset correction of a lower numbered PPB by a higher numbered PPB.
-

20.8.2 PPB Error Calculation

In many applications, an error from a set point or expected value must be computed from the digital output of an ADC conversion. In other cases, a bipolar signal is necessary or convenient for control calculations. The PPB can perform these functions automatically, reducing the sample to output latency and reducing software overhead.

Error calculation is accomplished by first pointing the ADCPPBxCONFIG.CONFIG to the desired SOC, then writing a value to the ADCPPBxOFFCAL.OFFREF register. The post-processing block automatically subtracts the value in the OFFREF register from the ADCRESULT value and stores the value in the ADCPPBxRESULT register. This subtraction produces a sign-extended 32-bit result. It is also possible to selectively invert the calculated value before storing in the ADCPPBxRESULT register by setting the TWOSCOMPEN bit in the ADCPPBxCONFIG register.

Note

- In 12-bit mode, do not write a value larger than 12 bits to the ADCPPBxOFFREF register.
 - Since the ADCPPBxRESULT register is unique for each PPB, to point multiple PPBs to the same SOC and get different results for each PPB is possible.
 - Writing a 0 to the ADCPPBxOFFREF register effectively disables the error calculation feature, passing the ADCRESULT value unchanged to the ADCPPBxRESULT register.
 - Writing a new value to ADCPPBxOFFREF causes an immediate update to the ADCPPBxRESULT register. However, the flags coming out of the PPB do not change until the next end-of-conversion (EOC). For instance, if changing the ADCPPBxOFFREF register causes ADCPPBxRESULT to change signs, but the next conversion brings the result back to the same sign as before the OFFREF change, no ADCPPBxZERO flag is set.
-

20.8.3 PPB Limit Detection and Zero-Crossing Detection

Many applications perform a limit check against the ADC conversion results. The PPB can automatically perform a check against high and low limits, or whenever ADCPPBxRESULT changes sign. Based on these comparisons, the PPB can generate a trip to the PWM and an interrupt automatically, lowering the sample to

ePWM latency and reducing software overhead. This functionality also enables safety-conscious applications to trip the ePWM based on an out-of-range ADC conversion without any CPU intervention.

To enable this functionality, first point the ADCPPBxCONFIG.CONFIG to the desired SOC, then write a value to one or both of the registers ADCPPBxTRIPHI.LIMITHI and ADCPPBxTRIPLO.LIMITLO (zero-crossing detection does not require further configuration). Whenever these limits are exceeded, the PPBxTRIPHI bit or PPBxTRIPLO bit is set in the ADCEVTSTAT register. Note that the PPBxZERO bit in the ADCEVTSTAT register is gated by end-of-conversion (EOC), not by the sign change in the ADCPPBxRESULT register. The ADCEVTCLR register has corresponding bits to clear these event flags. The ADCEVTSEL register has corresponding bits which allow the events to propagate through to the PWM. The ADCEVTINTSEL register has corresponding bits that allow the events to propagate through to the PIE.

One PIE interrupt is shared between all the PPBs for a given ADC module as shown in [Figure 20-10](#).

Note

- If different actions need to be taken for different PPB events from the same ADC module, then the ADCEVTINT ISR has to read the PPB event flags in the ADCEVTSTAT register to determine which event caused the interrupt.
- If different ePWM trips need to be generated separately for high compare, low compare, and zero-crossing, this can be achieved by pointing multiple PPBs to the same SOC.
- The zero-crossing detect circuit considers a result of zero to be positive.

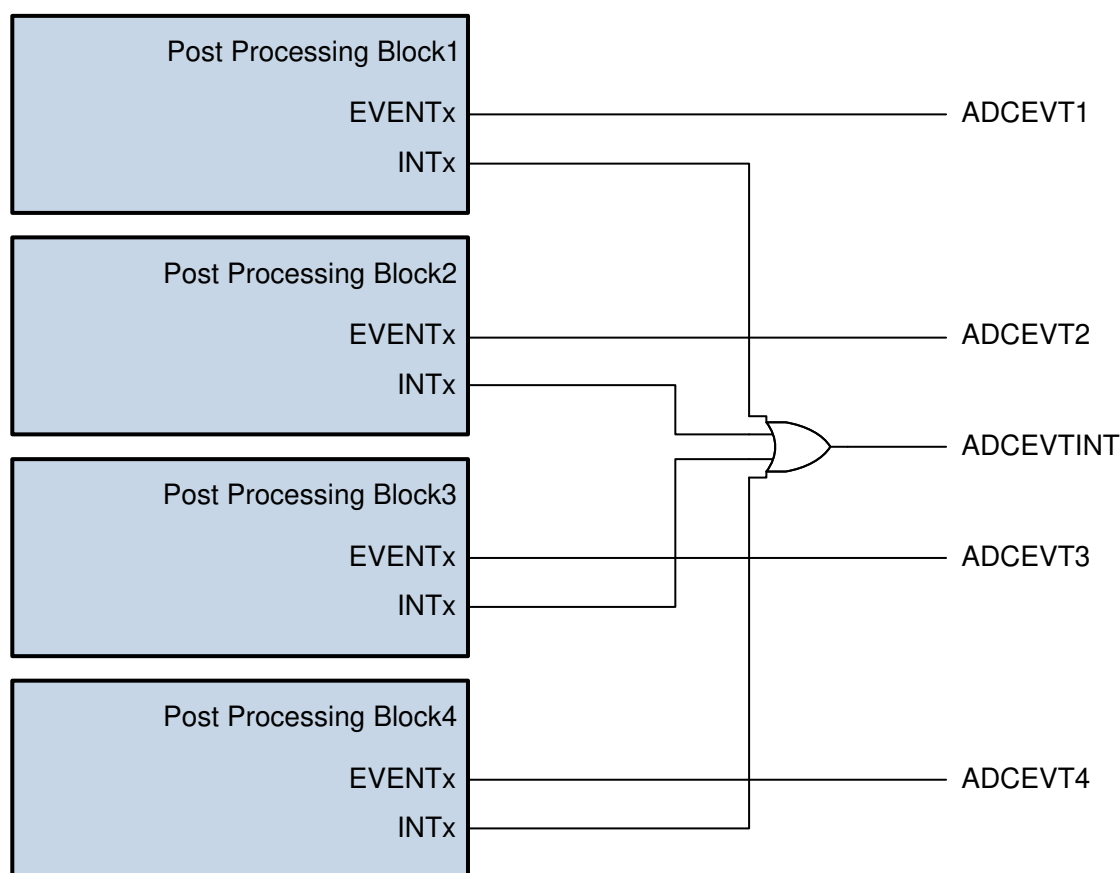


Figure 20-10. ADC PPB Interrupt Event

20.8.4 PPB Sample Delay Capture

When multiple control loops are running asynchronously on the same ADC, there is a chance that an ADC request from two or more loops collide, causing one of the samples to be delayed. This shows up as a measurement error in the system. By knowing when this delay occurs and the amount of delay that has occurred, software can employ extrapolation techniques to reduce the error.

To this effect, each PPB has the field DLYSTAMP in the ADCPPBxSTAMP register. This field contains the number of SYSCLK cycles between when the associated SOC was triggered and when the SOC began converting.

This is achieved by having a global 12-bit free running counter based off of SYSCLK, which is in the field FREECOUNT in the ADCCOUNTER register. When the trigger for the associated SOC arrives, the value of this counter is loaded into the bit field ADCPPBxTRIPLO.REQSTAMP. When the actual sample window for that SOC begins, the value in REQSTAMP is subtracted from the current FREECOUNT value and stored in DLYSTAMP.

Note

If more than 4096 SYSCLK cycles elapse between the SOC trigger and the actual start of the SOC acquisition, the FREECOUNT register can overflow more than once, leading to incorrect DLYSTAMP value. Be cautious when using very slow conversions to prevent this from happening.

The sample delay capture does not function, if the associated SOC is triggered using software. The sample delay capture, however, correctly records the delay, if the software triggering of a different SOC causes the SOC associated with the PPB to be delayed

20.9 Opens/Shorts Detection Circuit (OSDETECT)

The opens/shorts detection circuit (OSDETECT) can be used to detect pin faults in the system. The circuit connects to the ADC input after the channel select multiplexer but before the S+H circuit as shown in [Figure 20-11](#).

Note

- The divider resistance tolerances can vary widely; hence, this feature must not be used to check for conversion accuracy.
- See the data sheet for implementation and availability of analog input channels.
- Due to high drive impedance, a S+H duration much longer than the ADC minimum is needed.

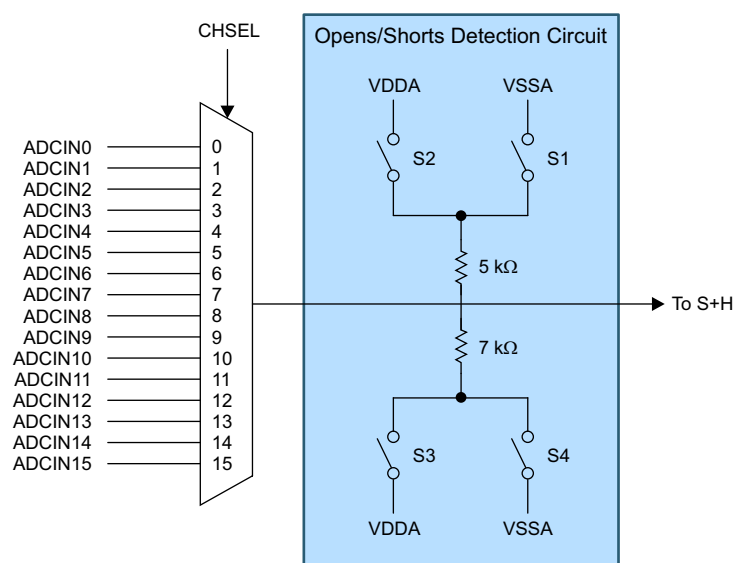


Figure 20-11. Opens/Shorts Detection Circuit

The circuit can be operated by writing a value to the DETECTCFG field in the ADCOSDETECT register. This causes the circuit to source a voltage onto the input during the S+H phase of any conversion. The voltage and drive strength of the OSDETECT circuit for different DETECTCFG settings is given in [Table 20-9](#).

Table 20-9. DETECTCFG Settings

ADCOSDETECT. DETECTCFG	Source Voltage	S4	S3	S2	S1	Drive Impedance
0	Off	Open	Open	Open	Open	Open
1	Zero Scale	Closed	Open	Open	Closed	5K 7K
2	Full Scale	Open	Closed	Closed	Open	5K 7K
3	5/12 VDDA	Open	Closed	Open	Closed	5K 7K
4	7/12 VDDA	Closed	Open	Closed	Open	5K 7K
5	Zero Scale	Open	Open	Open	Closed	5K
6	Full Scale	Open	Open	Closed	Open	5K
7	Zero Scale	Closed	Open	Open	Open	7K

20.9.1 Implementation

A representative circuit with the OSDETECT implementation consists of the signal source with series resistance R_S , shunt capacitor C_P , the equivalent OSDETECT resistance $R_{OSDETECT}$ and voltage $V_{OSDETECT}$ is shown in Figure 20-12 and can be used as a basis to calculate the signal level going in to the sampling capacitor. $R_{OSDETECT}$ and $V_{OSDETECT}$ are the equivalent input resistance and voltage source contributed by the OSDETECT circuit with values shown in Table 20-9 for the different configuration settings. Refer to Figure 20-12 when deriving the input signal to S/H if signal source V_S is driving while the OSDETECT feature is enabled.

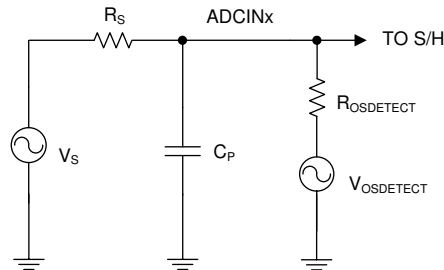


Figure 20-12. Input Circuit Equivalent with OSDETECT Enabled

The input impedance R_S and C_P are integral parts of the signal source or can have been implemented in the design to precondition the signal or to control signal settling time to meet S/H requirements. The input path has to be considered when using the OSDETECT feature, as this affects the conversion results. For instance, driving an input signal when this feature is enabled connects signal V_S to the OSDETECT circuit through R_S and affects the ADC results. Larger C_P values (in the order greater than hundreds of pF) require using higher ACQPS to make sure the signal at the input has settled prior to conversion.

To enable the circuit:

1. Configure the ADC for conversion (for example, channel, SOC, ACQPS, prescaler, trigger, and so on).
2. Set up the ADCOSDETECT register for the desired voltage divider connection as shown in Table 20-9.
3. Initiate a conversion and inspect the conversion result.

Interpret the results based on what is driving on the input side and what are the values of R_S and C_P . If the V_S signal can be disconnected from the input pin, the circuit can be used to detect open and shorted input pins as described in the following sections.

20.9.2 Detecting an Open Input Pin

By cycling through the various OSDETECT settings, the input signal is pulled towards the sourced voltages. An input with good drive strength (pin not open) is minimally affected. However, if the pin is open, the sampled voltages is close to the source voltages specified in Table 20-9.

20.9.3 Detecting a Shorted Input Pin

By cycling through the various OSDETECT settings, the input signal is pulled towards the sourced voltages. An input with finite drive strength (pin not shorted) is pulled toward each sourced voltage. However, if the pin is shorted, the signal remains at the same voltage.

20.10 Power-Up Sequence

Upon device power-up or system level reset, the ADC is powered down and disabled. When powering up the ADC, use the following sequence:

1. Set the bit to enable the desired ADC clock in the PCLKCR13 register.
2. Set the desired ADC clock divider in the PRESCALE field of ADCCTL2.
3. Power up the ADC by setting the ADCPWDNZ bit in ADCCTL1.
4. Allow a delay before sampling. See the data sheet for the necessary time.

If multiple ADCs are powered up simultaneously, steps 1 and step 3 can each be done for all ADCs in one write instruction. Also, only one delay is necessary as long as the delay occurs after all the ADCs have begun powering up.

20.11 ADC Calibration

During the fabrication and test process, Texas Instruments calibrates the gain, offset, and linearity of the ADCs and the offset of the buffered DACs. These trim settings are stored in TI reserved OTP memory, and can be loaded using C-callable functions.

- The Device_cal() function copies the trim values for ADC and DAC offset from OTP memory to the respective trim registers.
- The trim functions in Device_cal() are callable in C2000ware as ADC_setOFFSETTRIM(), ADC_setINLTRIM(), and DAC_setDACTRIM(). These functions fetch production test trim values from TI reserved OTP memory locations, and write the values to the corresponding analog module register destinations. Because these values are stored in the CPU1 Flash subsystem, the functions must be executed from CPU1 to fetch the correct values.

Until the appropriate factory trim is loaded, the ADC and other analog modules are not specified to operate within the data sheet specifications. Similarly, if trim values other than the factory settings are placed into the trim registers, the ADC (and other modules) is not specified to operate within the data sheet specifications.

The boot ROM calls the calibration functions, so trim values are initially populated without user intervention. However, if the trims are cleared due to a module reset or modified for some other reason, then the user must call the calibration functions (defined in the C2000Ware header files).

Note

Trim and calibration values are stored in the OTP region of CPU1 Flash, and cannot be accessed by CPU2. When calling user calibration functions to update ADC, DAC or temperature sensor values, the function must be called from CPU1 code.

Refer to [Section 20.11.2](#) for ADC registers being populated during ADC calibration.

20.11.1 ADC Zero Offset Calibration

ADC offset error is determined and calibrated during factory testing. However, the user still has the option to perform offset calibration if the end application specifically requires this. This section describes how to perform offset calibration using internal VREFLO connection for single-ended operation, and external channels for differential operation.

Zero offset error is defined as the difference from 0 that occurs when converting a voltage at VREFLO (for single-ended operation), or the difference from MAX_CODE/2 when converting ADCINxP = ADCINxN (for differential mode). The zero offset error can be positive or negative. To correct this error, an adjustment of equal magnitude and opposite polarity is written into the ADCOFFTRIM register. The value contained in this register is applied before the results are available in the ADC result registers. This operation is fully contained within the ADC core, so the timing of the results is not affected, and the full dynamic range of the ADC is maintained for any trim value.

Note

Regardless of the converter resolution, the size of each ADCOFFTRIM step is (VREFHI-VREFLO)/65536.

Use the following procedure to re-calibrate the ADC offset in 12-bit single-ended mode:

1. Set ADCOFFTRIM to +112 steps (0x70). This adds an artificial offset to account for negative offset that can reside in the ADC core.
2. Perform some multiple of 16 conversions on VREFLO (internal connection), accumulating the results (for example, 32*16 conversions = 512 conversions). Use the maximum value of ACQPS to make sure longer settling time to account for parasitic impedance of internal VREFLO connections.
3. Divide the accumulated result by the multiple of 16 (for example, for 512 conversions, divide by 32).
4. Set ADCOFFTRIM to 112 – result from step 3.

Use the following procedure to recalibrate the ADC offset in 16-bit differential mode:

1. Set ADCOFFTRIM to no adjustment (0x00).
2. Short ADCINxP and ADCINyN together (external connection) to a voltage near Vrefcm and accumulate some multiple of 16 conversions (for example, 32*16 conversions = 512 conversions).
3. Divide the accumulated result by the number of conversions (for example, for 512 conversions, divide by 512).
4. Set ADCOFFTRIM to 0 – result from step 3).

20.11.2 ADC Calibration Routines in OTP Memory

During factory testing, the calibration values for ADC offset (ADCOFFTRIM register) and ADC linearity correction (ADCINLTRIM1-6 registers) are measured per device. The resulting correction values for ADC offset and linearity and the associated functions to populate these into the corresponding ADC registers are stored in the OTP memory.

20.12 ADC Timings

The process of converting an analog voltage to a digital value is broken down into an S+H phase and a conversion phase. The ADC sample and hold circuits (S+H) are clocked by SYSCLK while the ADC conversion process is clocked by ADCCLK. ADCCLK is generated by dividing down SYSCLK based on the PRESCALE field in the ADCCTL2 register.

The S+H duration is the value of the ACQPS field of the SOC being converted, plus one, times the SYSCLK period. The user must make sure that this duration exceeds both 1 ADCCLK period and the minimum S+H duration specified in the data sheet. The conversion time is approximately 10.5 ADCCLK cycles in 12-bit mode and 29.5 ADCCLK cycles in 16-bit mode. The exact conversion time is always a whole number of SYSCLK cycles. See the timing diagrams and tables in [Section 20.12.1](#) for exact timings.

20.12.1 ADC Timing Diagrams

The following diagrams show the ADC conversion timings for two SOC's given the following assumptions:

- SOC0 and SOC1 are configured to use the same trigger.
- No other SOC's are converting or pending when the trigger occurs.
- The round robin pointer is in a state that causes SOC0 to convert first.
- ADCINTSEL is configured to set an ADCINT flag upon end of conversion for SOC0 (whether this flag propagates through to the CPU to cause an interrupt is determined by the configurations in the PIE module).

[Table 20-10](#) describes the parameters in the following timing diagrams. [Table 20-11](#) and [Table 20-12](#) list the ADC timings..

Table 20-10. ADC Timing Parameter Descriptions

Parameter	Description
t_{SH}	The duration of the S+H window. At the end of this window, the value on the S+H capacitor becomes the voltage to be converted into a digital value. The duration is given by (ACQPS + 1) SYSCLK cycles. ACQPS can be configured individually for each SOC, so t_{SH} is not necessarily the same for different SOC's. Note: The value on the S+H capacitor is captured approximately 5ns before the end of the S+H window regardless of device clock settings.
t_{LAT}	The time from the end of the S+H window until the ADC results latch in the ADCRESULTx register. If the ADCRESULTx register is read before this time, the previous conversion results are returned.
t_{EOC}	The time from the end of the S+H window until the S+H window for the next ADC conversion can begin. In 16-bit mode, this coincides with the latching of the conversion results, while in 12-bit mode, the subsequent sample can start before the conversion results are latched.
t_{INT}	The time from the end of the S+H window until an ADCINT flag is set (if configured). If the INTPULSEPOS bit in the ADCCTL1 register is set, t_{INT} coincides with the end of conversion (EOC) signal. If the INTPULSEPOS bit is 0, and the OFFSET field in the ADCINTCYCLE register is not 0, then there is a delay of OFFSET SYSCLK cycles before the ADCINT flag is set. This delay can be used to enter the ISR or trigger the DMA exactly when the sample is ready. If the INTPULSEPOS bit is 0, t_{INT} coincides with the end of the S+H window. If t_{INT} triggers a read of the ADC result register (directly through DMA or indirectly by triggering an ISR that reads the result), care must be taken to make sure the read occurs after the results latch (otherwise, the previous results are read).

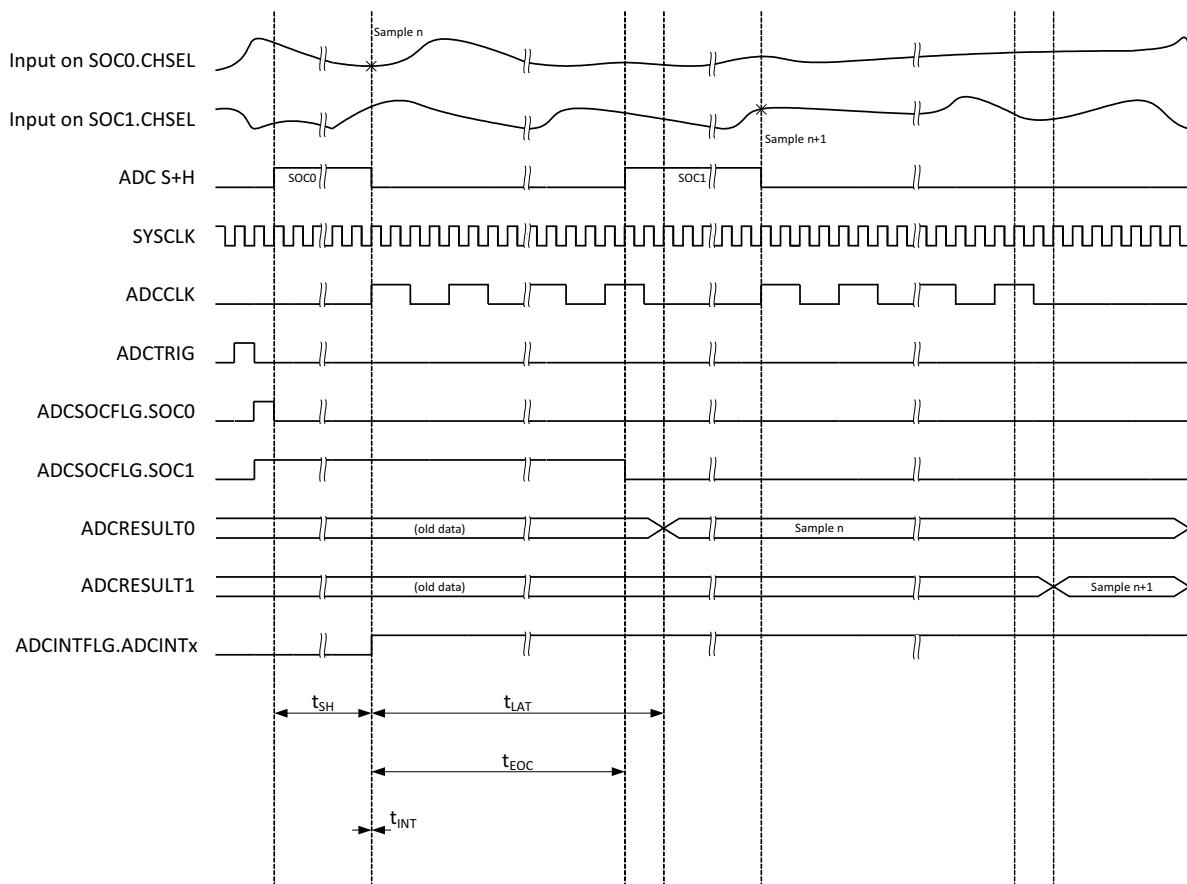


Figure 20-13. ADC Timings for 12-bit Mode in Early Interrupt Mode

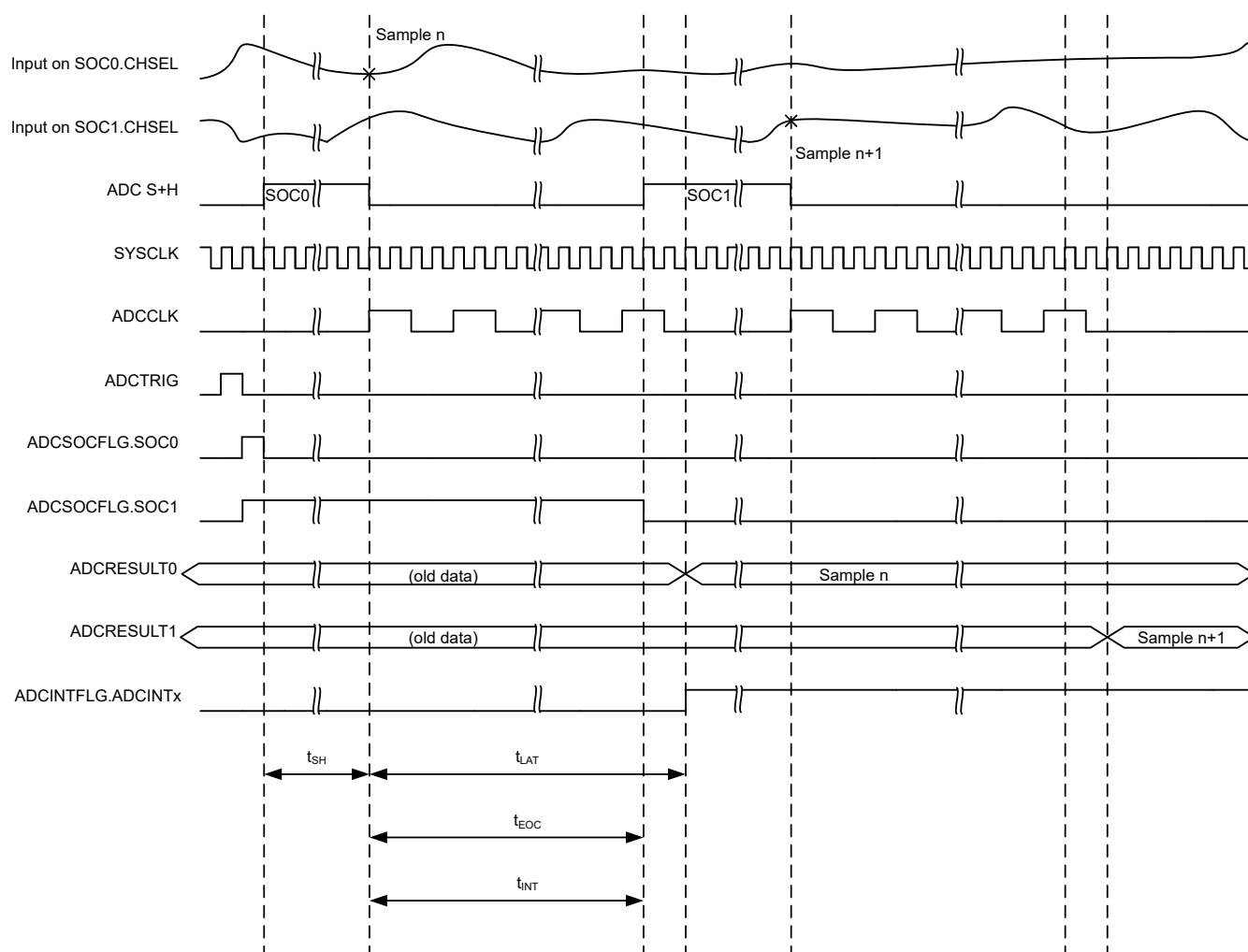


Figure 20-14. ADC Timings for 12-bit Mode in Late Interrupt Mode

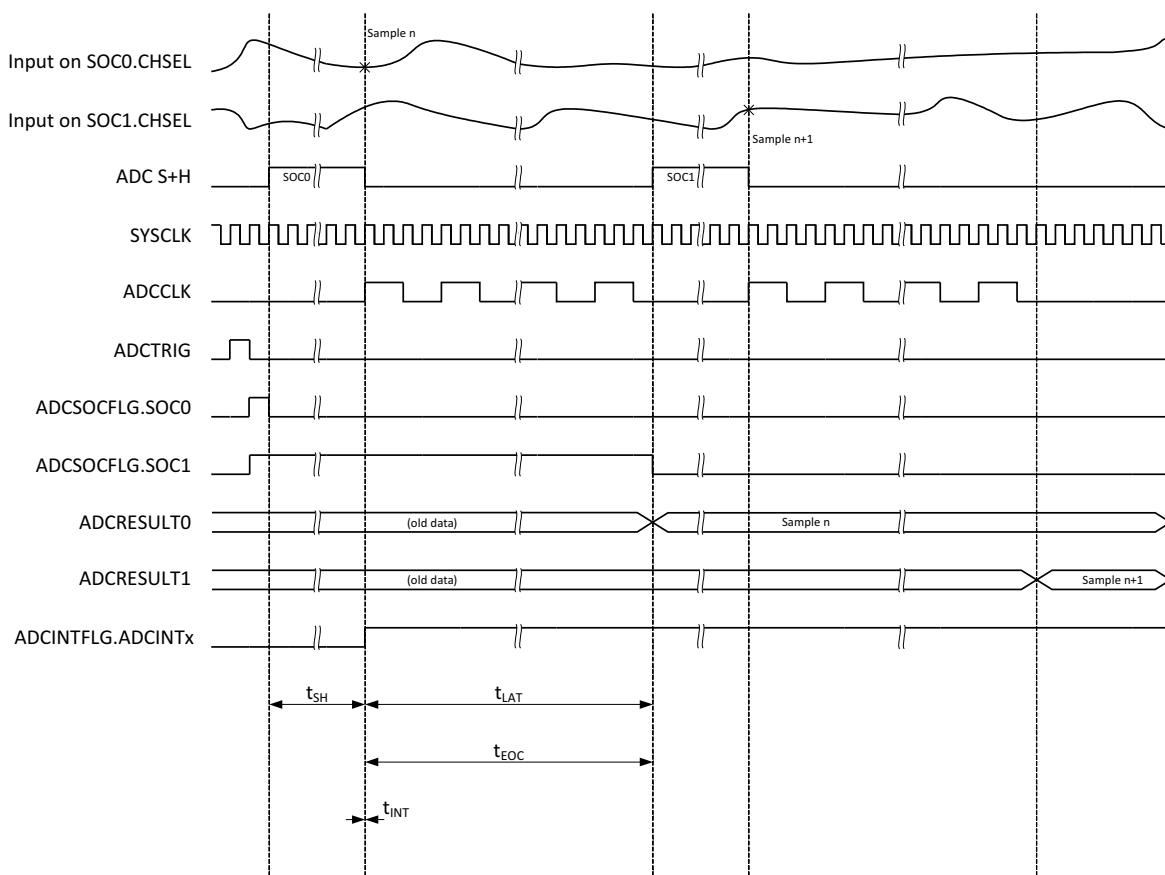


Figure 20-15. ADC Timings for 16-bit Mode in Early Interrupt Mode

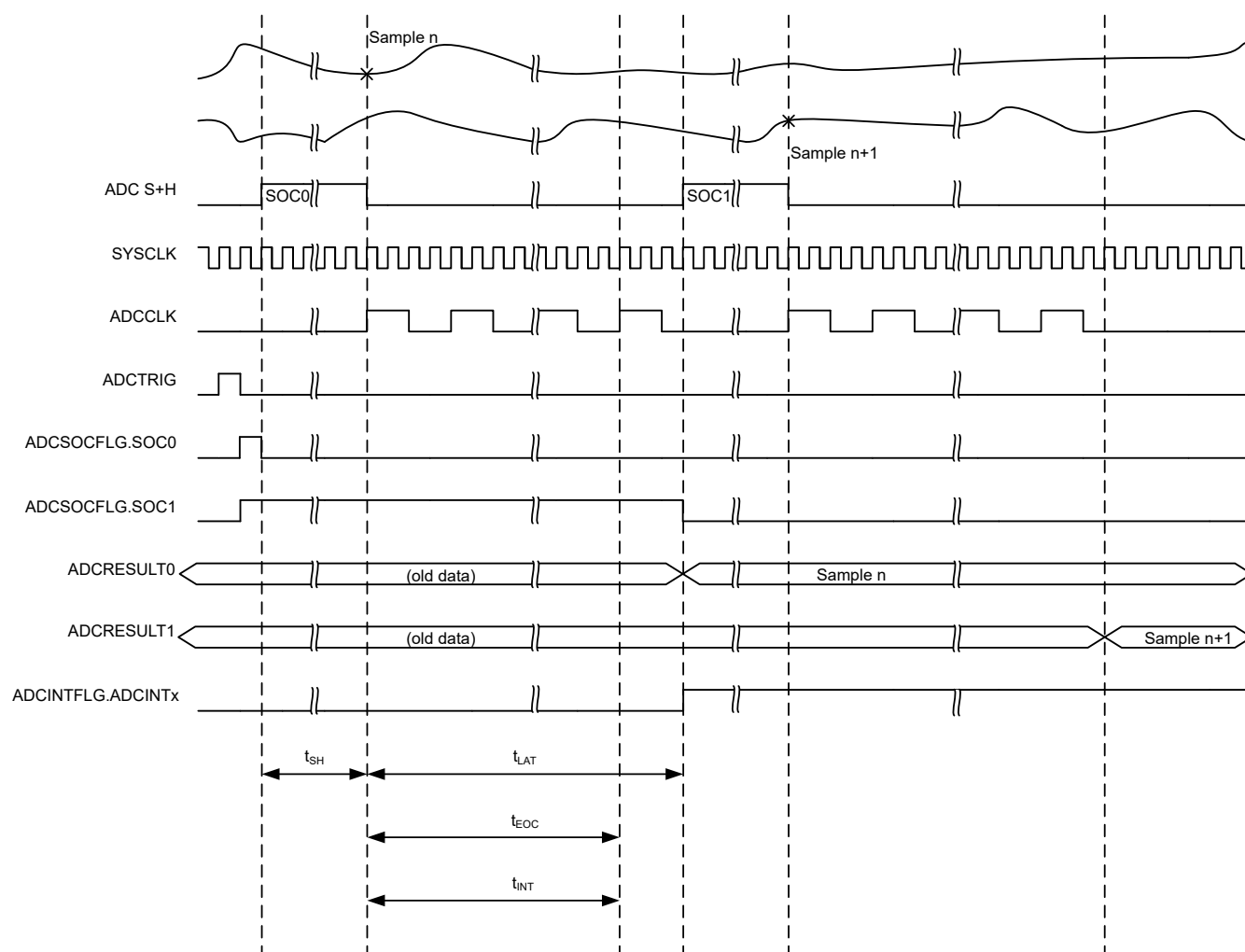


Figure 20-16. ADC Timings for 16-bit Mode in Late Interrupt Mode (SYSCLK Cycles)

Table 20-11. ADC Timings in 12-bit Mode

ADCCLK Prescale		SYSCLK Cycles			
ADCCTL2. PRESCALE	Prescale Ratio	t _{EOC}	t _{LAT} ⁽¹⁾	t _{INT} (Early) ⁽²⁾	t _{INT} (Late)
0	1	11	13	0	11
2	2	21	23	0	21
3	2.5	26	28	0	26
4	3	31	34	0	31
5	3.5	36	39	0	36
6	4	41	44	0	41
7	4.5	46	49	0	46
8	5	51	55	0	51
9	5.5	56	60	0	56
10	6	61	65	0	61
11	6.5	66	70	0	66
12	7	71	76	0	71
13	7.5	76	81	0	76
14	8	81	86	0	81
15	8.5	86	91	0	86

- (1) Refer to the "ADC: DMA Read of Stale Result" advisory in the [TMS320F2838x Real-Time MCUs Silicon Errata](#).
- (2) By default, t_{INT} occurs one SYSCLK cycle after the S+H window if INTPULSEPOS is 0. This can be changed by writing to the OFFSET field in the ADCINTCYCLE register.

Table 20-12. ADC Timings in 16-bit Mode

ADCCLK Prescale		SYSCLK Cycles			
ADCCTL2. PRESCALE	Prescale Ratio	t _{EOC}	t _{LAT} ⁽¹⁾	t _{INT} (Early) ⁽²⁾	t _{INT} (Late)
0	1	31	32	0	31
2	2	60	61	0	60
3	2.5	75	75	0	75
4	3	90	91	0	90
5	3.5	104	106	0	104
6	4	119	120	0	119
7	4.5	134	134	0	134
8	5	149	150	0	149
9	5.5	163	165	0	163
10	6	178	179	0	178
11	6.5	193	193	0	193
12	7	208	209	0	208
13	7.5	222	224	0	222
14	8	237	238	0	237
15	8.5	252	252	0	252

- (1) Refer to the "ADC: DMA Read of Stale Result" advisory in the [TMS320F2838x Real-Time MCUs Silicon Errata](#).
- (2) By default, t_{INT} occurs one SYSCLK cycle after the S+H window if INTPULSEPOS is 0. This can be changed by writing to the OFFSET field in the ADCINTCYCLE register.

20.13 Additional Information

The following sections contain additional practical information.

20.13.1 Ensuring Synchronous Operation

For best performance, all ADCs on the device must be operated synchronously. The device data sheet specifies the performance in both synchronous and asynchronous mode for those parameters which differ between the modes of operation.

To make sure synchronous operation, all ADCs on the device must operate in lockstep. This is accomplished by writing configurations to all ADCs that cause the sampling and conversion phases of all ADCs to be exactly aligned. The easiest way to accomplish this is to write identical values to the SOC configurations for each ADC for trigger select and ACQPS (S+H duration). In addition, synchronous ADCs must also configure identical values for the SOC priority control, burst mode, burst trigger, and burst size.

20.13.1.1 Basic Synchronous Operation

The following example configures two SOC's each on ADCA and ADCB with identical trigger select and ACQPS values. This results in synchronous operation between ADCA and ADCB. For devices with more than two ADCs, the same principles can be used to synchronize all the ADCs.

Example: Basic Synchronous Operation

```
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 4; //SOC0 converts ADCINA4
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 19; //SOC0 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 0; //SOC0 converts ADCINB0
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 19; //SOC0 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOCB

AdcaRegs.ADCSOC1CTL.bit.CHSEL = 4; //SOC1 converts ADCINA4
AdcaRegs.ADCSOC1CTL.bit.ACQPS = 30; //SOC1 uses sample duration of 31 SYSCLK cycles
AdcaRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC1CTL.bit.CHSEL = 1; //SOC1 converts ADCINB1
AdcbRegs.ADCSOC1CTL.bit.ACQPS = 30; //SOC1 uses sample duration of 31 SYSCLK cycles
AdcbRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOCB
```

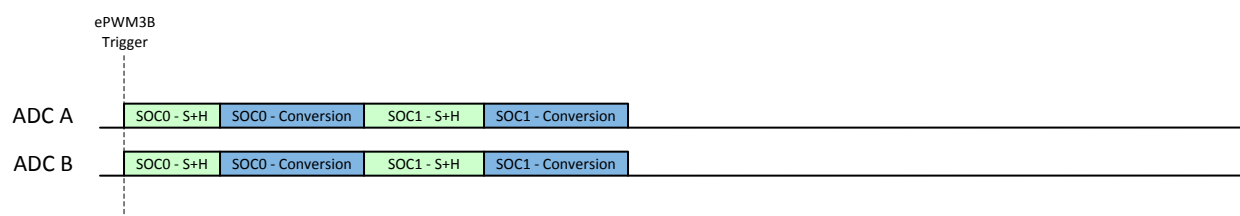


Figure 20-17. Example: Basic Synchronous Operation

Several things can be noted from [Figure 20-17](#). First, while the ACQPS values must be the same for SOC's with the same number, different ACQPS values can be used for SOC's with different numbers. Because of this, synchronous operation does not require a single global S+H time, but instead only channels sampled simultaneously require identical S+H durations. Another important point from this example is that any channel select value can be used for any SOC. Finally, this example assumes round-robin operation. If high-priority SOC's are to be used, the priority must be configured the same on all ADCs.

20.13.1.2 Synchronous Operation with Multiple Trigger Sources

As long as each set of SOC's has identical trigger select and ACQPS settings, multiple trigger sources can be used while still achieving synchronous operation.

The following example demonstrates synchronous operation between ADCA and ADCB while using three SOC's and two trigger sources. [Figure 20-18](#) demonstrates that any combination of relative trigger timings still results in synchronous operation.

Example: Synchronous Operation with Multiple Trigger Sources

```
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 4;    //SOC0 converts ADCINA4
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 0;    //SOC0 converts ADCINB0
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOCB

AdcaRegs.ADCSOC1CTL.bit.CHSEL = 4;    //SOC1 converts ADCINA4
AdcaRegs.ADCSOC1CTL.bit.ACQPS = 30;   //SOC1 uses sample duration of 31 SYSCLK cycles
AdcaRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC1CTL.bit.CHSEL = 1;    //SOC1 converts ADCINB1
AdcbRegs.ADCSOC1CTL.bit.ACQPS = 30;   //SOC1 uses sample duration of 31 SYSCLK cycles
AdcbRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOCB

AdcaRegs.ADCSOC2CTL.bit.CHSEL = 0;    //SOC2 converts ADCINA0
AdcaRegs.ADCSOC2CTL.bit.ACQPS = 19;   //SOC2 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC2CTL.bit.TRIGSEL = 2;  //SOC2 begins conversion on CPU Timer1
AdcbRegs.ADCSOC2CTL.bit.CHSEL = 2;    //SOC2 converts ADCINB2
AdcbRegs.ADCSOC2CTL.bit.ACQPS = 19;   //SOC2 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC2CTL.bit.TRIGSEL = 2;  //SOC2 begins conversion on CPU Timer1
```

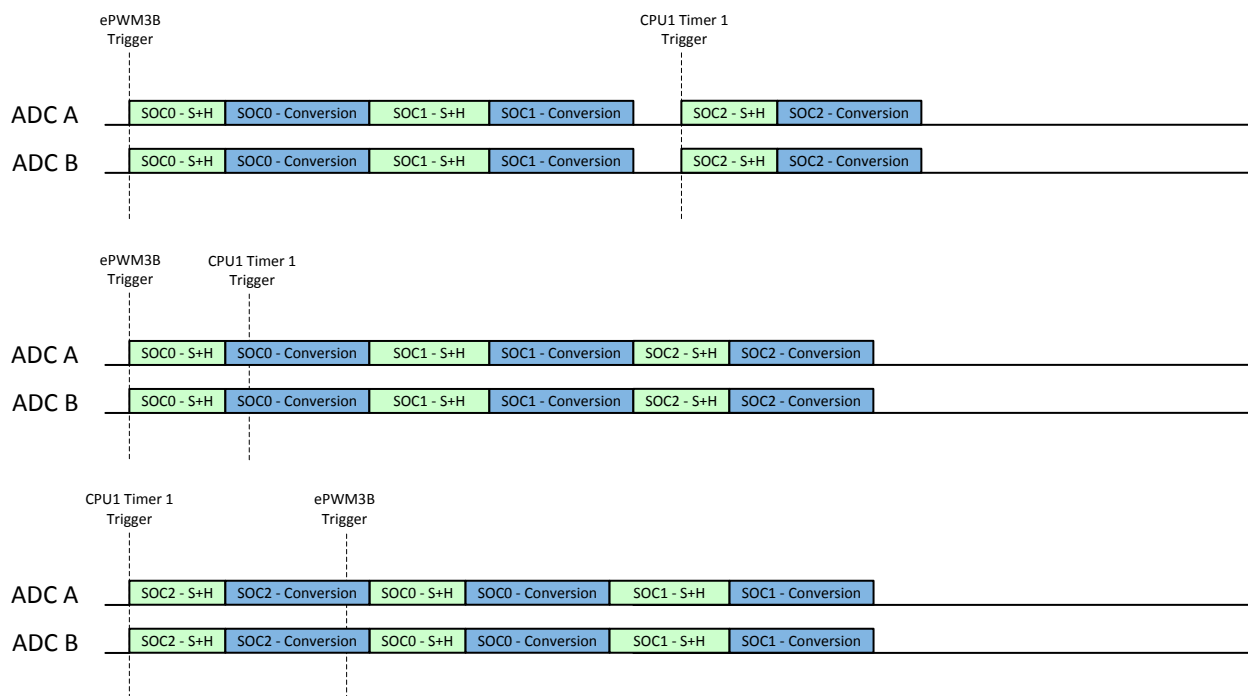


Figure 20-18. Example: Synchronous Operation with Multiple Trigger Sources

Note that any trigger source that can be selected in the TRIGSEL field can be used except for software triggering. There is no way to issue the software triggers for all ADCs simultaneously, so likely results in asynchronous operation. ADCINT1 or ADCINT2 can also be used as a trigger as long as the ADCINTSOCSEL1 and ADCINTSOCSEL2 registers are configured identically for all ADCs and software triggering is not used to start the chain of conversions.

20.13.1.3 Synchronous Operation with Uneven SOC Numbers

If only one trigger source is used, one ADC can use more SOC's than the other ADCs while still operating synchronously.

Example: Synchronous Operation with Uneven SOC Numbers

```

AdcaRegs.ADCSOC0CTL.bit.CHSEL = 4;    //SOC0 converts ADCINA4
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOC B
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 0;    //SOC0 converts ADCINB0
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOC B

AdcaRegs.ADCSOC1CTL.bit.CHSEL = 4;    //SOC1 converts ADCINA4
AdcaRegs.ADCSOC1CTL.bit.ACQPS = 30;   //SOC1 uses sample duration of 31 SYSCLK cycles
AdcaRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOC B
AdcbRegs.ADCSOC1CTL.bit.CHSEL = 1;    //SOC1 converts ADCINB1
AdcbRegs.ADCSOC1CTL.bit.ACQPS = 30;   //SOC1 uses sample duration of 31 SYSCLK cycles
AdcbRegs.ADCSOC1CTL.bit.TRIGSEL = 10; //SOC1 begins conversion on ePWM3 SOC B

AdcaRegs.ADCSOC2CTL.bit.CHSEL = 0;    //SOC2 converts ADCINA0
AdcaRegs.ADCSOC2CTL.bit.ACQPS = 19;   //SOC2 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC2CTL.bit.TRIGSEL = 10; //SOC2 begins conversion on ePWM3 SOC B

```

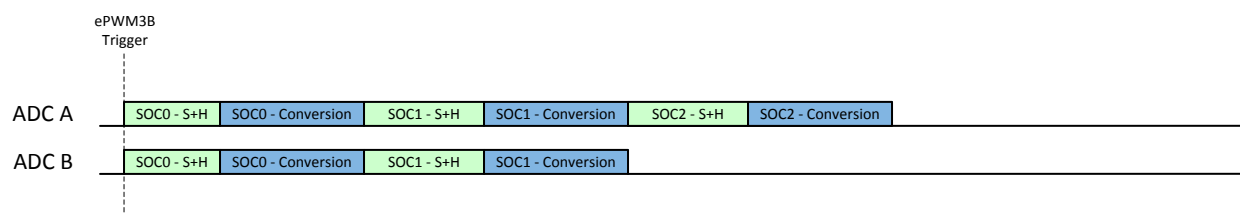


Figure 20-19. Example: Synchronous Operation with Uneven SOC Numbers

Note that if the trigger comes again before all SOC's have completed the conversions, ADCB begins converting immediately on SOC0 while ADCA does not start converting SOC0 again until SOC2 is complete. This results in asynchronous operation, so care must be taken to not overflow the trigger.

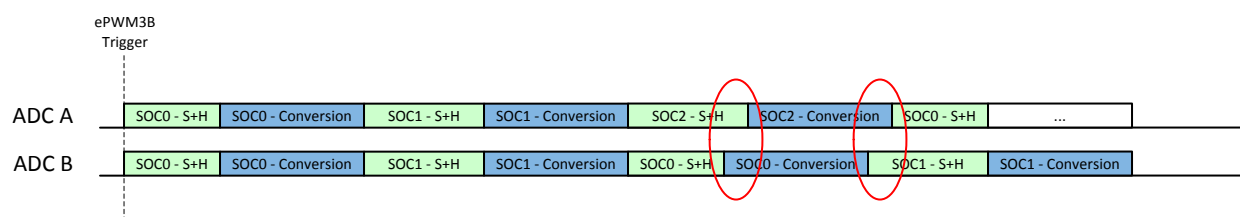


Figure 20-20. Example: Asynchronous Operation with Uneven SOC Numbers – Trigger Overflow

20.13.1.4 Synchronous Operation with Different Resolutions

Configuring different ADCs to use different resolutions results in asynchronous operation. This occurs because the conversion time for 12-bit mode and 16-bit mode are different. Synchronous operation requires both the start and end of the conversion phase to be aligned, so even using the same S+H window duration does not result in synchronous operation.

Example: Asynchronous Operation with Different Resolutions

```
//ADCA = 12-bit mode
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 4;    //SOC0 converts ADCINA4
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 50;   //SOC0 uses sample duration of 51 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOC0

//ADCB = 16-bit mode
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 0;    //SOC0 converts ADCINB0/B1
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 50;   //SOC0 uses sample duration of 51 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOC0
```

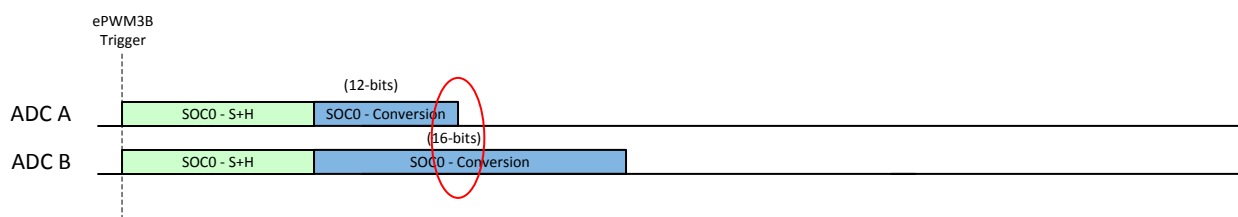


Figure 20-21. Example: Asynchronous Operation with Different Resolutions

To achieve synchronous operation while using both 12-bit and 16-bit resolution, conversions must be done in parallel at one resolution. Once conversions are complete at one resolution, the CPU must switch the resolution on all ADCs and then cause another trigger (this trigger must not be a software SOC force, as all ADCs cannot be started simultaneously using this method).

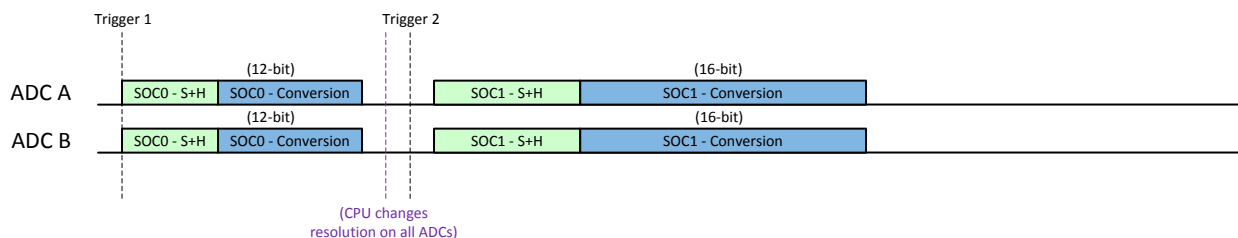


Figure 20-22. Example: Synchronous Operation with Different Resolutions

20.13.1.5 Non-overlapping Conversions

If conversion timings can be made sure to not overlap by the user, then it is not necessary to configure all SOC0s identically on all ADCs to achieve performance equivalent to synchronous operation. For example, if the two ADC triggers in a system come from two ePWM sources that are always 180-degrees out-of-phase, then SOC0 can be used for both ADCA and ADCB with different trigger sources and different ACQPS values.

Example: Operation with Non-overlapping Conversions

```
//ePWM3 SOCA and SOCB are 180 degrees out of phase
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 4;    //SOC0 converts ADCINA4
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10; //SOC0 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 0;    //SOC0 converts ADCINB0
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 19;   //SOC0 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 9;  //SOC0 begins conversion on ePWM3 SOCA
```

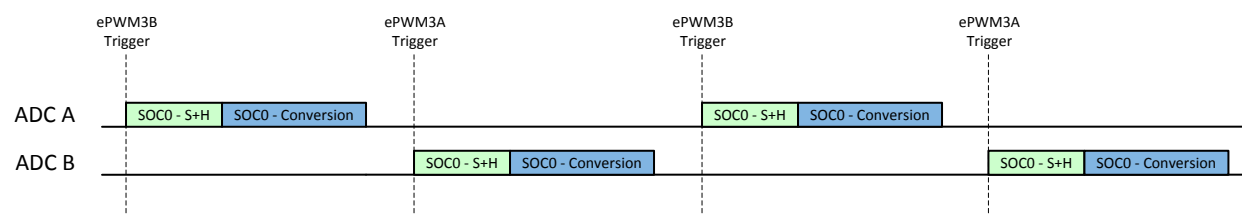


Figure 20-23. Example: Synchronous Equivalent Operation with Non-Overlapping Conversions

20.13.2 Choosing an Acquisition Window Duration

For correct operation, the input signal to the ADC must be allowed adequate time to charge the sample and hold capacitor, Ch. Typically, the S+H duration is chosen such that the sampling capacitor is charged to within ½ LSB or ¼ LSB of the final value, depending on the tolerable settling error.

The best methodology to determine the required settling time is to simulate the ADC and ADC driving circuits to make sure adequate settling performance. See [ADC Input Circuit Evaluation for C2000 MCUs](#) and [Charge-Sharing Driving Circuits for C2000 ADCs](#) for additional guidance on ADC signal conditioning circuit design and evaluation.

An approximation of the required settling time can also be determined using an RC settling model. The time constant for the model is given by the equation:

$$\tau = (R_S + R_{on}) \times C_h + R_S \times (C_S + C_p) \quad (14)$$

And the number of time constants needed is given by the equation:

$$k = \ln\left(\frac{2^n}{\text{settling error}}\right) - \ln\left(\frac{C_S + C_P}{CH}\right) \quad (15)$$

So the total S+H time must be set to at least:

$$t = k \cdot \tau \quad (16)$$

Where the following parameters are provided by the ADC input model in the device data sheet:

- n = ADC resolution (in bits)
- R_{ON} = ADC sampling switch resistance (provided in Ω)
- C_H = ADC sampling capacitor (provided in pF)
- C_p = ADC channel parasitic input capacitance (provided in pF)

And the following parameters are dependent on the application design:

- settling error = tolerable settling error (in LSBs)
- R_s = ADC driving circuit source impedance (typically in Ω or $k\Omega$)
- C_s = capacitance on ADC input pin (typically in pF or nF)

For example, assuming the following parameters:

- n = 12-bits
- R_{ON} = 500Ω
- C_H = 12.5pF
- C_p = 12.7pF
- settling error = $\frac{1}{4}$ LSB
- R_s = 180Ω
- C_s = 150pF

The time constant is calculated as:

$$\tau = (180\Omega + 500\Omega) \times 12.5\text{pF} + 180\Omega \times (150\text{pF} + 12.7\text{pF}) = 37.8\text{ns} \quad (17)$$

And the number of required time constants is:

$$k = \ln\left(\frac{2^{12}}{0.25}\right) - \ln\left(\frac{150\text{pF} + 12.7\text{pF}}{12.5\text{pF}}\right) = 9.70 - 2.57 = 7.13 \quad (18)$$

So the S+H time must be set to at least: $37.8\text{ns} \times 7.13 = 270\text{ns}$

If $\text{SYSCLK} = 200\text{MHz}$, then each SYSCLK cycle is 5ns . S+H duration is $270\text{ns}/5\text{ns} = 54$ SYSCLK cycles, so ACQPS for this input is set to at least $\text{CEILING}(54.0) - 1 = 53$.

While this gives a rough estimate of the required acquisition window, a better method is to setup a circuit with the ADC input model, a model of the source impedance/capacitance, and any board parasitics in SPICE (or similar software) and simulate to verify that the sampling capacitor settles to the desired accuracy.

Note

The device data sheet specifies a minimum ADC S+H window duration. Do not use an ACQPS value that gives a duration less than this specification.

20.13.3 Achieving Simultaneous Sampling

While each ADC does not have dual S+H circuits, achieving simultaneous sampling is accomplished by setting the SOC triggers on two or more ADC modules to use the same trigger source. The following example demonstrates simultaneous sampling on 4 ADCs based on an ePWM3 event. ADCINA3, ADCINB5, ADCINC5, and ADCIND2 are sampled. An acquisition window of 20 SYSCLK cycles is used, but different durations are possible.

```
AdcaRegs.ADCSOC0CTL.bit.CHSEL = 3;           //SOC0 convert ADCINA3
AdcaRegs.ADCSOC0CTL.bit.ACQPS = 19;          //SOC0 uses sample duration of 20 SYSCLK cycles
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL = 10;        //SOC0 begins conversion on ePWM3 SOCB
AdcbRegs.ADCSOC0CTL.bit.CHSEL = 5;           //SOC0 convert ADCINB5
AdcbRegs.ADCSOC0CTL.bit.ACQPS = 19;          //SOC0 uses sample duration of 20 SYSCLK cycles
AdcbRegs.ADCSOC0CTL.bit.TRIGSEL = 10;        //SOC0 begins conversion on ePWM3 SOCB
AdccRegs.ADCSOC0CTL.bit.CHSEL = 5;           //SOC0 convert ADCINC5
AdccRegs.ADCSOC0CTL.bit.ACQPS = 19;          //SOC0 uses sample duration of 20 SYSCLK cycles
AdccRegs.ADCSOC0CTL.bit.TRIGSEL = 10;        //SOC0 begins conversion on ePWM3 SOCB
AdcdRegs.ADCSOC0CTL.bit.CHSEL = 2;           //SOC0 convert ADCIND2
AdcdRegs.ADCSOC0CTL.bit.ACQPS = 19;          //SOC0 uses sample duration of 20 SYSCLK cycles
AdcdRegs.ADCSOC0CTL.bit.TRIGSEL = 10;        //SOC0 begins conversion on ePWM3 SOCB
```

When the ePWM3 trigger is received, all 4 ADCs begin converting in parallel immediately. All results are stored in the ADCRESULT0 register for each ADC. Note that this assumes that all ADCs are idle when the trigger is received. If one or more ADCs is busy, the samples do not happen at exactly the same time.

20.13.4 Result Register Mapping

The ADC results and the ADC PPB results are duplicated for each memory bus controller in the system. Bus controllers include all CPUs, DMAs, and CLAs present on the specific part family and part number. For each bus controller, no access configuration is needed to allow read access to the result registers, and no contention occurs in cases where multiple bus controllers try to read the ADC results simultaneously.

20.13.5 Internal Temperature Sensor

The internal temperature sensor measures the junction temperature of the device. The output of the sensor can be sampled with the ADC through an internal connection. This can be enabled on channel ADCIN13 on ADCA by setting the ENABLE bit in the TSNSCTL register.

To convert the temperature sensor reading into a temperature, pass the temperature sensor reading to the GetTemperatureC() function in F2838x_TempSensorConv.c.

Note

To sample the temperature sensor, the ADC must be in single-ended 12-bit mode.

If the temperature sensor is sampled in 16-bit mode, the internal temperature sensor is sampled with a 12-bit resolution, regardless of the ADC mode currently configured. This does not change the resolution of results obtained on other SOC's for the given ADC.

20.13.6 Designing an External Reference Circuit

Figure 20-24 shows the basic organization of the external voltage reference generation circuitry. TI recommends that a single reference voltage generation source is shared by all ADC modules. This minimizes reference voltage mismatch between ADC modules. For best performance, the externally generated reference voltage must be buffered by a precision op-amp with good bandwidth and low output impedance before being driven into the ADC reference pin. A capacitor between the high and low reference pins must be placed on the PCB as close to the pins as practical to help absorb high-frequency currents. A series resistor (typically <1Ω) in series with this capacitor is sometimes necessary to maintain op-amp stability.

To share two reference pins between one op-amp driver is possible. This organization is shown in [Figure 20-25](#). This can give slightly reduced performance compared to the case where each reference pin has a dedicated op-amp buffer, but can still be possible to achieve all ADC specifications in the data sheet.

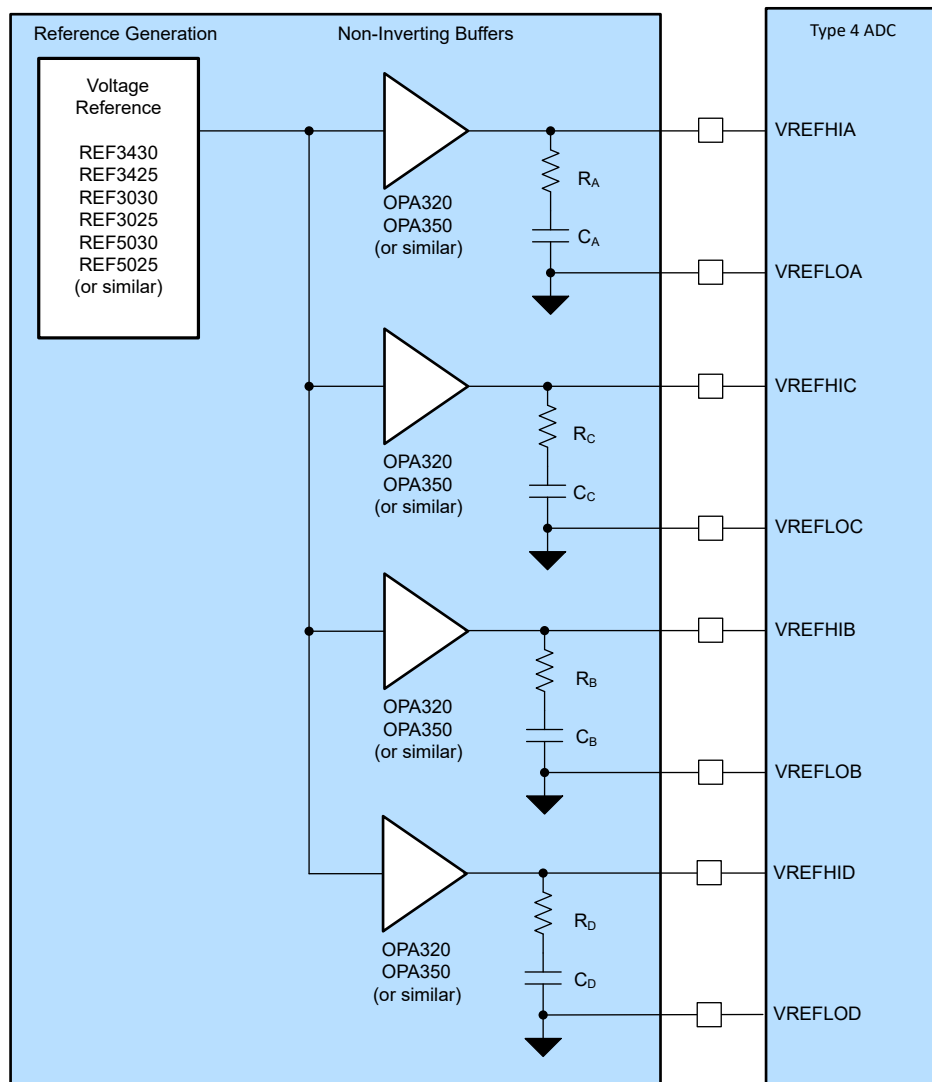


Figure 20-24. ADC Reference System

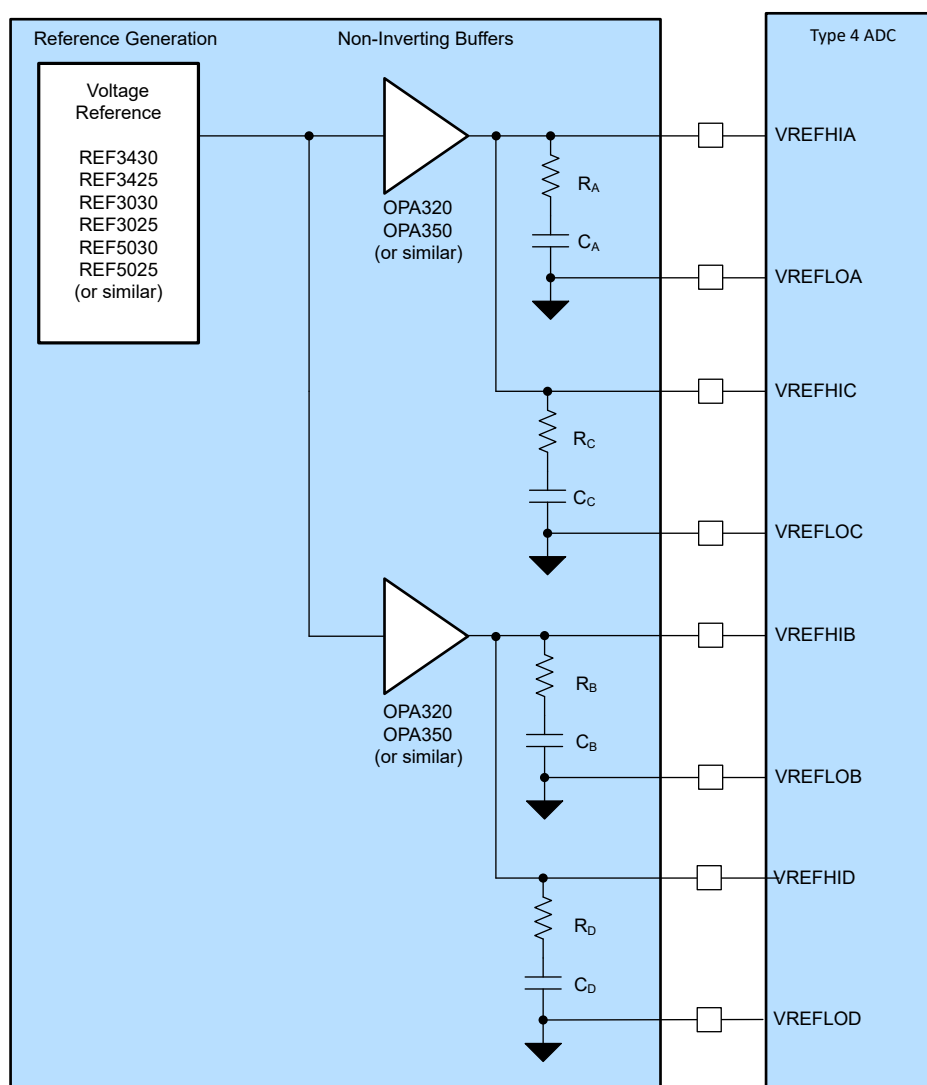


Figure 20-25. ADC Shared Reference System

20.14 Software

20.14.1 ADC Examples

NOTE: These examples are located in the [C2000Ware](#) installation at the following location:
C2000Ware_VERSION#/driverlib/DEVICE_GPN/examples/CORE_IF_MULTICORE/adc

Cloud access to these examples is available at the following link: dev.ti.com [C2000Ware Examples](#).

20.14.1.1 ADC Software Triggering

FILE: adc_ex1_soc_software.c

This example converts some voltages on ADCA and ADCC based on a software trigger.

The ADCC will not convert until ADCA is complete, so the ADCs will not run asynchronously. However, this is much less efficient than allowing the ADCs to convert synchronously in parallel (for example, by using an ePWM trigger).

External Connections

- A0, A1, C2, and C3 should be connected to signals to convert

Watch Variables

- *myADC0Result0* - Digital representation of the voltage on pin A0
- *myADC0Result1* - Digital representation of the voltage on pin A1
- *myADC1Result0* - Digital representation of the voltage on pin C2
- *myADC1Result1* - Digital representation of the voltage on pin C3

20.14.1.2 ADC ePWM Triggering

FILE: adc_ex2_soc_epwm.c

This example sets up ePWM1 to periodically trigger a conversion on ADCA.

External Connections

- A0 should be connected to a signal to convert

Watch Variables

- *myADC0Results* - A sequence of analog-to-digital conversion samples from pin A0. The time between samples is determined based on the period of the ePWM timer.

20.14.1.3 ADC Temperature Sensor Conversion

FILE: adc_ex3_temp_sensor.c

This example sets up the ePWM to periodically trigger the ADC. The ADC converts the internal connection to the temperature sensor, which is then interpreted as a temperature by calling the `ADC_getTemperatureC()` function.

Watch Variables

- *sensorSample* - The raw reading from the temperature sensor
- *sensorTemp* - The interpretation of the sensor sample as a temperature in degrees Celsius.

20.14.1.4 ADC Synchronous SOC Software Force (adc_soc_software_sync)

FILE: adc_ex4_soc_software_sync.c

This example converts some voltages on ADCA and ADCC using input 5 of the input X-BAR as a software force. Input 5 is triggered by toggling GPIO0, but any spare GPIO could be used. This method will ensure that both ADCs start converting at exactly the same time.

External Connections

- A2, A3, C2, C3 pins should be connected to signals to convert

Watch Variables

- *myADC0Result0* : a digital representation of the voltage on pin A2
- *myADC0Result1* : a digital representation of the voltage on pin A3
- *myADC1Result0*: a digital representation of the voltage on pin C2
- *myADC1Result1*: a digital representation of the voltage on pin C3

20.14.1.5 ADC Continuous Triggering (adc_soc_continuous)

FILE: adc_ex5_soc_continuous.c

This example sets up the ADC to convert continuously, achieving maximum sampling rate.

External Connections

- A0 pin should be connected to signal to convert

Watch Variables

- *adcAResults* - A sequence of analog-to-digital conversion samples from pin A0. The time between samples is the minimum possible based on the ADC speed.

20.14.1.6 ADC Continuous Conversions Read by DMA (adc_soc_continuous_dma)

FILE: adc_ex6_soc_continuous_dma.c

This example sets up two ADC channels to convert simultaneously. The results will be transferred by the DMA into a buffer in RAM.

External Connections

- A3 & C3 pins should be connected to signals to convert

Watch Variables

- *myADC0DataBuffer* : a digital representation of the voltage on pin A3
- *myADC1DataBuffer*: a digital representation of the voltage on pin C3

20.14.1.7 ADC PPB Offset (adc_ppb_offset)

FILE: adc_ex7_ppb_offset.c

This example software triggers the ADC. Some SOC's have automatic offset adjustment applied by the post-processing block. After the program runs, the memory will contain ADC & post-processing block(PPB) results.

External Connections

- A2, C2 pins should be connected to signals to convert

Watch Variables

- *myADC0Result* : a digital representation of the voltage on pin A2
- *myADC0PPBResult* : a digital representation of the voltage on pin A2, minus 100 LSBs of automatically added offset
- *myADC1Result*: a digital representation of the voltage on pin C2
- *myADC1PPBResult*: a digital representation of the voltage on pin C2 plus 100 LSBs of automatically added offset

20.14.1.8 ADC PPB Limits (adc_ppb_limits)

FILE: adc_ex8_ppb_limits.c

This example sets up the ePWM to periodically trigger the ADC. If the results are outside of the defined range, the post-processing block will generate an interrupt.

The default limits are 1000LSBs and 3000LSBs. With VREFHI set to 3.3V, the PPB will generate an interrupt if the input voltage goes above about 2.4V or below about 0.8V.

External Connections

- A0 should be connected to a signal to convert

Watch Variables

- None

20.14.1.9 ADC PPB Delay Capture (adc_ppb_delay)

FILE: adc_ex9_ppb_delay.c

This example demonstrates delay capture using the post-processing block.

Two asynchronous ADC triggers are setup:

- ePWM1, with period 2048, triggering SOC0 to convert on pin A0
 - ePWM2, with period 9999, triggering SOC1 to convert on pin A2
- Each conversion generates an ISR at the end of the conversion. In the ISR for SOC0, a conversion counter is incremented and the PPB is checked to determine if the sample was delayed.
- After the program runs, the memory will contain:

conversion : the sequence of conversions using SOC0 that were delayed

- *delay* : the corresponding delay of each of the delayed conversions

20.14.1.10 ADC ePWM Triggering Multiple SOC

FILE: adc_ex10_multiple_soc_epwm.c

This example sets up ePWM1 to periodically trigger a set of conversions on ADCA and ADCC. This example demonstrates multiple ADCs working together to process a batch of conversions using the available parallelism across multiple ADCs.

ADCA Interrupt ISRs are used to read results of both ADCA and ADCC.

External Connections

- A0, A1, A2 and C2, C3, C4 pins should be connected to signals to be

Watch Variables

- *adcAResult0* - Digital representation of the voltage on pin A0
- *adcAResult1* - Digital representation of the voltage on pin A1
- *adcAResult2* - Digital representation of the voltage on pin A2
- *adcCResult0* - Digital representation of the voltage on pin C2
- *adcCResult1* - Digital representation of the voltage on pin C3
- *adcCResult2* - Digital representation of the voltage on pin C4

20.14.1.11 ADC Burst Mode

FILE: adc_ex11_burst_mode_epwm.c

This example sets up ePWM1 to periodically trigger ADCA using burst mode. This allows for different channels to be sampled with each burst.

Each burst triggers 3 conversions. A0 and A1 are part of every burst while the third conversion rotates between A2, A3, and A4. This allows high importance signals to be sampled at high speed while lower priority signals can be sampled at a lower rate.

ADCA Interrupt ISRs are used to read results for ADCA.

External Connections

- A0, A1, A2, A3, A4

Watch Variables

- *adcAResult0* - Digital representation of the voltage on pin A0
- *adcAResult1* - Digital representation of the voltage on pin A1
- *adcAResult2* - Digital representation of the voltage on pin A2
- *adcAResult3* - Digital representation of the voltage on pin A3
- *adcAResult4* - Digital representation of the voltage on pin A4

20.14.1.12 ADC Burst Mode Oversampling

FILE: adc_ex12_burst_mode_oversampling.c

This example is an ADC oversampling example implemented with software. The ADC SOC's are configured in burst mode, triggered by the ePWM SOC A event trigger.

External Connection

- A2

Watch Variables

- *lv_results* - Array of digital values measured on pin A2 (oversampling is configured by Oversampling_Amount)

20.14.1.13 ADC SOC Oversampling

FILE: adc_ex13_soc_oversampling.c

This example sets up ePWM1 to periodically trigger a set of conversions on ADCA including multiple SOC's that all convert A2 to achieve oversampling on A2.

ADCA Interrupt ISRs are used to read results of ADCA.

External Connections

- A0, A1, A2 should be connected to signals to be converted.

Watch Variables

- *adcAResult0* - Digital representation of the voltage on pin A0
- *adcAResult1* - Digital representation of the voltage on pin A1
- *adcAResult2* - Digital representation of the voltage on pin A2

20.14.1.14 ADC PPB PWM trip (adc_ppb_pwm_trip)

FILE: adc_ex14_ppb_pwm_trip.c

This example demonstrates EPWM tripping through ADC limit detection PPB block. ADCAINT1 is configured to periodically trigger the ADCA channel 2 post initial software forced trigger. The limit detection post-processing block(PPB) is configured and if the ADC results are outside of the defined range, the post-processing block will generate an ADCxEVTy event. This event is configured as EPWM trip source through configuring EPWM XBAR and corresponding EPWM's trip zone and digital compare sub-modules. The example showcases

- one-shot
- cycle-by-cycle
- and direct tripping of PWMs through ADCAEVT1 source via Digital compare submodule.

The default limits are 0LSBs and 3600LSBs. With VREFHI set to 3.3V, the PPB will generate a trip event if the input voltage goes above about 2.9V.

External Connections

- A2 should be connected to a signal to convert
- Observe the following signals on an oscilloscope
 - ePWM1(GPIO0 - GPIO1)
 - ePWM2(GPIO2 - GPIO3)
 - ePWM3(GPIO4 - GPIO5)

Watch Variables

- *adcA2Results* - digital representation of the voltage on pin A2

20.14.1.15 ADC High Priority SOC (adc_high_priority_soc)

FILE: adc_ex15_high_priority_soc.c

This example demonstrates configuration of ADC high priority SOC's in order to sample fastest control loop signals with high priority while low priority signals are sampled with default round robin priority.

ADC PPB block is configured to capture delay between SOC trigger and actual start of the SOC sampling in order to quantify the jitters in sampling high priority signals due to low priority signals. The delay in processing an SOC is captured in ADCPPBxSTAMP.DLYSTAMP register field and the total delay in sampling an SOC is equal to (DLYSTAMP - 2) cycles. In an optimal design the high priority SOC is expected to have less delay between SOC trigger and actual start of the sample.

In this example ADCA, ADCB and ADCD are configured to sample both high and low priority signals. SOC0-3 are configured as high priority SOC's while rest are configured with default round-robin priority. ADCA SOC0-3 are configured as high priority SOC's sampling channels A0-A3, ADCB SOC0-1 are configured as high priority SOC's sampling channels B0-B1 and ADCD SOC0-1 are configured as high priority SOC's sampling channels D0-D1. For sampling low priority signals SOC4-SOC5 are configured sampling channel 4 and channel 13 respectively for ADCA and channel 4 and 5 for ADCB and ADCD. For ADCA, channel 13 is connected to internal temperature sensor output and hence no signal needs to be connected to channel A13. High priority SOC results are read in ADCINT1 ISR while low priority SOC results are read in idle loop.

This example has two modes of operation as follows. Desired mode can be selected by configuring the EX_ADC_LP_SOC_TRIGGER macro accordingly. Mode 0: ADCINT as round robin SOC trigger Mode 1: EPWM2 as round robin SOC trigger

In mode 0, EPWM1 is configured as trigger for high priority SOC's while ADCINT1 is configured as low priority SOC trigger. ADCINT1 is configured to be triggered on completion of SOC1 conversion which in turn trigger low priority SOC's 4 and 5 and ADCINT2 is configured to be triggered on completion of SOC5. ADC result for high priority SOC's are read in ADCINT ISR while low priority SOC results are read in idle loop. ADCINT3 is configured to be triggered on completion of SOC3 and hence SOC2-SOC3 results for ADCA are read post checking if the conversion is complete in ADCINT1 ISR.

In mode 0, SOC0-SOC1 for all ADCs will experience minimal delay(0 and 1 conversions) in processing due to the high priority configuration. For ADCA, SOC4-SOC5 are triggered when SOC2-SOC3 conversion is ongoing, hence SOC4-SOC5 will see some delay(2 and 3 conversions) in processing as expected. For ADCB and ADCD, SOC4-SOC5 will see minimal delay (0 and 1 conversions) in processing as expected.

In mode 1, EPWM1 is configured as trigger for high priority SOC's while EPWM2 is configured as low priority SOC trigger. ADCINT1 is configured to be triggered on completion of SOC3 conversion and ADC result for high priority SOC's are read in the ADCINT ISR. Low priority SOC's 4 and 5 are triggered through EPWM2 and ADCINT2 is configured to be triggered on completion of SOC5. The result for low priority SOC's are read in background loop. Since, SOC4-5 are triggered post SOC2-3 conversion(due to configured EPWM1 and EPWM2 trigger frequency and duty), SOC4-SOC5 will have minimal delay(0 and 1 conversions respectively) in SOC processing as expected.

Optimization Level: Example is expected to run with opt level = O2.

To view ADC results, put breakpoint at the statement where indexB is reset to zero in idle loop.

External Connections

- A0-A4, B0-B1, B4-B5, D0-D1 and D4-D5 pins should be connected to signals to be converted.
- Observe ePWM1, ePWM2 signals on oscilloscope

Watch Variables

ADC Results:

- adcA0Results - adcA4Results - ADC result of channels A0-A4
- adcA13Results - ADC result of channels A13(Temp Sensor output)
- adcB0Results - adcB1Results - ADC result of channels B0-B1
- adcB4Results - ADC result of channel B4
- adcB5Results - ADC result of channel B5
- adcD0Results - adcD1Results - ADC result of channels D0-D1
- adcD4Results - ADC result of channel D4
- adcD5Results - ADC result of channel D5

SOC conversion delays:

- delaySocA0 - Delay in sampling ADCA SOC0
- delaySocA3 - Delay in sampling ADCA SOC3
- delaySocA4-delaySocA5 - Delay in sampling ADCA SOC4-SOC5
- delaySocB0-delaySocB1 - Delay in sampling ADCB SOC0-SOC1
- delaySocB4-delaySocB5 - Delay in sampling ADCB SOC4-SOC5
- delaySocD0-delaySocD1 - Delay in sampling ADCD SOC0-SOC1
- delaySocD4-delaySocD5 - Delay in sampling ADCD SOC4-SOC5

20.14.1.16 ADC Interleaved Averaging in Software

FILE: adc_ex16_sw_interleaved_averaging.c

This example demonstrates software interleaved averaging of ADC input channels. ADCA/B channel 0 and 1 are sampled one after another in order to achieve interleaved averaging. SOC0-15 of ADCA and ADCB are configured to sample channel 0 and 1 alternatively with channel 0 being sampled by even SOC0s namely SOC0, 2, 4, 6, 8, 10, 12 & 14 and channel 1 being sampled by odd SOC0s namely SOC1, 3, 5, 7, 9, 11, 13 & 15.

Sampling is initially triggered through external GPIO signal through enabling ADCEXTSOC signal via input XBAR and thereafter through ADCINT1. GPIO33 is configured to trigger ADCA and B SOC0s for the first time in order to ensure synchronous operation. GPIO33 needs to be connected to GPIO32 which in turn is driven by software to trigger the respective ADC SOC0s.

ADCA Interrupt ISRs are used to read results of both ADCA and ADCB to demonstrate parallel operation of multiple ADCs. ADCINT2 is configured to be triggered after completion of SOC7 while ADCINT1 is configured to be triggered after completion of SOC15 conversion and respective SOC results of ADCA and ADCB are read in ADCAINT2 and ADCAINT1 ISRs. Read SOC0-SOC15 ADC results are then averaged in ADCINT1 ISR to get the filtered ADC output.

Early interrupt mode is configured to trigger the ADC interrupt just at the end of acquisition window in order to save cycles (~42 cycles) since ADCA result of SOC7 in ADCINT2 and SOC15 in ADCINT1 are read post around 60 and 65 cycles respectively. Also, Fast ISRs are configured in the example to save some cycles during compiler specific context save and restore.

Optimization level: This example is expected to be run with opt level = 2

Sampling rate related calculations:

- ADC acquisition cycles programmed(S + H) = 15 SYSCLKS
- Conversion time for 12-bit data = 10.5 ADCCLKS = N = 42 SYSCLKS
- Time from 1st SOC trigger to interrupt trigger: $15 * 57 + 15 = 870$ SYSCLKS
- Next ADC trigger will come after 870 SYSCLKS
- Approximate ISR trigger frequency = $1/(870 * 5ns) = 1/4350ns = 229$ KSPS
- Time taken in ADCINT1 ISR : 132 SYSCLKS (Fast ISR)(O2)
- Time taken in ADCINT2 ISR : 76 SYSCLKS (Fast ISR)(O2)

To view results in graph window, add breakpoint inside while(1) loop in main() where bufferFull flag is cleared and plot the watch variables.

External Connections

- A0, A1, B0 and B1 pins should be connected to signals to be converted
- Connect GPIO32 to GPIO33 to trigger ADC channels for the first time

Watch Variables

- *adcA0Results* - A sequence of analog-to-digital conversion samples from pin A0.
- *adcA1Results* - A sequence of analog-to-digital conversion samples from pin A1.
- *adcB0Results* - A sequence of analog-to-digital conversion samples from pin B0.
- *adcB1Results* - A sequence of analog-to-digital conversion samples from pin B1.

20.14.1.17 ADC Open Shorts Detection (adc_open_shorts_detection)

FILE: adc_ex17_open_shorts_detection.c

This example demonstrates the ADC open/shorts detection(ADCOSDETECT) circuit configuration for detecting pin faults in the system. The example enables the open/shorts detection circuit along with mandatory ADC configurations and diagnoses ADCA A0 input pin state before starting normal ADC conversions.

To enable the ADC OSDetect circuit:

1. Configure the ADC for conversion (E.g. channel, SOC, ACQPS, prescaler, trigger etc). The OSDetect functionality is available in 12-bit only, hence ADCA is configured in 12-bit mode in the example.
2. Set up the ADCOSDETECT register for the desired voltage divider connection. Refer device TRM for details on available OSDetect configurations.
3. Initiate a conversion and inspect the conversion result. Note: The results must be interpreted based on what is driving on the input side and what are the values of Rs and Cp. If the Vs signal can be disconnected from the input pin, the circuit can be used to detect open and shorted input pins. In the example, after configuring ADCA A0 channel in 12-bit mode along with other required ADC configurations, following algorithm is used to check the A0 pin status: Step 1: Configure full scale OSDetect mode & capture ADC results(resultHi) Step 2: Configure zero scale OSDetect mode & capture ADC results(resultLo) Step 3: Disable OSDetect mode and capture ADC results(resultNormal) Step 4: Determine the state of the ADC pin a. If the pin is open, resultLo would be equal to Vreflo and resultHi would be equal to Vrefhi b. If the pin is shorted to Vrefhi, resultLo should be approximately equal to Vrefhi and resultHi should be equal to Vrefhi c. If the pin is shorted to Vreflo, resultLo should be equal to Vreflo and resultHi should be approximately equal to Vreflo d. If the pin is connected to a valid signal, resultLo should be greater than osdLoLimit but less than resultNormal while resultHi should be less than osdHiLimit but greater than result

Normal Input | Full-Scale output | Zero-scale Output | Pin Status Unknown| VREFHI | VREFLO | Open VREFHI | VREFHI | approx. VREFHI | Shorted to VREFHI VREFLO | approx. VREFLO | VREFLO | Shorted to VREFLO Vn | Vn < resultHi < VREFHI | VREFLO < resultLo < Vn | Good

Step 5: osDetectStatusVal of value greater than 4 would mean that there is no pin fault. a. If osDetectStatusVal == 1, means pin A0 is OPEN b. If osDetectStatusVal == 2, means pin A0 is shorted to VREFLO c. If osDetectStatusVal == 4, means pin A0 is shorted to VREFHI d. If osDetectStatusVal == 8, means pin A0 is in GOOD/VALID state e. Any value of osDetectStatusVal > 4, means pin A0 is in VALID state

Following points should be noted while configuring the ADC in OSDetect mode.

1. The divider resistance tolerances can vary widely, hence this feature should not be used to check for conversion accuracy.
2. Consult the device data manual for implementation and availability of analog input channels.
3. Due to high drive impedance, a S+H duration much longer than the ADC minimum will be needed.

External Connections

- A0 pin should be connected to signals to convert

Watch Variables

- *osDetectStatusVal* : OS detection status of voltage on pin A0
- *adcAResult0* : a digital representation of the voltage on pin A0

20.15 ADC Registers

This section describes the Analog-to-Digital Converter Registers.

20.15.1 ADC Base Address Table (C28)

Table 20-13. ADC Base Address Table (C28)

Bit Field Name		DriverLib Name	Base Address	CPU1	CPU2	DMA	CLA	Pipeline Protected
Instance	Structure							
AdcaRegs	ADC_REGS	ADCA_BASE	0x0000_7400	YES	YES	-	YES	YES
AdcbRegs	ADC_REGS	ADCB_BASE	0x0000_7480	YES	YES	-	YES	YES
AdccRegs	ADC_REGS	ADCC_BASE	0x0000_7500	YES	YES	-	YES	YES
AdcdRegs	ADC_REGS	ADCD_BASE	0x0000_7580	YES	YES	-	YES	YES
AdcaResultRegs	ADC_RESULT_REGS	ADCARESULT_BASE	0x0000_0B00	YES	YES	YES	YES	-
AdcbResultRegs	ADC_RESULT_REGS	ADCBRESULT_BASE	0x0000_0B20	YES	YES	YES	YES	-
AdccResultRegs	ADC_RESULT_REGS	ADCCRESULT_BASE	0x0000_0B40	YES	YES	YES	YES	-
AdcdResultRegs	ADC_RESULT_REGS	ADCDRESULT_BASE	0x0000_0B60	YES	YES	YES	YES	-

20.15.2 ADC_REGS Registers

Table 20-14 lists the memory-mapped registers for the ADC_REGS registers. All register offset addresses not listed in Table 20-14 should be considered as reserved locations and the register contents should not be modified.

Table 20-14. ADC_REGS Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	ADCCTL1	ADC Control 1 Register	EALLOW	Go
1h	ADCCTL2	ADC Control 2 Register	EALLOW	Go
2h	ADCBURSTCTL	ADC Burst Control Register	EALLOW	Go
3h	ADCINTFLG	ADC Interrupt Flag Register		Go
4h	ADCINTFLGCLR	ADC Interrupt Flag Clear Register		Go
5h	ADCINTOVF	ADC Interrupt Overflow Register		Go
6h	ADCINTOVFCLR	ADC Interrupt Overflow Clear Register		Go
7h	ADCINTSEL1N2	ADC Interrupt 1 and 2 Selection Register	EALLOW	Go
8h	ADCINTSEL3N4	ADC Interrupt 3 and 4 Selection Register	EALLOW	Go
9h	ADCSOCPRCTL	ADC SOC Priority Control Register	EALLOW	Go
Ah	ADCINTSOCSEL1	ADC Interrupt SOC Selection 1 Register	EALLOW	Go
Bh	ADCINTSOCSEL2	ADC Interrupt SOC Selection 2 Register	EALLOW	Go
Ch	ADCSOCFLG1	ADC SOC Flag 1 Register		Go
Dh	ADCSOCFRC1	ADC SOC Force 1 Register		Go
Eh	ADCSOCOVF1	ADC SOC Overflow 1 Register		Go
Fh	ADCSOCOVFCLR1	ADC SOC Overflow Clear 1 Register		Go
10h	ADCSOC0CTL	ADC SOC0 Control Register	EALLOW	Go
12h	ADCSOC1CTL	ADC SOC1 Control Register	EALLOW	Go
14h	ADCSOC2CTL	ADC SOC2 Control Register	EALLOW	Go
16h	ADCSOC3CTL	ADC SOC3 Control Register	EALLOW	Go
18h	ADCSOC4CTL	ADC SOC4 Control Register	EALLOW	Go
1Ah	ADCSOC5CTL	ADC SOC5 Control Register	EALLOW	Go
1Ch	ADCSOC6CTL	ADC SOC6 Control Register	EALLOW	Go
1Eh	ADCSOC7CTL	ADC SOC7 Control Register	EALLOW	Go
20h	ADCSOC8CTL	ADC SOC8 Control Register	EALLOW	Go
22h	ADCSOC9CTL	ADC SOC9 Control Register	EALLOW	Go
24h	ADCSOC10CTL	ADC SOC10 Control Register	EALLOW	Go
26h	ADCSOC11CTL	ADC SOC11 Control Register	EALLOW	Go
28h	ADCSOC12CTL	ADC SOC12 Control Register	EALLOW	Go
2Ah	ADCSOC13CTL	ADC SOC13 Control Register	EALLOW	Go
2Ch	ADCSOC14CTL	ADC SOC14 Control Register	EALLOW	Go
2Eh	ADCSOC15CTL	ADC SOC15 Control Register	EALLOW	Go
30h	ADCEVTSTAT	ADC Event Status Register		Go
32h	ADCEVTCLR	ADC Event Clear Register		Go
34h	ADCEVTSEL	ADC Event Selection Register	EALLOW	Go
36h	ADCEVTINTSEL	ADC Event Interrupt Selection Register	EALLOW	Go
38h	ADCOSDETECT	ADC Open and Shorts Detect Register	EALLOW	Go
39h	ADCCOUNTER	ADC Counter Register		Go
3Ah	ADCREV	ADC Revision Register		Go
3Bh	ADCOFFTRIM	ADC Offset Trim Register	EALLOW	Go

Table 20-14. ADC_REGS Registers (continued)

Offset	Acronym	Register Name	Write Protection	Section
40h	ADCPPB1CONFIG	ADC PPB1 Config Register	EALLOW	Go
41h	ADCPPB1STAMP	ADC PPB1 Sample Delay Time Stamp Register		Go
42h	ADCPPB1OFFCAL	ADC PPB1 Offset Calibration Register	EALLOW	Go
43h	ADCPPB1OFFREF	ADC PPB1 Offset Reference Register		Go
44h	ADCPPB1TRIPHI	ADC PPB1 Trip High Register	EALLOW	Go
46h	ADCPPB1TRIPLO	ADC PPB1 Trip Low/Trigger Time Stamp Register	EALLOW	Go
48h	ADCPPB2CONFIG	ADC PPB2 Config Register	EALLOW	Go
49h	ADCPPB2STAMP	ADC PPB2 Sample Delay Time Stamp Register		Go
4Ah	ADCPPB2OFFCAL	ADC PPB2 Offset Calibration Register	EALLOW	Go
4Bh	ADCPPB2OFFREF	ADC PPB2 Offset Reference Register		Go
4Ch	ADCPPB2TRIPHI	ADC PPB2 Trip High Register	EALLOW	Go
4Eh	ADCPPB2TRIPLO	ADC PPB2 Trip Low/Trigger Time Stamp Register	EALLOW	Go
50h	ADCPPB3CONFIG	ADC PPB3 Config Register	EALLOW	Go
51h	ADCPPB3STAMP	ADC PPB3 Sample Delay Time Stamp Register		Go
52h	ADCPPB3OFFCAL	ADC PPB3 Offset Calibration Register	EALLOW	Go
53h	ADCPPB3OFFREF	ADC PPB3 Offset Reference Register		Go
54h	ADCPPB3TRIPHI	ADC PPB3 Trip High Register	EALLOW	Go
56h	ADCPPB3TRIPLO	ADC PPB3 Trip Low/Trigger Time Stamp Register	EALLOW	Go
58h	ADCPPB4CONFIG	ADC PPB4 Config Register	EALLOW	Go
59h	ADCPPB4STAMP	ADC PPB4 Sample Delay Time Stamp Register		Go
5Ah	ADCPPB4OFFCAL	ADC PPB4 Offset Calibration Register	EALLOW	Go
5Bh	ADCPPB4OFFREF	ADC PPB4 Offset Reference Register		Go
5Ch	ADCPPB4TRIPHI	ADC PPB4 Trip High Register	EALLOW	Go
5Eh	ADCPPB4TRIPLO	ADC PPB4 Trip Low/Trigger Time Stamp Register	EALLOW	Go
6Fh	ADCINTCYCLE	ADC Early Interrupt Generation Cycle	EALLOW	Go
70h	ADCINLTRIM1	ADC Linearity Trim 1 Register	EALLOW	Go
72h	ADCINLTRIM2	ADC Linearity Trim 2 Register	EALLOW	Go
74h	ADCINLTRIM3	ADC Linearity Trim 3 Register	EALLOW	Go
76h	ADCINLTRIM4	ADC Linearity Trim 4 Register	EALLOW	Go
78h	ADCINLTRIM5	ADC Linearity Trim 5 Register	EALLOW	Go
7Ah	ADCINLTRIM6	ADC Linearity Trim 6 Register	EALLOW	Go

Complex bit access types are encoded to fit into small table cells. [Table 20-15](#) shows the codes that are used for access types in this section.

Table 20-15. ADC_REGS Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
R-0	R -0	Read Returns 0s
Write Type		
W	W	Write
W1C	W 1C	Write 1 to clear
W1S	W 1S	Write 1 to set

Table 20-15. ADC_REGS Access Type Codes (continued)

Access Type	Code	Description
Reset or Default Value		
-n		Value after reset or the default value
Register Array Variables		
i,j,k,l,m,n		When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula.
y		When this variable is used in a register name, an offset, or an address it refers to the value of a register array.

20.15.2.1 ADCCTL1 Register (Offset = 0h) [Reset = 0000h]

ADCCTL1 is shown in [Figure 20-26](#) and described in [Table 20-16](#).

Return to the [Summary Table](#).

ADC Control 1 Register

Figure 20-26. ADCCTL1 Register

15	14	13	12	11	10	9	8
RESERVED		ADCBSY	RESERVED	ADCBSYCHN			
R-0h		R-0h	R-0h	R-0h			
7	6	5	4	3	2	1	0
ADCPWDNZ	RESERVED				INTPULSEPOS	RESERVED	
R/W-0h	R-0h				R/W-0h	R-0h	

Table 20-16. ADCCTL1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	RESERVED	R	0h	Reserved
13	ADCBSY	R	0h	ADC Busy. Set when ADC SOC is generated, cleared by hardware four ADC clocks after negative edge of S/H pulse. Used by the ADC state machine to determine if ADC is available to sample. 0 ADC is available to sample next channel 1 ADC is busy and cannot sample another channel Reset type: SYSRSn
12	RESERVED	R	0h	Reserved
11-8	ADCBSYCHN	R	0h	ADC Busy Channel. Set when an ADC Start of Conversion (SOC) is generated. When ADCBSY=0: holds the value of the last converted SOC When ADCBSY=1: reflects the SOC currently being processed 0h SOC0 is currently processing or was last SOC converted 1h SOC1 is currently processing or was last SOC converted 2h SOC2 is currently processing or was last SOC converted 3h SOC3 is currently processing or was last SOC converted 4h SOC4 is currently processing or was last SOC converted 5h SOC5 is currently processing or was last SOC converted 6h SOC6 is currently processing or was last SOC converted 7h SOC7 is currently processing or was last SOC converted 8h SOC8 is currently processing or was last SOC converted 9h SOC9 is currently processing or was last SOC converted Ah SOC10 is currently processing or was last SOC converted Bh SOC11 is currently processing or was last SOC converted Ch SOC12 is currently processing or was last SOC converted Dh SOC13 is currently processing or was last SOC converted Eh SOC14 is currently processing or was last SOC converted Fh SOC15 is currently processing or was last SOC converted Reset type: SYSRSn
7	ADCPWDNZ	R/W	0h	ADC Power Down (active low). This bit controls the power up and power down of all the analog circuitry inside the analog core. 0 All analog circuitry inside the core is powered down 1 All analog circuitry inside the core is powered up Reset type: SYSRSn
6-3	RESERVED	R	0h	Reserved
2	INTPULSEPOS	R/W	0h	ADC Interrupt Pulse Position. 0 Interrupt pulse generation occurs when ADC begins conversion (at the end of the acquisition window) plus a number of SYSCLK cycles as specified in the ADCINTCYCLE.OFFSET register. 1 Interrupt pulse generation occurs at the end of the conversion, 1 cycle prior to the ADC result latching into its result register Reset type: SYSRSn

Table 20-16. ADCCTL1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1-0	RESERVED	R	0h	Reserved

20.15.2.2 ADCCTL2 Register (Offset = 1h) [Reset = 0000h]

ADCCTL2 is shown in [Figure 20-27](#) and described in [Table 20-17](#).

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ADC Control 2 Register

Figure 20-27. ADCCTL2 Register

15	14	13	12	11	10	9	8
RESERVED				RESERVED			
R-0h				R-0h			
7	6	5	4	3	2	1	0
SIGNALMODE	RESOLUTION	RESERVED		PRESCALE			
R/W-0h	R/W-0h	R-0h		R/W-0h			

Table 20-17. ADCCTL2 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-13	RESERVED	R	0h	Reserved
12-8	RESERVED	R	0h	Reserved
7	SIGNALMODE	R/W	0h	SOC Signaling Mode. Selects the input mode of the converter. Use the <code>AdcSetMode</code> function to change the signal mode. 0 Single-ended 1 Differential Reset type: SYSRSn
6	RESOLUTION	R/W	0h	SOC Conversion Resolution. Selects the resolution of the converter. Use the <code>AdcSetMode</code> function to change the resolution. 0 12-bit resolution 1 16-bit resolution Reset type: SYSRSn
5-4	RESERVED	R	0h	Reserved
3-0	PRESCALE	R/W	0h	ADC Clock Prescaler. 0000 ADCCLK = Input Clock / 1.0 0001 Invalid 0010 ADCCLK = Input Clock / 2.0 0011 ADCCLK = Input Clock / 2.5 0100 ADCCLK = Input Clock / 3.0 0101 ADCCLK = Input Clock / 3.5 0110 ADCCLK = Input Clock / 4.0 0111 ADCCLK = Input Clock / 4.5 1000 ADCCLK = Input Clock / 5.0 1001 ADCCLK = Input Clock / 5.5 1010 ADCCLK = Input Clock / 6.0 1011 ADCCLK = Input Clock / 6.5 1100 ADCCLK = Input Clock / 7.0 1101 ADCCLK = Input Clock / 7.5 1110 ADCCLK = Input Clock / 8.0 1111 ADCCLK = Input Clock / 8.5 Reset type: SYSRSn

20.15.2.3 ADCBURSTCTL Register (Offset = 2h) [Reset = 0000h]

ADCBURSTCTL is shown in [Figure 20-28](#) and described in [Table 20-18](#).

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ADC Burst Control Register

Figure 20-28. ADCBURSTCTL Register

15	14	13	12	11	10	9	8
BURSTEN	RESERVED				BURSTSIZE		
R/W-0h	R-0h				R/W-0h		
7	6	5	4	3	2	1	0
RESERVED		BURSTTRIGSEL					
R-0h		R/W-0h					

Table 20-18. ADCBURSTCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	BURSTEN	R/W	0h	SOC Burst Mode Enable. This bit enables the SOC Burst Mode of operation. 0 Burst mode is disabled. 1 Burst mode is enabled. Reset type: SYSRSn
14-12	RESERVED	R	0h	Reserved
11-8	BURSTSIZE	R/W	0h	SOC Burst Size Select. This bit field determines how many SOC's are converted when a burst conversion sequence is started. The first SOC converted is defined by the round robin pointer, which is advanced as each SOC is converted. 0h 1 SOC converted 1h 2 SOC's converted 2h 3 SOC's converted 3h 4 SOC's converted 4h 5 SOC's converted 5h 6 SOC's converted 6h 7 SOC's converted 7h 8 SOC's converted 8h 9 SOC's converted 9h 10 SOC's converted Ah 11 SOC's converted Bh 12 SOC's converted Ch 13 SOC's converted Dh 14 SOC's converted Eh 15 SOC's converted Fh 16 SOC's converted Reset type: SYSRSn
7-6	RESERVED	R	0h	Reserved

Table 20-18. ADCBURSTCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5-0	BURSTTRIGSEL	R/W	0h	SOC Burst Trigger Source Select. Configures which trigger will start a burst conversion sequence. 00h BURSTTRIG0 - Software only 01h BURSTTRIG1 - CPU1 Timer 0, TINT0n 02h BURSTTRIG2 - CPU1 Timer 1, TINT1n 03h BURSTTRIG3 - CPU1 Timer 2, TINT2n 04h BURSTTRIG4 - GPIO, Input X-Bar INPUT5 05h BURSTTRIG5 - ePWM1, ADCSOCA 06h BURSTTRIG6 - ePWM1, ADCSOCA 07h BURSTTRIG7 - ePWM2, ADCSOCA 08h BURSTTRIG8 - ePWM2, ADCSOCA 09h BURSTTRIG9 - ePWM3, ADCSOCA 0Ah BURSTTRIG10 - ePWM3, ADCSOCA 0Bh BURSTTRIG11 - ePWM4, ADCSOCA 0Ch BURSTTRIG12 - ePWM4, ADCSOCA 0Dh BURSTTRIG13 - ePWM5, ADCSOCA 0Eh BURSTTRIG14 - ePWM5, ADCSOCA 0Fh BURSTTRIG15 - ePWM6, ADCSOCA 10h BURSTTRIG16 - ePWM6, ADCSOCA 11h BURSTTRIG17 - ePWM7, ADCSOCA 12h BURSTTRIG18 - ePWM7, ADCSOCA 13h BURSTTRIG19 - ePWM8, ADCSOCA 14h BURSTTRIG20 - ePWM8, ADCSOCA 15h BURSTTRIG21 - ePWM9, ADCSOCA 16h BURSTTRIG22 - ePWM9, ADCSOCA 17h BURSTTRIG23 - ePWM10, ADCSOCA 18h BURSTTRIG24 - ePWM10, ADCSOCA 19h BURSTTRIG25 - ePWM11, ADCSOCA 1Ah BURSTTRIG26 - ePWM11, ADCSOCA 1Bh BURSTTRIG27 - ePWM12, ADCSOCA 1Ch BURSTTRIG28 - ePWM12, ADCSOCA 1Dh BURSTTRIG29 - CPU2 Timer 0, TINT0n 1Eh BURSTTRIG30 - CPU2 Timer 1, TINT1n 1Fh BURSTTRIG31 - CPU2 Timer 2, TINT2n 20h BURSTTRIG32 - ePWM13, ADCSOCA 21h BURSTTRIG33 - ePWM13, ADCSOCA 22h BURSTTRIG34 - ePWM14, ADCSOCA 23h BURSTTRIG35 - ePWM14, ADCSOCA 24h BURSTTRIG36 - ePWM15, ADCSOCA 25h BURSTTRIG37 - ePWM15, ADCSOCA 26h BURSTTRIG38 - ePWM16, ADCSOCA 27h BURSTTRIG39 - ePWM16, ADCSOCA 28h - 3Fh - Reserved Reset type: SYSRSn

20.15.2.4 ADCINTFLG Register (Offset = 3h) [Reset = 0000h]

ADCINTFLG is shown in [Figure 20-29](#) and described in [Table 20-19](#).

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ADC Interrupt Flag Register

Figure 20-29. ADCINTFLG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				ADCINT4	ADCINT3	ADCINT2	ADCINT1
R-0h				R-0h	R-0h	R-0h	R-0h

Table 20-19. ADCINTFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	ADCINT4	R	0h	ADC Interrupt 4 Flag. Reading these flags indicates if the associated ADCINT pulse was generated since the last clear. 0 No ADC interrupt pulse generated 1 ADC interrupt pulse generated If the ADC interrupt is placed in continue to interrupt mode (INTSELxNy register) then further interrupt pulses are generated whenever a selected EOC event occurs even if the flag bit is set. If the continuous mode is not enabled, then no further interrupt pulses are generated until the user clears this flag bit using the ADCINTFLGCLR register. Rather, an ADC interrupt overflow event occurs in the ADCINTOVF register. Reset type: SYSRSn
2	ADCINT3	R	0h	ADC Interrupt 3 Flag. Reading these flags indicates if the associated ADCINT pulse was generated since the last clear. 0 No ADC interrupt pulse generated 1 ADC interrupt pulse generated If the ADC interrupt is placed in continue to interrupt mode (INTSELxNy register) then further interrupt pulses are generated whenever a selected EOC event occurs even if the flag bit is set. If the continuous mode is not enabled, then no further interrupt pulses are generated until the user clears this flag bit using the ADCINTFLGCLR register. Rather, an ADC interrupt overflow event occurs in the ADCINTOVF register. Reset type: SYSRSn
1	ADCINT2	R	0h	ADC Interrupt 2 Flag. Reading these flags indicates if the associated ADCINT pulse was generated since the last clear. 0 No ADC interrupt pulse generated 1 ADC interrupt pulse generated If the ADC interrupt is placed in continue to interrupt mode (INTSELxNy register) then further interrupt pulses are generated whenever a selected EOC event occurs even if the flag bit is set. If the continuous mode is not enabled, then no further interrupt pulses are generated until the user clears this flag bit using the ADCINTFLGCLR register. Rather, an ADC interrupt overflow event occurs in the ADCINTOVF register. Reset type: SYSRSn

Table 20-19. ADCINTFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	ADCINT1	R	0h	ADC Interrupt 1 Flag. Reading these flags indicates if the associated ADCINT pulse was generated since the last clear. 0 No ADC interrupt pulse generated 1 ADC interrupt pulse generated If the ADC interrupt is placed in continue to interrupt mode (INTSELxNy register) then further interrupt pulses are generated whenever a selected EOC event occurs even if the flag bit is set. If the continuous mode is not enabled, then no further interrupt pulses are generated until the user clears this flag bit using the ADCINTFLGCLR register. Rather, an ADC interrupt overflow event occurs in the ADCINTOVF register. Reset type: SYSRStn

20.15.2.5 ADCINTFLGCLR Register (Offset = 4h) [Reset = 0000h]

ADCINTFLGCLR is shown in [Figure 20-30](#) and described in [Table 20-20](#).

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ADC Interrupt Flag Clear Register

Figure 20-30. ADCINTFLGCLR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				ADCINT4	ADCINT3	ADCINT2	ADCINT1
R-0h				R-0/W1C-0h	R-0/W1C-0h	R-0/W1C-0h	R-0/W1C-0h

Table 20-20. ADCINTFLGCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	ADCINT4	R-0/W1C	0h	ADC Interrupt 4 Flag Clear. Reads return 0. 0 No action 1 Clears respective flag bit in the ADCINTFLG register. If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority and the overflow bit will not be set Reset type: SYSRSn
2	ADCINT3	R-0/W1C	0h	ADC Interrupt 3 Flag Clear. Reads return 0. 0 No action 1 Clears respective flag bit in the ADCINTFLG register. If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority and the overflow bit will not be set Reset type: SYSRSn
1	ADCINT2	R-0/W1C	0h	ADC Interrupt 2 Flag Clear. Reads return 0. 0 No action 1 Clears respective flag bit in the ADCINTFLG register. If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority and the overflow bit will not be set Reset type: SYSRSn
0	ADCINT1	R-0/W1C	0h	ADC Interrupt 1 Flag Clear. Reads return 0. 0 No action 1 Clears respective flag bit in the ADCINTFLG register. If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority and the overflow bit will not be set Reset type: SYSRSn

20.15.2.6 ADCINTOVF Register (Offset = 5h) [Reset = 0000h]

ADCINTOVF is shown in [Figure 20-31](#) and described in [Table 20-21](#).

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ADC Interrupt Overflow Register

Figure 20-31. ADCINTOVF Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				ADCINT4	ADCINT3	ADCINT2	ADCINT1
R-0h				R-0h	R-0h	R-0h	R-0h

Table 20-21. ADCINTOVF Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	ADCINT4	R	0h	ADC Interrupt 4 Overflow Flags Indicates if an overflow occurred when generating ADCINT pulses. If the respective ADCINTFLG bit is set and a selected additional EOC trigger is generated, then an overflow condition occurs. 0 No ADC interrupt overflow event detected. 1 ADC Interrupt overflow event detected. The overflow bit does not care about the continuous mode bit state. An overflow condition is generated irrespective of this mode selection. Reset type: SYSRSn
2	ADCINT3	R	0h	ADC Interrupt 3 Overflow Flags Indicates if an overflow occurred when generating ADCINT pulses. If the respective ADCINTFLG bit is set and a selected additional EOC trigger is generated, then an overflow condition occurs. 0 No ADC interrupt overflow event detected. 1 ADC Interrupt overflow event detected. The overflow bit does not care about the continuous mode bit state. An overflow condition is generated irrespective of this mode selection. Reset type: SYSRSn
1	ADCINT2	R	0h	ADC Interrupt 2 Overflow Flags Indicates if an overflow occurred when generating ADCINT pulses. If the respective ADCINTFLG bit is set and a selected additional EOC trigger is generated, then an overflow condition occurs. 0 No ADC interrupt overflow event detected. 1 ADC Interrupt overflow event detected. The overflow bit does not care about the continuous mode bit state. An overflow condition is generated irrespective of this mode selection. Reset type: SYSRSn
0	ADCINT1	R	0h	ADC Interrupt 1 Overflow Flags Indicates if an overflow occurred when generating ADCINT pulses. If the respective ADCINTFLG bit is set and a selected additional EOC trigger is generated, then an overflow condition occurs. 0 No ADC interrupt overflow event detected. 1 ADC Interrupt overflow event detected. The overflow bit does not care about the continuous mode bit state. An overflow condition is generated irrespective of this mode selection. Reset type: SYSRSn

20.15.2.7 ADCINTOVFCLR Register (Offset = 6h) [Reset = 0000h]

ADCINTOVFCLR is shown in [Figure 20-32](#) and described in [Table 20-22](#).

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ADC Interrupt Overflow Clear Register

Figure 20-32. ADCINTOVFCLR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				ADCINT4	ADCINT3	ADCINT2	ADCINT1
R-0h				R-0/W1C-0h	R-0/W1C-0h	R-0/W1C-0h	R-0/W1C-0h

Table 20-22. ADCINTOVFCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	ADCINT4	R-0/W1C	0h	ADC Interrupt 4 Overflow Clear Bits 0 No action. 1 Clears the respective overflow bit in the ADCINTOVF register. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCINTOVF register, then hardware has priority and the ADCINTOVF bit will be set. Reset type: SYSRSn
2	ADCINT3	R-0/W1C	0h	ADC Interrupt 3 Overflow Clear Bits 0 No action. 1 Clears the respective overflow bit in the ADCINTOVF register. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCINTOVF register, then hardware has priority and the ADCINTOVF bit will be set. Reset type: SYSRSn
1	ADCINT2	R-0/W1C	0h	ADC Interrupt 2 Overflow Clear Bits 0 No action. 1 Clears the respective overflow bit in the ADCINTOVF register. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCINTOVF register, then hardware has priority and the ADCINTOVF bit will be set. Reset type: SYSRSn
0	ADCINT1	R-0/W1C	0h	ADC Interrupt 1 Overflow Clear Bits 0 No action. 1 Clears the respective overflow bit in the ADCINTOVF register. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCINTOVF register, then hardware has priority and the ADCINTOVF bit will be set. Reset type: SYSRSn

20.15.2.8 ADCINTSEL1N2 Register (Offset = 7h) [Reset = 0000h]

ADCINTSEL1N2 is shown in [Figure 20-33](#) and described in [Table 20-23](#).

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ADC Interrupt 1 and 2 Selection Register

Figure 20-33. ADCINTSEL1N2 Register

15	14	13	12	11	10	9	8
RESERVED	INT2CONT	INT2E	RESERVED	INT2SEL			
R-0h	R/W-0h	R/W-0h	R-0h	R/W-0h			
7	6	5	4	3	2	1	0
RESERVED	INT1CONT	INT1E	RESERVED	INT1SEL			
R-0h	R/W-0h	R/W-0h	R-0h	R/W-0h			

Table 20-23. ADCINTSEL1N2 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	INT2CONT	R/W	0h	ADCINT2 Continue to Interrupt Mode 0 No further ADCINT2 pulses are generated until ADCINT2 flag (in ADCINTFLG register) is cleared by user. 1 ADCINT2 pulses are generated whenever an EOC pulse is generated irrespective of whether the flag bit is cleared or not. Reset type: SYSRSn
13	INT2E	R/W	0h	ADCINT2 Interrupt Enable 0 ADCINT2 is disabled 1 ADCINT2 is enabled Reset type: SYSRSn
12	RESERVED	R	0h	Reserved
11-8	INT2SEL	R/W	0h	ADCINT2 EOC Source Select 0h EOC0 is trigger for ADCINT2 1h EOC1 is trigger for ADCINT2 2h EOC2 is trigger for ADCINT2 3h EOC3 is trigger for ADCINT2 4h EOC4 is trigger for ADCINT2 5h EOC5 is trigger for ADCINT2 6h EOC6 is trigger for ADCINT2 7h EOC7 is trigger for ADCINT2 8h EOC8 is trigger for ADCINT2 9h EOC9 is trigger for ADCINT2 Ah EOC10 is trigger for ADCINT2 Bh EOC11 is trigger for ADCINT2 Ch EOC12 is trigger for ADCINT2 Dh EOC13 is trigger for ADCINT2 Eh EOC14 is trigger for ADCINT2 Fh EOC15 is trigger for ADCINT2 Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	INT1CONT	R/W	0h	ADCINT1 Continue to Interrupt Mode 0 No further ADCINT1 pulses are generated until ADCINT1 flag (in ADCINTFLG register) is cleared by user. 1 ADCINT1 pulses are generated whenever an EOC pulse is generated irrespective of whether the flag bit is cleared or not. Reset type: SYSRSn
5	INT1E	R/W	0h	ADCINT1 Interrupt Enable 0 ADCINT1 is disabled 1 ADCINT1 is enabled Reset type: SYSRSn

Table 20-23. ADCINTSEL1N2 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4	RESERVED	R	0h	Reserved
3-0	INT1SEL	R/W	0h	ADCINT1 EOC Source Select 0h EOC0 is trigger for ADCINT1 1h EOC1 is trigger for ADCINT1 2h EOC2 is trigger for ADCINT1 3h EOC3 is trigger for ADCINT1 4h EOC4 is trigger for ADCINT1 5h EOC5 is trigger for ADCINT1 6h EOC6 is trigger for ADCINT1 7h EOC7 is trigger for ADCINT1 8h EOC8 is trigger for ADCINT1 9h EOC9 is trigger for ADCINT1 Ah EOC10 is trigger for ADCINT1 Bh EOC11 is trigger for ADCINT1 Ch EOC12 is trigger for ADCINT1 Dh EOC13 is trigger for ADCINT1 Eh EOC14 is trigger for ADCINT1 Fh EOC15 is trigger for ADCINT1 Reset type: SYSRStn

20.15.2.9 ADCINTSEL3N4 Register (Offset = 8h) [Reset = 0000h]

ADCINTSEL3N4 is shown in [Figure 20-34](#) and described in [Table 20-24](#).

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ADC Interrupt 3 and 4 Selection Register

Figure 20-34. ADCINTSEL3N4 Register

15	14	13	12	11	10	9	8
RESERVED	INT4CONT	INT4E	RESERVED	INT4SEL			
R-0h	R/W-0h	R/W-0h	R-0h	R/W-0h			
7	6	5	4	3	2	1	0
RESERVED	INT3CONT	INT3E	RESERVED	INT3SEL			
R-0h	R/W-0h	R/W-0h	R-0h	R/W-0h			

Table 20-24. ADCINTSEL3N4 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	INT4CONT	R/W	0h	ADCINT4 Continue to Interrupt Mode 0 No further ADCINT4 pulses are generated until ADCINT4 flag (in ADCINTFLG register) is cleared by user. 1 ADCINT4 pulses are generated whenever an EOC pulse is generated irrespective of whether the flag bit is cleared or not. Reset type: SYSRSn
13	INT4E	R/W	0h	ADCINT4 Interrupt Enable 0 ADCINT4 is disabled 1 ADCINT4 is enabled Reset type: SYSRSn
12	RESERVED	R	0h	Reserved
11-8	INT4SEL	R/W	0h	ADCINT4 EOC Source Select 0h EOC0 is trigger for ADCINT4 1h EOC1 is trigger for ADCINT4 2h EOC2 is trigger for ADCINT4 3h EOC3 is trigger for ADCINT4 4h EOC4 is trigger for ADCINT4 5h EOC5 is trigger for ADCINT4 6h EOC6 is trigger for ADCINT4 7h EOC7 is trigger for ADCINT4 8h EOC8 is trigger for ADCINT4 9h EOC9 is trigger for ADCINT4 Ah EOC10 is trigger for ADCINT4 Bh EOC11 is trigger for ADCINT4 Ch EOC12 is trigger for ADCINT4 Dh EOC13 is trigger for ADCINT4 Eh EOC14 is trigger for ADCINT4 Fh EOC15 is trigger for ADCINT4 Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	INT3CONT	R/W	0h	ADCINT3 Continue to Interrupt Mode 0 No further ADCINT3 pulses are generated until ADCINT3 flag (in ADCINTFLG register) is cleared by user. 1 ADCINT3 pulses are generated whenever an EOC pulse is generated irrespective of whether the flag bit is cleared or not. Reset type: SYSRSn
5	INT3E	R/W	0h	ADCINT3 Interrupt Enable 0 ADCINT3 is disabled 1 ADCINT3 is enabled Reset type: SYSRSn

Table 20-24. ADCINTSEL3N4 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4	RESERVED	R	0h	Reserved
3-0	INT3SEL	R/W	0h	ADCINT3 EOC Source Select 0h EOC0 is trigger for ADCINT3 1h EOC1 is trigger for ADCINT3 2h EOC2 is trigger for ADCINT3 3h EOC3 is trigger for ADCINT3 4h EOC4 is trigger for ADCINT3 5h EOC5 is trigger for ADCINT3 6h EOC6 is trigger for ADCINT3 7h EOC7 is trigger for ADCINT3 8h EOC8 is trigger for ADCINT3 9h EOC9 is trigger for ADCINT3 Ah EOC10 is trigger for ADCINT3 Bh EOC11 is trigger for ADCINT3 Ch EOC12 is trigger for ADCINT3 Dh EOC13 is trigger for ADCINT3 Eh EOC14 is trigger for ADCINT3 Fh EOC15 is trigger for ADCINT3 Reset type: SYSRStn

20.15.2.10 ADCSOCPRCTL Register (Offset = 9h) [Reset = 0200h]

ADCSOCPRCTL is shown in [Figure 20-35](#) and described in [Table 20-25](#).

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ADC SOC Priority Control Register

Figure 20-35. ADCSOCPRCTL Register

15	14	13	12	11	10	9	8
RESERVED						RRPOINTER	
R-0h						R-10h	
7	6	5	4	3	2	1	0
RRPOINTER				SOCPRIORITY			
R-10h				R/W-0h			

Table 20-25. ADCSOCPRCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-5	RRPOINTER	R	10h	Round Robin Pointer. Holds the value of the last converted round robin SOCx to be used by the round robin scheme to determine order of conversions. 00h SOC0 was last round robin SOC to convert, SOC1 is highest round robin priority. 01h SOC1 was last round robin SOC to convert, SOC2 is highest round robin priority. 02h SOC2 was last round robin SOC to convert, SOC3 is highest round robin priority. 03h SOC3 was last round robin SOC to convert, SOC4 is highest round robin priority. 04h SOC4 was last round robin SOC to convert, SOC5 is highest round robin priority. 05h SOC5 was last round robin SOC to convert, SOC6 is highest round robin priority. 06h SOC6 was last round robin SOC to convert, SOC7 is highest round robin priority. 07h SOC7 was last round robin SOC to convert, SOC8 is highest round robin priority. 08h SOC8 was last round robin SOC to convert, SOC9 is highest round robin priority. 09h SOC9 was last round robin SOC to convert, SOC10 is highest round robin priority. 0Ah SOC10 was last round robin SOC to convert, SOC11 is highest round robin priority. 0Bh SOC11 was last round robin SOC to convert, SOC12 is highest round robin priority. 0Ch SOC12 was last round robin SOC to convert, SOC13 is highest round robin priority. 0Dh SOC13 was last round robin SOC to convert, SOC14 is highest round robin priority. 0Eh SOC14 was last round robin SOC to convert, SOC15 is highest round robin priority. 0Fh SOC15 was last round robin SOC to convert, SOC0 is highest round robin priority. 10h Reset value to indicate no SOC has been converted. SOC0 is highest round robin priority. Set to this value when the ADC module is reset by SOFTPRES or when the ADCSOCPRCTL register is written. In the latter case, if a conversion is currently in progress, it will complete and then the new priority will take effect. Others Invalid value. Reset type: SYSRSn

Table 20-25. ADCSOCPRCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4-0	SOC PRIORITY	R/W	0h	SOC Priority Determines the cutoff point for priority mode and round robin arbitration for SOCx 00h SOC priority is handled in round robin mode for all channels. 01h SOC0 is high priority, rest of channels are in round robin mode. 02h SOC0-SOC1 are high priority, SOC2-SOC15 are in round robin mode. 03h SOC0-SOC2 are high priority, SOC3-SOC15 are in round robin mode. 04h SOC0-SOC3 are high priority, SOC4-SOC15 are in round robin mode. 05h SOC0-SOC4 are high priority, SOC5-SOC15 are in round robin mode. 06h SOC0-SOC5 are high priority, SOC6-SOC15 are in round robin mode. 07h SOC0-SOC6 are high priority, SOC7-SOC15 are in round robin mode. 08h SOC0-SOC7 are high priority, SOC8-SOC15 are in round robin mode. 09h SOC0-SOC8 are high priority, SOC9-SOC15 are in round robin mode. 0Ah SOC0-SOC9 are high priority, SOC10-SOC15 are in round robin mode. 0Bh SOC0-SOC10 are high priority, SOC11-SOC15 are in round robin mode. 0Ch SOC0-SOC11 are high priority, SOC12-SOC15 are in round robin mode. 0Dh SOC0-SOC12 are high priority, SOC13-SOC15 are in round robin mode. 0Eh SOC0-SOC13 are high priority, SOC14-SOC15 are in round robin mode. 0Fh SOC0-SOC14 are high priority, SOC15 is in round robin mode. 10h All SOCx are in high priority mode, arbitrated by SOC number. Others Invalid selection. Reset type: SYSRSn

20.15.2.11 ADCINTSOCSEL1 Register (Offset = Ah) [Reset = 0000h]

ADCINTSOCSEL1 is shown in [Figure 20-36](#) and described in [Table 20-26](#).

Return to the [Summary Table](#).

ADC Interrupt SOC Selection 1 Register

Figure 20-36. ADCINTSOCSEL1 Register

15	14	13	12	11	10	9	8
SOC7		SOC6		SOC5		SOC4	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
SOC3		SOC2		SOC1		SOC0	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	

Table 20-26. ADCINTSOCSEL1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	SOC7	R/W	0h	SOC7 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC7. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC7. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC7. 10 ADCINT2 will trigger SOC7. 11 Invalid selection. Reset type: SYSRSn
13-12	SOC6	R/W	0h	SOC6 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC6. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC6. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC6. 10 ADCINT2 will trigger SOC6. 11 Invalid selection. Reset type: SYSRSn
11-10	SOC5	R/W	0h	SOC5 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC5. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC5. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC5. 10 ADCINT2 will trigger SOC5. 11 Invalid selection. Reset type: SYSRSn
9-8	SOC4	R/W	0h	SOC4 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC4. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC4. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC4. 10 ADCINT2 will trigger SOC4. 11 Invalid selection. Reset type: SYSRSn

Table 20-26. ADCINTSOCSEL1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	SOC3	R/W	0h	SOC3 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC3. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC3. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC3. 10 ADCINT2 will trigger SOC3. 11 Invalid selection. Reset type: SYSRSn
5-4	SOC2	R/W	0h	SOC2 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC2. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC2. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC2. 10 ADCINT2 will trigger SOC2. 11 Invalid selection. Reset type: SYSRSn
3-2	SOC1	R/W	0h	SOC1 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC1. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC1. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC1. 10 ADCINT2 will trigger SOC1. 11 Invalid selection. Reset type: SYSRSn
1-0	SOC0	R/W	0h	SOC0 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC0. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC0. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC0. 10 ADCINT2 will trigger SOC0. 11 Invalid selection. Reset type: SYSRSn

20.15.2.12 ADCINTSOCSEL2 Register (Offset = Bh) [Reset = 0000h]

ADCINTSOCSEL2 is shown in [Figure 20-37](#) and described in [Table 20-27](#).

Return to the [Summary Table](#).

ADC Interrupt SOC Selection 2 Register

Figure 20-37. ADCINTSOCSEL2 Register

15	14	13	12	11	10	9	8
SOC15		SOC14		SOC13		SOC12	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
SOC11		SOC10		SOC9		SOC8	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	

Table 20-27. ADCINTSOCSEL2 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	SOC15	R/W	0h	SOC15 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC15. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC15. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC15. 10 ADCINT2 will trigger SOC15. 11 Invalid selection. Reset type: SYSRSn
13-12	SOC14	R/W	0h	SOC14 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC14. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC14. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC14. 10 ADCINT2 will trigger SOC14. 11 Invalid selection. Reset type: SYSRSn
11-10	SOC13	R/W	0h	SOC13 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC13. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC13. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC13. 10 ADCINT2 will trigger SOC13. 11 Invalid selection. Reset type: SYSRSn
9-8	SOC12	R/W	0h	SOC12 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC12. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC12. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC12. 10 ADCINT2 will trigger SOC12. 11 Invalid selection. Reset type: SYSRSn

Table 20-27. ADCINTSOCSEL2 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	SOC11	R/W	0h	SOC11 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC11. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC11. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC11. 10 ADCINT2 will trigger SOC11. 11 Invalid selection. Reset type: SYSRSn
5-4	SOC10	R/W	0h	SOC10 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC10. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC10. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC10. 10 ADCINT2 will trigger SOC10. 11 Invalid selection. Reset type: SYSRSn
3-2	SOC9	R/W	0h	SOC9 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC9. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC9. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC9. 10 ADCINT2 will trigger SOC9. 11 Invalid selection. Reset type: SYSRSn
1-0	SOC8	R/W	0h	SOC8 ADC Interrupt Trigger Select. Selects which, if any, ADCINT triggers SOC8. The trigger selected in this field is in addition to the TRIGSEL field in the ADCSOCxCTL register. 00 No ADCINT will trigger SOC8. TRIGSEL field alone determines SOC0 trigger. 01 ADCINT1 will trigger SOC8. 10 ADCINT2 will trigger SOC8. 11 Invalid selection. Reset type: SYSRSn

20.15.2.13 ADCSOCFLG1 Register (Offset = Ch) [Reset = 0000h]

ADCSOCFLG1 is shown in [Figure 20-38](#) and described in [Table 20-28](#).

Return to the [Summary Table](#).

ADC SOC Flag 1 Register

Figure 20-38. ADCSOCFLG1 Register

15	14	13	12	11	10	9	8
SOC15	SOC14	SOC13	SOC12	SOC11	SOC10	SOC9	SOC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
SOC7	SOC6	SOC5	SOC4	SOC3	SOC2	SOC1	SOC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 20-28. ADCSOCFLG1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	SOC15	R	0h	SOC15 Start of Conversion Flag. Indicates the state of SOC15 conversions. 0 No sample pending for SOC15. 1 Trigger has been received and sample is pending for SOC15. This bit will be automatically cleared when the SOC15 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
14	SOC14	R	0h	SOC14 Start of Conversion Flag. Indicates the state of SOC14 conversions. 0 No sample pending for SOC14. 1 Trigger has been received and sample is pending for SOC14. This bit will be automatically cleared when the SOC14 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
13	SOC13	R	0h	SOC13 Start of Conversion Flag. Indicates the state of SOC13 conversions. 0 No sample pending for SOC13. 1 Trigger has been received and sample is pending for SOC13. This bit will be automatically cleared when the SOC13 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn

Table 20-28. ADCSOCFLG1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
12	SOC12	R	0h	SOC12 Start of Conversion Flag. Indicates the state of SOC12 conversions. 0 No sample pending for SOC12. 1 Trigger has been received and sample is pending for SOC12. This bit will be automatically cleared when the SOC12 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
11	SOC11	R	0h	SOC11 Start of Conversion Flag. Indicates the state of SOC11 conversions. 0 No sample pending for SOC11. 1 Trigger has been received and sample is pending for SOC11. This bit will be automatically cleared when the SOC11 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
10	SOC10	R	0h	SOC10 Start of Conversion Flag. Indicates the state of SOC10 conversions. 0 No sample pending for SOC10. 1 Trigger has been received and sample is pending for SOC10. This bit will be automatically cleared when the SOC10 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
9	SOC9	R	0h	SOC9 Start of Conversion Flag. Indicates the state of SOC9 conversions. 0 No sample pending for SOC9. 1 Trigger has been received and sample is pending for SOC9. This bit will be automatically cleared when the SOC9 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
8	SOC8	R	0h	SOC8 Start of Conversion Flag. Indicates the state of SOC8 conversions. 0 No sample pending for SOC8. 1 Trigger has been received and sample is pending for SOC8. This bit will be automatically cleared when the SOC8 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn

Table 20-28. ADCSOCFLG1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7	SOC7	R	0h	SOC7 Start of Conversion Flag. Indicates the state of SOC7 conversions. 0 No sample pending for SOC7. 1 Trigger has been received and sample is pending for SOC7. This bit will be automatically cleared when the SOC7 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
6	SOC6	R	0h	SOC6 Start of Conversion Flag. Indicates the state of SOC6 conversions. 0 No sample pending for SOC6. 1 Trigger has been received and sample is pending for SOC6. This bit will be automatically cleared when the SOC6 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
5	SOC5	R	0h	SOC5 Start of Conversion Flag. Indicates the state of SOC5 conversions. 0 No sample pending for SOC5. 1 Trigger has been received and sample is pending for SOC5. This bit will be automatically cleared when the SOC5 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
4	SOC4	R	0h	SOC4 Start of Conversion Flag. Indicates the state of SOC4 conversions. 0 No sample pending for SOC4. 1 Trigger has been received and sample is pending for SOC4. This bit will be automatically cleared when the SOC4 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn
3	SOC3	R	0h	SOC3 Start of Conversion Flag. Indicates the state of SOC3 conversions. 0 No sample pending for SOC3. 1 Trigger has been received and sample is pending for SOC3. This bit will be automatically cleared when the SOC3 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRSn

Table 20-28. ADCSOCFLG1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
2	SOC2	R	0h	SOC2 Start of Conversion Flag. Indicates the state of SOC2 conversions. 0 No sample pending for SOC2. 1 Trigger has been received and sample is pending for SOC2. This bit will be automatically cleared when the SOC2 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRStn
1	SOC1	R	0h	SOC1 Start of Conversion Flag. Indicates the state of SOC1 conversions. 0 No sample pending for SOC1. 1 Trigger has been received and sample is pending for SOC1. This bit will be automatically cleared when the SOC1 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRStn
0	SOC0	R	0h	SOC0 Start of Conversion Flag. Indicates the state of SOC0 conversions. 0 No sample pending for SOC0. 1 Trigger has been received and sample is pending for SOC0. This bit will be automatically cleared when the SOC0 conversion is started. If contention exists where this bit receives both a request to set and a request to clear on the same cycle, regardless of the source of either, this bit will be set and the request to clear will be ignored. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether this bit was previously set or not. Reset type: SYSRStn

20.15.2.14 ADCSOCFRC1 Register (Offset = Dh) [Reset = 0000h]

ADCSOCFRC1 is shown in [Figure 20-39](#) and described in [Table 20-29](#).

Return to the [Summary Table](#).

ADC SOC Force 1 Register

Figure 20-39. ADCSOCFRC1 Register

15	14	13	12	11	10	9	8
SOC15	SOC14	SOC13	SOC12	SOC11	SOC10	SOC9	SOC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
SOC7	SOC6	SOC5	SOC4	SOC3	SOC2	SOC1	SOC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 20-29. ADCSOCFRC1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	SOC15	R-0/W1S	0h	SOC15 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC15 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC15 flag bit to 1. This will cause a conversion to start once priority is given to SOC15. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC15 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
14	SOC14	R-0/W1S	0h	SOC14 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC14 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC14 flag bit to 1. This will cause a conversion to start once priority is given to SOC14. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC14 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
13	SOC13	R-0/W1S	0h	SOC13 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC13 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC13 flag bit to 1. This will cause a conversion to start once priority is given to SOC13. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC13 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn

Table 20-29. ADCSOCFRC1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
12	SOC12	R-0/W1S	0h	SOC12 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC12 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC12 flag bit to 1. This will cause a conversion to start once priority is given to SOC12. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC12 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
11	SOC11	R-0/W1S	0h	SOC11 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC11 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC11 flag bit to 1. This will cause a conversion to start once priority is given to SOC11. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC11 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
10	SOC10	R-0/W1S	0h	SOC10 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC10 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC10 flag bit to 1. This will cause a conversion to start once priority is given to SOC10. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC10 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
9	SOC9	R-0/W1S	0h	SOC9 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC9 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC9 flag bit to 1. This will cause a conversion to start once priority is given to SOC9. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC9 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn

Table 20-29. ADCSOCFRC1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
8	SOC8	R-0/W1S	0h	SOC8 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC8 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC8 flag bit to 1. This will cause a conversion to start once priority is given to SOC8. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC8 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
7	SOC7	R-0/W1S	0h	SOC7 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC7 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC7 flag bit to 1. This will cause a conversion to start once priority is given to SOC7. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC7 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
6	SOC6	R-0/W1S	0h	SOC6 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC6 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC6 flag bit to 1. This will cause a conversion to start once priority is given to SOC6. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC6 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
5	SOC5	R-0/W1S	0h	SOC5 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC5 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC5 flag bit to 1. This will cause a conversion to start once priority is given to SOC5. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC5 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn

Table 20-29. ADCSOCFRC1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4	SOC4	R-0/W1S	0h	SOC4 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC4 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC4 flag bit to 1. This will cause a conversion to start once priority is given to SOC4. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC4 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
3	SOC3	R-0/W1S	0h	SOC3 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC3 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC3 flag bit to 1. This will cause a conversion to start once priority is given to SOC3. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC3 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
2	SOC2	R-0/W1S	0h	SOC2 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC2 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC2 flag bit to 1. This will cause a conversion to start once priority is given to SOC2. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC2 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn
1	SOC1	R-0/W1S	0h	SOC1 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC1 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC1 flag bit to 1. This will cause a conversion to start once priority is given to SOC1. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC1 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRSn

Table 20-29. ADCSOCFRC1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	SOC0	R-0/W1S	0h	SOC0 Force Start of Conversion Bit. Writing a 1 will force to 1 the SOC0 flag in the ADCSOCFLG1 register. This can be used to initiate a software initiated conversion. Writes of 0 are ignored. This bit will always read as a 0. 0 No action. 1 Force SOC0 flag bit to 1. This will cause a conversion to start once priority is given to SOC0. If software tries to set this bit on the same clock cycle that hardware tries to clear the SOC0 bit in the ADCSOCFLG1 register, then software has priority and the ADCSOCFLG1 bit will be set. In this case the overflow bit in the ADCSOCOVF1 register will not be affected regardless of whether the ADCSOCFLG1 bit was previously set or not. Reset type: SYSRStn

20.15.2.15 ADCSOCOVF1 Register (Offset = Eh) [Reset = 0000h]

ADCSOCOVF1 is shown in [Figure 20-40](#) and described in [Table 20-30](#).

Return to the [Summary Table](#).

ADC SOC Overflow 1 Register

Figure 20-40. ADCSOCOVF1 Register

15	14	13	12	11	10	9	8
SOC15	SOC14	SOC13	SOC12	SOC11	SOC10	SOC9	SOC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
SOC7	SOC6	SOC5	SOC4	SOC3	SOC2	SOC1	SOC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 20-30. ADCSOCOVF1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	SOC15	R	0h	SOC15 Start of Conversion Overflow Flag. Indicates an SOC15 event was generated in hardware while an existing SOC15 event was already pending. 0 No SOC15 event overflow. 1 SOC15 event overflow. An overflow condition does not stop SOC15 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
14	SOC14	R	0h	SOC14 Start of Conversion Overflow Flag. Indicates an SOC14 event was generated in hardware while an existing SOC14 event was already pending. 0 No SOC14 event overflow. 1 SOC14 event overflow. An overflow condition does not stop SOC14 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
13	SOC13	R	0h	SOC13 Start of Conversion Overflow Flag. Indicates an SOC13 event was generated in hardware while an existing SOC13 event was already pending. 0 No SOC13 event overflow. 1 SOC13 event overflow. An overflow condition does not stop SOC13 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
12	SOC12	R	0h	SOC12 Start of Conversion Overflow Flag. Indicates an SOC12 event was generated in hardware while an existing SOC12 event was already pending. 0 No SOC12 event overflow. 1 SOC12 event overflow. An overflow condition does not stop SOC12 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn

Table 20-30. ADCSOCOVF1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
11	SOC11	R	0h	SOC11 Start of Conversion Overflow Flag. Indicates an SOC11 event was generated in hardware while an existing SOC11 event was already pending. 0 No SOC11 event overflow. 1 SOC11 event overflow. An overflow condition does not stop SOC11 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
10	SOC10	R	0h	SOC10 Start of Conversion Overflow Flag. Indicates an SOC10 event was generated in hardware while an existing SOC10 event was already pending. 0 No SOC10 event overflow. 1 SOC10 event overflow. An overflow condition does not stop SOC10 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
9	SOC9	R	0h	SOC9 Start of Conversion Overflow Flag. Indicates an SOC9 event was generated in hardware while an existing SOC9 event was already pending. 0 No SOC9 event overflow. 1 SOC9 event overflow. An overflow condition does not stop SOC9 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
8	SOC8	R	0h	SOC8 Start of Conversion Overflow Flag. Indicates an SOC8 event was generated in hardware while an existing SOC8 event was already pending. 0 No SOC8 event overflow. 1 SOC8 event overflow. An overflow condition does not stop SOC8 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
7	SOC7	R	0h	SOC7 Start of Conversion Overflow Flag. Indicates an SOC7 event was generated in hardware while an existing SOC7 event was already pending. 0 No SOC7 event overflow. 1 SOC7 event overflow. An overflow condition does not stop SOC7 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
6	SOC6	R	0h	SOC6 Start of Conversion Overflow Flag. Indicates an SOC6 event was generated in hardware while an existing SOC6 event was already pending. 0 No SOC6 event overflow. 1 SOC6 event overflow. An overflow condition does not stop SOC6 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn

Table 20-30. ADCSOCOVF1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5	SOC5	R	0h	SOC5 Start of Conversion Overflow Flag. Indicates an SOC5 event was generated in hardware while an existing SOC5 event was already pending. 0 No SOC5 event overflow. 1 SOC5 event overflow. An overflow condition does not stop SOC5 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
4	SOC4	R	0h	SOC4 Start of Conversion Overflow Flag. Indicates an SOC4 event was generated in hardware while an existing SOC4 event was already pending. 0 No SOC4 event overflow. 1 SOC4 event overflow. An overflow condition does not stop SOC4 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
3	SOC3	R	0h	SOC3 Start of Conversion Overflow Flag. Indicates an SOC3 event was generated in hardware while an existing SOC3 event was already pending. 0 No SOC3 event overflow. 1 SOC3 event overflow. An overflow condition does not stop SOC3 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
2	SOC2	R	0h	SOC2 Start of Conversion Overflow Flag. Indicates an SOC2 event was generated in hardware while an existing SOC2 event was already pending. 0 No SOC2 event overflow. 1 SOC2 event overflow. An overflow condition does not stop SOC2 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
1	SOC1	R	0h	SOC1 Start of Conversion Overflow Flag. Indicates an SOC1 event was generated in hardware while an existing SOC1 event was already pending. 0 No SOC1 event overflow. 1 SOC1 event overflow. An overflow condition does not stop SOC1 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn
0	SOC0	R	0h	SOC0 Start of Conversion Overflow Flag. Indicates an SOC0 event was generated in hardware while an existing SOC0 event was already pending. 0 No SOC0 event overflow. 1 SOC0 event overflow. An overflow condition does not stop SOC0 events from being processed. It simply is an indication that a hardware trigger was missed. A write to the ADCSOCFRC1 register does not affect this bit. Reset type: SYSRSn

20.15.2.16 ADCSOCOVFCLR1 Register (Offset = Fh) [Reset = 0000h]

ADCSOCOVFCLR1 is shown in [Figure 20-41](#) and described in [Table 20-31](#).

Return to the [Summary Table](#).

ADC SOC Overflow Clear 1 Register

Figure 20-41. ADCSOCOVFCLR1 Register

15	14	13	12	11	10	9	8
SOC15	SOC14	SOC13	SOC12	SOC11	SOC10	SOC9	SOC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
SOC7	SOC6	SOC5	SOC4	SOC3	SOC2	SOC1	SOC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 20-31. ADCSOCOVFCLR1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15	SOC15	R-0/W1S	0h	SOC15 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC15 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC15 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
14	SOC14	R-0/W1S	0h	SOC14 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC14 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC14 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
13	SOC13	R-0/W1S	0h	SOC13 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC13 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC13 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
12	SOC12	R-0/W1S	0h	SOC12 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC12 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC12 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn

Table 20-31. ADCSOCOVFCLR1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
11	SOC11	R-0/W1S	0h	SOC11 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC11 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC11 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
10	SOC10	R-0/W1S	0h	SOC10 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC10 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC10 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
9	SOC9	R-0/W1S	0h	SOC9 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC9 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC9 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
8	SOC8	R-0/W1S	0h	SOC8 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC8 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC8 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
7	SOC7	R-0/W1S	0h	SOC7 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC7 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC7 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
6	SOC6	R-0/W1S	0h	SOC6 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC6 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC6 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn

Table 20-31. ADCSOCOVFCLR1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5	SOC5	R-0/W1S	0h	SOC5 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC5 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC5 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
4	SOC4	R-0/W1S	0h	SOC4 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC4 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC4 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
3	SOC3	R-0/W1S	0h	SOC3 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC3 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC3 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
2	SOC2	R-0/W1S	0h	SOC2 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC2 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC2 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
1	SOC1	R-0/W1S	0h	SOC1 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC1 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC1 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn
0	SOC0	R-0/W1S	0h	SOC0 Clear Start of Conversion Overflow Bit. Writing a 1 will clear the SOC0 overflow flag in the ADCSOCOVF1 register. Writes of 0 are ignored. Reads will always return a 0. 0 No action. 1 Clear SOC0 overflow flag. If software tries to set this bit on the same clock cycle that hardware tries to set the overflow bit in the ADCSOCOVF1 register, then hardware has priority and the ADCSOCOVF1 bit will be set.. Reset type: SYSRSn

20.15.2.17 ADCSOC0CTL Register (Offset = 10h) [Reset = 00000000h]

ADCSOC0CTL is shown in [Figure 20-42](#) and described in [Table 20-32](#).

Return to the [Summary Table](#).

ADC SOC0 Control Register

Figure 20-42. ADCSOC0CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-32. ADCSOC0CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-32. ADCSOC0CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC0 Trigger Source Select. Along with the SOC0 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC0 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-32. ADCSOC0CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC0 Channel Select. Selects the channel to be converted when SOC0 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC0 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.18 ADCSOC1CTL Register (Offset = 12h) [Reset = 00000000h]

ADCSOC1CTL is shown in [Figure 20-43](#) and described in [Table 20-33](#).

Return to the [Summary Table](#).

ADC SOC1 Control Register

Figure 20-43. ADCSOC1CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-33. ADCSOC1CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-33. ADCSOC1CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC1 Trigger Source Select. Along with the SOC1 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC1 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-33. ADCSOC1CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC1 Channel Select. Selects the channel to be converted when SOC1 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC1 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.19 ADCSOC2CTL Register (Offset = 14h) [Reset = 00000000h]

ADCSOC2CTL is shown in [Figure 20-44](#) and described in [Table 20-34](#).

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ADC SOC2 Control Register

Figure 20-44. ADCSOC2CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-34. ADCSOC2CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-34. ADCSOC2CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC2 Trigger Source Select. Along with the SOC2 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC2 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-34. ADCSOC2CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC2 Channel Select. Selects the channel to be converted when SOC2 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC2 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.20 ADCSOC3CTL Register (Offset = 16h) [Reset = 00000000h]

ADCSOC3CTL is shown in [Figure 20-45](#) and described in [Table 20-35](#).

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ADC SOC3 Control Register

Figure 20-45. ADCSOC3CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-35. ADCSOC3CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-35. ADCSOC3CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC3 Trigger Source Select. Along with the SOC3 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC3 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-35. ADCSOC3CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC3 Channel Select. Selects the channel to be converted when SOC3 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC3 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.21 ADCSOC4CTL Register (Offset = 18h) [Reset = 00000000h]

ADCSOC4CTL is shown in [Figure 20-46](#) and described in [Table 20-36](#).

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ADC SOC4 Control Register

Figure 20-46. ADCSOC4CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-36. ADCSOC4CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-36. ADCSOC4CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC4 Trigger Source Select. Along with the SOC4 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC4 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-36. ADCSOC4CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC4 Channel Select. Selects the channel to be converted when SOC4 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC4 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.22 ADCSOC5CTL Register (Offset = 1Ah) [Reset = 00000000h]

ADCSOC5CTL is shown in [Figure 20-47](#) and described in [Table 20-37](#).

Return to the [Summary Table](#).

ADC SOC5 Control Register

Figure 20-47. ADCSOC5CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-37. ADCSOC5CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-37. ADCSOC5CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC5 Trigger Source Select. Along with the SOC5 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC5 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-37. ADCSOC5CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC5 Channel Select. Selects the channel to be converted when SOC5 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC5 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.23 ADCSOC6CTL Register (Offset = 1Ch) [Reset = 00000000h]

ADCSOC6CTL is shown in [Figure 20-48](#) and described in [Table 20-38](#).

Return to the [Summary Table](#).

ADC SOC6 Control Register

Figure 20-48. ADCSOC6CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-38. ADCSOC6CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-38. ADCSOC6CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC6 Trigger Source Select. Along with the SOC6 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC6 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-38. ADCSOC6CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC6 Channel Select. Selects the channel to be converted when SOC6 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC6 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.24 ADCSOC7CTL Register (Offset = 1Eh) [Reset = 00000000h]

ADCSOC7CTL is shown in [Figure 20-49](#) and described in [Table 20-39](#).

Return to the [Summary Table](#).

ADC SOC7 Control Register

Figure 20-49. ADCSOC7CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-39. ADCSOC7CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-39. ADCSOC7CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC7 Trigger Source Select. Along with the SOC7 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC7 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-39. ADCSOC7CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC7 Channel Select. Selects the channel to be converted when SOC7 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC7 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.25 ADCSOC8CTL Register (Offset = 20h) [Reset = 00000000h]

ADCSOC8CTL is shown in [Figure 20-50](#) and described in [Table 20-40](#).

Return to the [Summary Table](#).

ADC SOC8 Control Register

Figure 20-50. ADCSOC8CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-40. ADCSOC8CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-40. ADCSOC8CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC8 Trigger Source Select. Along with the SOC8 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC8 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-40. ADCSOC8CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC8 Channel Select. Selects the channel to be converted when SOC8 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC8 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.26 ADCSOC9CTL Register (Offset = 22h) [Reset = 00000000h]

ADCSOC9CTL is shown in [Figure 20-51](#) and described in [Table 20-41](#).

Return to the [Summary Table](#).

ADC SOC9 Control Register

Figure 20-51. ADCSOC9CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-41. ADCSOC9CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-41. ADCSOC9CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC9 Trigger Source Select. Along with the SOC9 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC9 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-41. ADCSOC9CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC9 Channel Select. Selects the channel to be converted when SOC9 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC9 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.27 ADCSOC10CTL Register (Offset = 24h) [Reset = 00000000h]

ADCSOC10CTL is shown in [Figure 20-52](#) and described in [Table 20-42](#).

Return to the [Summary Table](#).

ADC SOC10 Control Register

Figure 20-52. ADCSOC10CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-42. ADCSOC10CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-42. ADCSOC10CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC10 Trigger Source Select. Along with the SOC10 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC10 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-42. ADCSOC10CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC10 Channel Select. Selects the channel to be converted when SOC10 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC10 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.28 ADCSOC11CTL Register (Offset = 26h) [Reset = 00000000h]

ADCSOC11CTL is shown in [Figure 20-53](#) and described in [Table 20-43](#).

Return to the [Summary Table](#).

ADC SOC11 Control Register

Figure 20-53. ADCSOC11CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-43. ADCSOC11CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-43. ADCSOC11CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC11 Trigger Source Select. Along with the SOC11 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC11 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-43. ADCSOC11CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC11 Channel Select. Selects the channel to be converted when SOC11 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC11 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.29 ADCSOC12CTL Register (Offset = 28h) [Reset = 00000000h]

ADCSOC12CTL is shown in [Figure 20-54](#) and described in [Table 20-44](#).

Return to the [Summary Table](#).

ADC SOC12 Control Register

Figure 20-54. ADCSOC12CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-44. ADCSOC12CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-44. ADCSOC12CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC12 Trigger Source Select. Along with the SOC12 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC12 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-44. ADCSOC12CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC12 Channel Select. Selects the channel to be converted when SOC12 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC12 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.30 ADCSOC13CTL Register (Offset = 2Ah) [Reset = 00000000h]

ADCSOC13CTL is shown in [Figure 20-55](#) and described in [Table 20-45](#).

Return to the [Summary Table](#).

ADC SOC13 Control Register

Figure 20-55. ADCSOC13CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-45. ADCSOC13CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-45. ADCSOC13CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC13 Trigger Source Select. Along with the SOC13 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC13 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-45. ADCSOC13CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC13 Channel Select. Selects the channel to be converted when SOC13 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC13 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.31 ADCSOC14CTL Register (Offset = 2Ch) [Reset = 00000000h]

ADCSOC14CTL is shown in [Figure 20-56](#) and described in [Table 20-46](#).

Return to the [Summary Table](#).

ADC SOC14 Control Register

Figure 20-56. ADCSOC14CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-46. ADCSOC14CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-46. ADCSOC14CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC14 Trigger Source Select. Along with the SOC14 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC14 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-46. ADCSOC14CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC14 Channel Select. Selects the channel to be converted when SOC14 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC14 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.32 ADCSOC15CTL Register (Offset = 2Eh) [Reset = 00000000h]

ADCSOC15CTL is shown in [Figure 20-57](#) and described in [Table 20-47](#).

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ADC SOC15 Control Register

Figure 20-57. ADCSOC15CTL Register

31	30	29	28	27	26	25	24
RESERVED						TRIGSEL	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
TRIGSEL				RESERVED	CHSEL		
R/W-0h				R-0h	R/W-0h		
15	14	13	12	11	10	9	8
CHSEL	RESERVED						ACQPS
R/W-0h	R-0h						R/W-0h
7	6	5	4	3	2	1	0
ACQPS							
R/W-0h							

Table 20-47. ADCSOC15CTL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-26	RESERVED	R	0h	Reserved

Table 20-47. ADCSOC15CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25-20	TRIGSEL	R/W	0h	SOC15 Trigger Source Select. Along with the SOC15 field in the ADCINTSOCSEL1 register, this bit field configures which trigger will set the SOC15 flag in the ADCSOCFLG1 register to initiate a conversion to start once priority is given to it. 00h ADCTRIG0 - Software only 01h ADCTRIG1 - CPU1 Timer 0, TINT0n 02h ADCTRIG2 - CPU1 Timer 1, TINT1n 03h ADCTRIG3 - CPU1 Timer 2, TINT2n 04h ADCTRIG4 - GPIO, ADCEXTSOC 05h ADCTRIG5 - ePWM1, ADCSOCA 06h ADCTRIG6 - ePWM1, ADCSOCA 07h ADCTRIG7 - ePWM2, ADCSOCA 08h ADCTRIG8 - ePWM2, ADCSOCA 09h ADCTRIG9 - ePWM3, ADCSOCA 0Ah ADCTRIG10 - ePWM3, ADCSOCA 0Bh ADCTRIG11 - ePWM4, ADCSOCA 0Ch ADCTRIG12 - ePWM4, ADCSOCA 0Dh ADCTRIG13 - ePWM5, ADCSOCA 0Eh ADCTRIG14 - ePWM5, ADCSOCA 0Fh ADCTRIG15 - ePWM6, ADCSOCA 10h ADCTRIG16 - ePWM6, ADCSOCA 11h ADCTRIG17 - ePWM7, ADCSOCA 12h ADCTRIG18 - ePWM7, ADCSOCA 13h ADCTRIG19 - ePWM8, ADCSOCA 14h ADCTRIG20 - ePWM8, ADCSOCA 15h ADCTRIG21 - ePWM9, ADCSOCA 16h ADCTRIG22 - ePWM9, ADCSOCA 17h ADCTRIG23 - ePWM10, ADCSOCA 18h ADCTRIG24 - ePWM10, ADCSOCA 19h ADCTRIG25 - ePWM11, ADCSOCA 1Ah ADCTRIG26 - ePWM11, ADCSOCA 1Bh ADCTRIG27 - ePWM12, ADCSOCA 1Ch ADCTRIG28 - ePWM12, ADCSOCA 1Dh ADCTRIG29 - CPU2 Timer 0, TINT0n 1Eh ADCTRIG30 - CPU2 Timer 1, TINT1n 1Fh ADCTRIG31 - CPU2 Timer 2, TINT2n 20h ADCTRIM32 - ePWM13, ADCSOCA 21h ADCTRIM33 - ePWM13, ADCSOCA 22h ADCTRIM34 - ePWM14, ADCSOCA 23h ADCTRIM35 - ePWM14, ADCSOCA 24h ADCTRIM36 - ePWM15, ADCSOCA 25h ADCTRIM37 - ePWM15, ADCSOCA 26h ADCTRIM38 - ePWM16, ADCSOCA 27h ADCTRIM39 - ePWM16, ADCSOCA 28h - 3Fh Reserved Reset type: SYSRSn
19	RESERVED	R	0h	Reserved

Table 20-47. ADCSOC15CTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
18-15	CHSEL	R/W	0h	SOC15 Channel Select. Selects the channel to be converted when SOC15 is received by the ADC. Single-ended Signaling Mode (SIGNALMODE = 0): 0h ADCIN0 1h ADCIN1 2h ADCIN2 3h ADCIN3 4h ADCIN4 5h ADCIN5 6h ADCIN6 7h ADCIN7 8h ADCIN8 9h ADCIN9 Ah ADCIN10 Bh ADCIN11 Ch ADCIN12 Dh ADCIN13 Eh ADCIN14 Fh ADCIN15 Differential Signaling Mode (SIGNALMODE = 1): 0h ADCIN0 (non-inverting) and ADCIN1 (inverting) 1h ADCIN0 (non-inverting) and ADCIN1 (inverting) 2h ADCIN2 (non-inverting) and ADCIN3 (inverting) 3h ADCIN2 (non-inverting) and ADCIN3 (inverting) 4h ADCIN4 (non-inverting) and ADCIN5 (inverting) 5h ADCIN4 (non-inverting) and ADCIN5 (inverting) 6h ADCIN6 (non-inverting) and ADCIN7 (inverting) 7h ADCIN6 (non-inverting) and ADCIN7 (inverting) 8h ADCIN8 (non-inverting) and ADCIN9 (inverting) 9h ADCIN8 (non-inverting) and ADCIN9 (inverting) Ah ADCIN10 (non-inverting) and ADCIN11 (inverting) Bh ADCIN10 (non-inverting) and ADCIN11 (inverting) Ch ADCIN12 (non-inverting) and ADCIN13 (inverting) Dh ADCIN12 (non-inverting) and ADCIN13 (inverting) Eh ADCIN14 (non-inverting) and ADCIN15 (inverting) Fh ADCIN14 (non-inverting) and ADCIN15 (inverting) Reset type: SYSRSn
14-9	RESERVED	R	0h	Reserved
8-0	ACQPS	R/W	0h	SOC15 Acquisition Prescale. Controls the sample and hold window for this SOC. The configured acquisition time must be at least as long as one ADCCLK cycle for correct ADC operation. The device datasheet will also specify a minimum sample and hold window duration. 000h Sample window is 1 system clock cycle wide 001h Sample window is 2 system clock cycles wide 002h Sample window is 3 system clock cycles wide ... 1FFh Sample window is 512 system clock cycles wide Reset type: SYSRSn

20.15.2.33 ADCEVTSTAT Register (Offset = 30h) [Reset = 0000h]

ADCEVTSTAT is shown in [Figure 20-58](#) and described in [Table 20-48](#).

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ADC Event Status Register

Figure 20-58. ADCEVTSTAT Register

15	14	13	12	11	10	9	8
RESERVED	PPB4ZERO	PPB4TRIPLO	PPB4TRIPHI	RESERVED	PPB3ZERO	PPB3TRIPLO	PPB3TRIPHI
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
RESERVED	PPB2ZERO	PPB2TRIPLO	PPB2TRIPHI	RESERVED	PPB1ZERO	PPB1TRIPLO	PPB1TRIPHI
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 20-48. ADCEVTSTAT Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	PPB4ZERO	R	0h	Post Processing Block 4 Zero Crossing Flag. When set indicates the ADCPPB4RESULT register has changed sign. This bit is gated by EOC signal. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
13	PPB4TRIPLO	R	0h	Post Processing Block 4 Trip Low Flag. When set indicates a digital compare trip low event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
12	PPB4TRIPHI	R	0h	Post Processing Block 4 Trip High Flag. When set indicates a digital compare trip high event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
11	RESERVED	R	0h	Reserved
10	PPB3ZERO	R	0h	Post Processing Block 3 Zero Crossing Flag. When set indicates the ADCPPB3RESULT register has changed sign. This bit is gated by EOC signal. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn

Table 20-48. ADCEVTSTAT Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
9	PPB3TRIPLO	R	0h	Post Processing Block 3 Trip Low Flag. When set indicates a digital compare trip low event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
8	PPB3TRIPHI	R	0h	Post Processing Block 3 Trip High Flag. When set indicates a digital compare trip high event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	PPB2ZERO	R	0h	Post Processing Block 2 Zero Crossing Flag. When set indicates the ADCPPB2RESULT register has changed sign. This bit is gated by EOC signal. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
5	PPB2TRIPLO	R	0h	Post Processing Block 2 Trip Low Flag. When set indicates a digital compare trip low event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
4	PPB2TRIPHI	R	0h	Post Processing Block 2 Trip High Flag. When set indicates a digital compare trip high event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
3	RESERVED	R	0h	Reserved
2	PPB1ZERO	R	0h	Post Processing Block 1 Zero Crossing Flag. When set indicates the ADCPPB1RESULT register has changed sign. This bit is gated by EOC signal. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn

Table 20-48. ADCEVTSTAT Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1	PPB1TRIPLO	R	0h	Post Processing Block 1 Trip Low Flag. When set indicates a digital compare trip low event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
0	PPB1TRIPHI	R	0h	Post Processing Block 1 Trip High Flag. When set indicates a digital compare trip high event has occurred. Note: these bits are set even when the corresponding enable in ADCEVTINTSEL is not set. Because of this, an ISR may need to examine both the ADCEVTSTAT and ADCEVTINTSEL registers to determine the source of the interrupt. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn

20.15.2.34 ADCEVTCLR Register (Offset = 32h) [Reset = 0000h]

ADCEVTCLR is shown in [Figure 20-59](#) and described in [Table 20-49](#).

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ADC Event Clear Register

Figure 20-59. ADCEVTCLR Register

15	14	13	12	11	10	9	8
RESERVED	PPB4ZERO	PPB4TRIPLO	PPB4TRIPHI	RESERVED	PPB3ZERO	PPB3TRIPLO	PPB3TRIPHI
R-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
RESERVED	PPB2ZERO	PPB2TRIPLO	PPB2TRIPHI	RESERVED	PPB1ZERO	PPB1TRIPLO	PPB1TRIPHI
R-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 20-49. ADCEVTCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	PPB4ZERO	R-0/W1S	0h	Post Processing Block 4 Zero Crossing Clear. Clears the corresponding zero crossing flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
13	PPB4TRIPLO	R-0/W1S	0h	Post Processing Block 4 Trip Low Clear. Clears the corresponding trip low flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
12	PPB4TRIPHI	R-0/W1S	0h	Post Processing Block 4 Trip High Clear. Clears the corresponding trip high flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
11	RESERVED	R	0h	Reserved
10	PPB3ZERO	R-0/W1S	0h	Post Processing Block 3 Zero Crossing Clear. Clears the corresponding zero crossing flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
9	PPB3TRIPLO	R-0/W1S	0h	Post Processing Block 3 Trip Low Clear. Clears the corresponding trip low flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
8	PPB3TRIPHI	R-0/W1S	0h	Post Processing Block 3 Trip High Clear. Clears the corresponding trip high flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	PPB2ZERO	R-0/W1S	0h	Post Processing Block 2 Zero Crossing Clear. Clears the corresponding zero crossing flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn

Table 20-49. ADCEVTCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5	PPB2TRIPLO	R-0/W1S	0h	Post Processing Block 2 Trip Low Clear. Clears the corresponding trip low flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
4	PPB2TRIPHI	R-0/W1S	0h	Post Processing Block 2 Trip High Clear. Clears the corresponding trip high flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
3	RESERVED	R	0h	Reserved
2	PPB1ZERO	R-0/W1S	0h	Post Processing Block 1 Zero Crossing Clear. Clears the corresponding zero crossing flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
1	PPB1TRIPLO	R-0/W1S	0h	Post Processing Block 1 Trip Low Clear. Clears the corresponding trip low flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn
0	PPB1TRIPHI	R-0/W1S	0h	Post Processing Block 1 Trip High Clear. Clears the corresponding trip high flag in the ADCEVTSTAT register. Note: If software sets the clear bit on the same cycle that hardware is trying to set the flag bit, then hardware has priority Reset type: SYSRSn

20.15.2.35 ADCEVTSEL Register (Offset = 34h) [Reset = 0000h]

ADCEVTSEL is shown in [Figure 20-60](#) and described in [Table 20-50](#).

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ADC Event Selection Register

Figure 20-60. ADCEVTSEL Register

15	14	13	12	11	10	9	8
RESERVED	PPB4ZERO	PPB4TRIPLO	PPB4TRIPHI	RESERVED	PPB3ZERO	PPB3TRIPLO	PPB3TRIPHI
R-0h	R/W-0h	R/W-0h	R/W-0h	R-0h	R/W-0h	R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
RESERVED	PPB2ZERO	PPB2TRIPLO	PPB2TRIPHI	RESERVED	PPB1ZERO	PPB1TRIPLO	PPB1TRIPHI
R-0h	R/W-0h	R/W-0h	R/W-0h	R-0h	R/W-0h	R/W-0h	R/W-0h

Table 20-50. ADCEVTSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	PPB4ZERO	R/W	0h	Post Processing Block 4 Zero Crossing Event Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
13	PPB4TRIPLO	R/W	0h	Post Processing Block 4 Trip Low Event Enable. Setting this bit allows the corresponding rising trip low flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
12	PPB4TRIPHI	R/W	0h	Post Processing Block 4 Trip High Event Enable. Setting this bit allows the corresponding rising trip high flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
11	RESERVED	R	0h	Reserved
10	PPB3ZERO	R/W	0h	Post Processing Block 3 Zero Crossing Event Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
9	PPB3TRIPLO	R/W	0h	Post Processing Block 3 Trip Low Event Enable. Setting this bit allows the corresponding rising trip low flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
8	PPB3TRIPHI	R/W	0h	Post Processing Block 3 Trip High Event Enable. Setting this bit allows the corresponding rising trip high flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	PPB2ZERO	R/W	0h	Post Processing Block 2 Zero Crossing Event Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn

Table 20-50. ADCEVTSEL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5	PPB2TRIPLO	R/W	0h	Post Processing Block 2 Trip Low Event Enable. Setting this bit allows the corresponding rising trip low flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
4	PPB2TRIPHI	R/W	0h	Post Processing Block 2 Trip High Event Enable. Setting this bit allows the corresponding rising trip high flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
3	RESERVED	R	0h	Reserved
2	PPB1ZERO	R/W	0h	Post Processing Block 1 Zero Crossing Event Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
1	PPB1TRIPLO	R/W	0h	Post Processing Block 1 Trip Low Event Enable. Setting this bit allows the corresponding rising trip low flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn
0	PPB1TRIPHI	R/W	0h	Post Processing Block 1 Trip High Event Enable. Setting this bit allows the corresponding rising trip high flag to activate the event signal to the PWM blocks. The flag must be cleared before it can produce additional events to the PWM blocks. Reset type: SYSRSn

20.15.2.36 ADCEVTINTSEL Register (Offset = 36h) [Reset = 0000h]

ADCEVTINTSEL is shown in [Figure 20-61](#) and described in [Table 20-51](#).

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ADC Event Interrupt Selection Register

Figure 20-61. ADCEVTINTSEL Register

15	14	13	12	11	10	9	8
RESERVED	PPB4ZERO	PPB4TRIPLO	PPB4TRIPHI	RESERVED	PPB3ZERO	PPB3TRIPLO	PPB3TRIPHI
R-0h	R/W-0h	R/W-0h	R/W-0h	R-0h	R/W-0h	R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
RESERVED	PPB2ZERO	PPB2TRIPLO	PPB2TRIPHI	RESERVED	PPB1ZERO	PPB1TRIPLO	PPB1TRIPHI
R-0h	R/W-0h	R/W-0h	R/W-0h	R-0h	R/W-0h	R/W-0h	R/W-0h

Table 20-51. ADCEVTINTSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	PPB4ZERO	R/W	0h	Post Processing Block 4 Zero Crossing Interrupt Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
13	PPB4TRIPLO	R/W	0h	Post Processing Block 4 Trip Low Interrupt Enable. Setting this bit allows the corresponding rising trip low flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
12	PPB4TRIPHI	R/W	0h	Post Processing Block 4 Trip High Interrupt Enable. Setting this bit allows the corresponding rising trip high flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
11	RESERVED	R	0h	Reserved
10	PPB3ZERO	R/W	0h	Post Processing Block 3 Zero Crossing Interrupt Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
9	PPB3TRIPLO	R/W	0h	Post Processing Block 3 Trip Low Interrupt Enable. Setting this bit allows the corresponding rising trip low flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
8	PPB3TRIPHI	R/W	0h	Post Processing Block 3 Trip High Interrupt Enable. Setting this bit allows the corresponding rising trip high flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
7	RESERVED	R	0h	Reserved
6	PPB2ZERO	R/W	0h	Post Processing Block 2 Zero Crossing Interrupt Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn

Table 20-51. ADCEVTINTSEL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
5	PPB2TRIPLO	R/W	0h	Post Processing Block 2 Trip Low Interrupt Enable. Setting this bit allows the corresponding rising trip low flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
4	PPB2TRIPHI	R/W	0h	Post Processing Block 2 Trip High Interrupt Enable. Setting this bit allows the corresponding rising trip high flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
3	RESERVED	R	0h	Reserved
2	PPB1ZERO	R/W	0h	Post Processing Block 1 Zero Crossing Interrupt Enable. Setting this bit allows the corresponding rising zero crossing flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
1	PPB1TRIPLO	R/W	0h	Post Processing Block 1 Trip Low Interrupt Enable. Setting this bit allows the corresponding rising trip low flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn
0	PPB1TRIPHI	R/W	0h	Post Processing Block 1 Trip High Interrupt Enable. Setting this bit allows the corresponding rising trip high flag to activate the event interrupt signal to the PIE. The flag must be cleared before it can produce additional interrupts to the PIE. Reset type: SYSRSn

20.15.2.37 ADCOSDETECT Register (Offset = 38h) [Reset = 0000h]

ADCOSDETECT is shown in [Figure 20-62](#) and described in [Table 20-52](#).

Return to the [Summary Table](#).

ADC Open and Shorts Detect Register

Figure 20-62. ADCOSDETECT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					DETECTCFG		
R-0h					R/W-0h		

Table 20-52. ADCOSDETECT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-3	RESERVED	R	0h	Reserved
2-0	DETECTCFG	R/W	0h	ADC Opens and Shorts Detect Configuration. This bit field defines the open/shorts detection circuit state. 0h Open/Shorts detection circuit is disabled. 1h Open/Shorts detection circuit is enabled at zero scale. 2h Open/Shorts detection circuit is enabled at full scale. 3h Open/Shorts detection circuit is enabled at (nominal) 5/12 scale. 4h Open/Shorts detection circuit is enabled at (nominal) 7/12 scale. 5h Open/Shorts detection circuit is enabled with a (nominal) 5K pulldown to VSSA. 6h Open/Shorts detection circuit is enabled with a (nominal) 5K pullup to VDDA. 7h Open/Shorts detection circuit is enabled with a (nominal) 7K pulldown to VSSA. Reset type: SYSRStn

20.15.2.38 ADCCOUNTER Register (Offset = 39h) [Reset = 0000h]

ADCCOUNTER is shown in [Figure 20-63](#) and described in [Table 20-53](#).

Return to the [Summary Table](#).

ADC Counter Register

Figure 20-63. ADCCOUNTER Register

15	14	13	12	11	10	9	8
RESERVED				FREECOUNT			
R-0h				R-0h			
7	6	5	4	3	2	1	0
FREECOUNT							
R-0h							

Table 20-53. ADCCOUNTER Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-0	FREECOUNT	R	0h	ADC Free Running Counter Value. This bit field reflects the status of the free running ADC counter. Reset type: SYSRSn

20.15.2.39 ADCREV Register (Offset = 3Ah) [Reset = 0105h]

ADCREV is shown in [Figure 20-64](#) and described in [Table 20-54](#).

Return to the [Summary Table](#).

ADC Revision Register

Figure 20-64. ADCREV Register

15	14	13	12	11	10	9	8
REV							
R-1h							
7	6	5	4	3	2	1	0
TYPE							
R-5h							

Table 20-54. ADCREV Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	REV	R	1h	ADC Revision. To allow documentation of differences between revisions. First version is labeled as 00h. Reset type: SYSRSn
7-0	TYPE	R	5h	ADC Type. Always set to 5 for this ADC. Reset type: SYSRSn

20.15.2.40 ADCOFFTRIM Register (Offset = 3Bh) [Reset = 0000h]

ADCOFFTRIM is shown in [Figure 20-65](#) and described in [Table 20-55](#).

Return to the [Summary Table](#).

ADC Offset Trim Register

Figure 20-65. ADCOFFTRIM Register

15	14	13	12	11	10	9	8
RESERVED				RESERVED			
R-0h				R/W-0h			
7	6	5	4	3	2	1	0
OFFTRIM							
R/W-0h							

Table 20-55. ADCOFFTRIM Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-8	RESERVED	R/W	0h	Reserved
7-0	OFFTRIM	R/W	0h	ADC Offset Trim. Adjusts the conversion results of the converter up or down to account for offset error in the ADC. A different offset trim is required for each combination of resolution and signal mode. Using the <code>AdcSetMode</code> function to set the resolution and signal mode will ensure that the correct offset trim is loaded. Range is +127 steps to -128 steps (2's compliment format). Regardless of the converter resolution, the size of each trim step is $(VREFHI-VREFLO)/65536$. Reset type: SYSRSn

20.15.2.41 ADCPPB1CONFIG Register (Offset = 40h) [Reset = 0000h]

ADCPPB1CONFIG is shown in [Figure 20-66](#) and described in [Table 20-56](#).

Return to the [Summary Table](#).

ADC PPB1 Config Register

Figure 20-66. ADCPPB1CONFIG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED		CBCEN	TWOSCOMPEN	CONFIG			
R-0h		R/W-0h	R/W-0h	R/W-0h			

Table 20-56. ADCPPB1CONFIG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-6	RESERVED	R	0h	Reserved
5	CBCEN	R/W	0h	ADC Post Processing Block Cycle By Cycle Enable. When set, this bit enables the post conversion hardware processing circuit to automatically clear the ADCEVTSTAT on a conversion if the event condition is no longer present. Reset type: SYSRSn
4	TWOSCOMPEN	R/W	0h	ADC Post Processing Block 1 Two's Complement Enable. When set this bit enables the post conversion hardware processing circuit that performs a two's complement on the output of the offset/reference subtraction unit before storing the result in the ADCPPB1RESULT register. 0 ADCPPB1RESULT = ADCRESULTx - ADCPPB1OFFREF 1 ADCPPB1RESULT = ADCPPB1OFFREF - ADCRESULTx Reset type: SYSRSn

Table 20-56. ADCPPB1CONFIG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-0	CONFIG	R/W	0h	ADC Post Processing Block 1 Configuration. This bit field defines which SOC/EOC/RESULT is associated with this post processing block. 0000 SOC0/EOC0/RESULT0 is associated with post processing block 1 0001 SOC1/EOC1/RESULT1 is associated with post processing block 1 0010 SOC2/EOC2/RESULT2 is associated with post processing block 1 0011 SOC3/EOC3/RESULT3 is associated with post processing block 1 0100 SOC4/EOC4/RESULT4 is associated with post processing block 1 0101 SOC5/EOC5/RESULT5 is associated with post processing block 1 0110 SOC6/EOC6/RESULT6 is associated with post processing block 1 0111 SOC7/EOC7/RESULT7 is associated with post processing block 1 1000 SOC8/EOC8/RESULT8 is associated with post processing block 1 1001 SOC9/EOC9/RESULT9 is associated with post processing block 1 1010 SOC10/EOC10/RESULT10 is associated with post processing block 1 1011 SOC11/EOC11/RESULT11 is associated with post processing block 1 1100 SOC12/EOC12/RESULT12 is associated with post processing block 1 1101 SOC13/EOC13/RESULT13 is associated with post processing block 1 1110 SOC14/EOC14/RESULT14 is associated with post processing block 1 1111 SOC15/EOC15/RESULT15 is associated with post processing block 1 Reset type: SYSRSn

20.15.2.42 ADCPPB1STAMP Register (Offset = 41h) [Reset = 0000h]

ADCPPB1STAMP is shown in [Figure 20-67](#) and described in [Table 20-57](#).

Return to the [Summary Table](#).

ADC PPB1 Sample Delay Time Stamp Register

Figure 20-67. ADCPPB1STAMP Register

15	14	13	12	11	10	9	8
RESERVED				DLYSTAMP			
R-0h				R-0h			
7	6	5	4	3	2	1	0
DLYSTAMP							
R-0h							

Table 20-57. ADCPPB1STAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-0	DLYSTAMP	R	0h	ADC Post Processing Block 1 Delay Time Stamp. When an SOC starts sampling the value contained in REQSTAMP is subtracted from the value in ADCCOUNTER.FREECOUNT and loaded into this bit field, thereby giving the number of system clock cycles delay between the SOC trigger and the actual start of the sample. Reset type: SYSRSn

20.15.2.43 ADCPPB1OFFCAL Register (Offset = 42h) [Reset = 0000h]

ADCPPB1OFFCAL is shown in [Figure 20-68](#) and described in [Table 20-58](#).

Return to the [Summary Table](#).

ADC PPB1 Offset Calibration Register

Figure 20-68. ADCPPB1OFFCAL Register

15	14	13	12	11	10	9	8
RESERVED						OFFCAL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
OFFCAL							
R/W-0h							

Table 20-58. ADCPPB1OFFCAL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	OFFCAL	R/W	0h	ADC Post Processing Block 1 Offset Correction. This bit field can be used to digitally remove any system level offset inherent in the ADCIN circuit. This 10-bit signed value is subtracted from the ADC output before being stored in the ADCRESULT register. 000h No change. The ADC output is stored directly into ADCRESULT. 001h ADC output - 1 is stored into ADCRESULT. 002h ADC output - 2 is stored into ADCRESULT. ... 200h ADC output + 512 is stored into ADCRESULT. ... 3FFh ADC output + 1 is stored into ADCRESULT. NOTE: In 16-bit mode, the subtraction will saturate at 0000h and FFFFh before being stored into the ADCRESULT register. In 12-bit mode, the subtraction will saturate at 0000h and 0FFFh before being stored into the ADCRESULT register. Note: in the case that multiple PPBs point to the same SOC, only the OFFCAL of the highest numbered PPB will be applied. Reset type: SYSRSn

20.15.2.44 ADCPPB1OFFREF Register (Offset = 43h) [Reset = 0000h]

ADCPPB1OFFREF is shown in [Figure 20-69](#) and described in [Table 20-59](#).

Return to the [Summary Table](#).

ADC PPB1 Offset Reference Register

Figure 20-69. ADCPPB1OFFREF Register

15	14	13	12	11	10	9	8
OFFREF							
R/W-0h							
7	6	5	4	3	2	1	0
OFFREF							
R/W-0h							

Table 20-59. ADCPPB1OFFREF Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFREF	R/W	0h	ADC Post Processing Block 1 Offset Correction. This bit field can be used to either calculate the feedback error or convert a unipolar signal to bipolar by subtracting a reference value. This 16-bit unsigned value is subtracted from the ADCRESULT register before being passed through an optional two's complement function and stored in the ADCPPB1RESULT register. This subtraction is not saturated. 0000h No change. The ADCRESULT value is passed on. 0001h ADCRESULT - 1 is passed on. 0002h ADCRESULT - 2 is passed on. ... 8000h ADCRESULT - 32,768 is passed on. ... FFFFh ADCRESULT - 65,535 is passed on. NOTE: In 12-bit mode the size of this register does not change from 16-bits. It is the user's responsibility to ensure that only a 12-bit value is written to this register when in 12-bit mode. Reset type: SYSRSn

20.15.2.45 ADCPPB1TRIPHI Register (Offset = 44h) [Reset = 00000000h]

ADCPPB1TRIPHI is shown in [Figure 20-70](#) and described in [Table 20-60](#).

Return to the [Summary Table](#).

ADC PPB1 Trip High Register

Figure 20-70. ADCPPB1TRIPHI Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							HSIGN
R-0h							R/W-0h
15	14	13	12	11	10	9	8
LIMITHI							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITHI							
R/W-0h							

Table 20-60. ADCPPB1TRIPHI Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reserved
16	HSIGN	R/W	0h	High Limit Sign Bit. This is the sign bit (17th bit) to the LIMITHI bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITHI	R/W	0h	ADC Post Processing Block 1 Trip High Limit. This value sets the digital comparator trip high limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB1RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB1RESULT register. Reset type: SYSRSn

20.15.2.46 ADCPPB1TRIPLO Register (Offset = 46h) [Reset = 00000000h]

ADCPPB1TRIPLO is shown in [Figure 20-71](#) and described in [Table 20-61](#).

Return to the [Summary Table](#).

ADC PPB1 Trip Low/Trigger Time Stamp Register

Figure 20-71. ADCPPB1TRIPLO Register

31	30	29	28	27	26	25	24
REQSTAMP							
R-0h							
23	22	21	20	19	18	17	16
REQSTAMP				RESERVED		LSIGN	
R-0h				R-0h		R/W-0h	
15	14	13	12	11	10	9	8
LIMITLO							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITLO							
R/W-0h							

Table 20-61. ADCPPB1TRIPLO Register Field Descriptions

Bit	Field	Type	Reset	Description
31-20	REQSTAMP	R	0h	ADC Post Processing Block 1 Request Time Stamp. When a trigger sets the associated SOC flag in the ADCSOCFLG1 register the value of ADCCOUNTER.FREECOUNT is loaded into this bit field. Reset type: SYSRSn
19-17	RESERVED	R	0h	Reserved
16	LSIGN	R/W	0h	Low Limit Sign Bit. This is the sign bit (17th bit) to the LIMITLO bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITLO	R/W	0h	ADC Post Processing Block 1 Trip Low Limit. This value sets the digital comparator trip low limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB1RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB1RESULT register. Reset type: SYSRSn

20.15.2.47 ADCPPB2CONFIG Register (Offset = 48h) [Reset = 0000h]

ADCPPB2CONFIG is shown in [Figure 20-72](#) and described in [Table 20-62](#).

Return to the [Summary Table](#).

ADC PPB2 Config Register

Figure 20-72. ADCPPB2CONFIG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED		CBCEN	TWOSCOMPEN	CONFIG			
R-0h		R/W-0h	R/W-0h	R/W-0h			

Table 20-62. ADCPPB2CONFIG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-6	RESERVED	R	0h	Reserved
5	CBCEN	R/W	0h	ADC Post Processing Block Cycle By Cycle Enable. When set, this bit enables the post conversion hardware processing circuit to automatically clear the ADCEVTSTAT on a conversion if the event condition is no longer present. Reset type: SYSRSn
4	TWOSCOMPEN	R/W	0h	ADC Post Processing Block 2 Two's Complement Enable. When set this bit enables the post conversion hardware processing circuit that performs a two's complement on the output of the offset/reference subtraction unit before storing the result in the ADCPPB2RESULT register. 0 ADCPPB2RESULT = ADCRESULTx - ADCPPB2OFFREF 1 ADCPPB2RESULT = ADCPPB2OFFREF - ADCRESULTx Reset type: SYSRSn

Table 20-62. ADCPPB2CONFIG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-0	CONFIG	R/W	0h	ADC Post Processing Block 2 Configuration. This bit field defines which SOC/EOC/RESULT is associated with this post processing block. 0000 SOC0/EOC0/RESULT0 is associated with post processing block 2 0001 SOC1/EOC1/RESULT1 is associated with post processing block 2 0010 SOC2/EOC2/RESULT2 is associated with post processing block 2 0011 SOC3/EOC3/RESULT3 is associated with post processing block 2 0100 SOC4/EOC4/RESULT4 is associated with post processing block 2 0101 SOC5/EOC5/RESULT5 is associated with post processing block 2 0110 SOC6/EOC6/RESULT6 is associated with post processing block 2 0111 SOC7/EOC7/RESULT7 is associated with post processing block 2 1000 SOC8/EOC8/RESULT8 is associated with post processing block 2 1001 SOC9/EOC9/RESULT9 is associated with post processing block 2 1010 SOC10/EOC10/RESULT10 is associated with post processing block 2 1011 SOC11/EOC11/RESULT11 is associated with post processing block 2 1100 SOC12/EOC12/RESULT12 is associated with post processing block 2 1101 SOC13/EOC13/RESULT13 is associated with post processing block 2 1110 SOC14/EOC14/RESULT14 is associated with post processing block 2 1111 SOC15/EOC15/RESULT15 is associated with post processing block 2 Reset type: SYSRSn

20.15.2.48 ADCPPB2STAMP Register (Offset = 49h) [Reset = 0000h]

ADCPPB2STAMP is shown in [Figure 20-73](#) and described in [Table 20-63](#).

Return to the [Summary Table](#).

ADC PPB2 Sample Delay Time Stamp Register

Figure 20-73. ADCPPB2STAMP Register

15	14	13	12	11	10	9	8
RESERVED				DLYSTAMP			
R-0h				R-0h			
7	6	5	4	3	2	1	0
DLYSTAMP							
R-0h							

Table 20-63. ADCPPB2STAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-0	DLYSTAMP	R	0h	ADC Post Processing Block 2 Delay Time Stamp. When an SOC starts sampling the value contained in REQSTAMP is subtracted from the value in ADCCOUNTER.FREECOUNT and loaded into this bit field, thereby giving the number of system clock cycles delay between the SOC trigger and the actual start of the sample. Reset type: SYSRSn

20.15.2.49 ADCPPB2OFFCAL Register (Offset = 4Ah) [Reset = 0000h]

ADCPPB2OFFCAL is shown in [Figure 20-74](#) and described in [Table 20-64](#).

Return to the [Summary Table](#).

ADC PPB2 Offset Calibration Register

Figure 20-74. ADCPPB2OFFCAL Register

15	14	13	12	11	10	9	8
RESERVED						OFFCAL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
OFFCAL							
R/W-0h							

Table 20-64. ADCPPB2OFFCAL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	OFFCAL	R/W	0h	ADC Post Processing Block 2 Offset Correction. This bit field can be used to digitally remove any system level offset inherent in the ADCIN circuit. This 10-bit signed value is subtracted from the ADC output before being stored in the ADCRESULT register. 000h No change. The ADC output is stored directly into ADCRESULT. 001h ADC output - 1 is stored into ADCRESULT. 002h ADC output - 2 is stored into ADCRESULT. ... 200h ADC output + 512 is stored into ADCRESULT. ... 3FFh ADC output + 1 is stored into ADCRESULT. NOTE: In 16-bit mode, the subtraction will saturate at 0000h and FFFFh before being stored into the ADCRESULT register. In 12-bit mode, the subtraction will saturate at 0000h and 0FFFh before being stored into the ADCRESULT register. Note: in the case that multiple PPBs point to the same SOC, only the OFFCAL of the highest numbered PPB will be applied. Reset type: SYSRSn

20.15.2.50 ADCPPB2OFFREF Register (Offset = 4Bh) [Reset = 0000h]

ADCPPB2OFFREF is shown in [Figure 20-75](#) and described in [Table 20-65](#).

Return to the [Summary Table](#).

ADC PPB2 Offset Reference Register

Figure 20-75. ADCPPB2OFFREF Register

15	14	13	12	11	10	9	8
OFFREF							
R/W-0h							
7	6	5	4	3	2	1	0
OFFREF							
R/W-0h							

Table 20-65. ADCPPB2OFFREF Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFREF	R/W	0h	ADC Post Processing Block 2 Offset Correction. This bit field can be used to either calculate the feedback error or convert a unipolar signal to bipolar by subtracting a reference value. This 16-bit unsigned value is subtracted from the ADCRESULT register before being passed through an optional two's complement function and stored in the ADCPPB2RESULT register. This subtraction is not saturated. 0000h No change. The ADCRESULT value is passed on. 0001h ADCRESULT - 1 is passed on. 0002h ADCRESULT - 2 is passed on. ... 8000h ADCRESULT - 32,768 is passed on. ... FFFFh ADCRESULT - 65,535 is passed on. NOTE: In 12-bit mode the size of this register does not change from 16-bits. It is the user's responsibility to ensure that only a 12-bit value is written to this register when in 12-bit mode. Reset type: SYSRStn

20.15.2.51 ADCPPB2TRIPHI Register (Offset = 4Ch) [Reset = 00000000h]

ADCPPB2TRIPHI is shown in [Figure 20-76](#) and described in [Table 20-66](#).

Return to the [Summary Table](#).

ADC PPB2 Trip High Register

Figure 20-76. ADCPPB2TRIPHI Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							HSIGN
R-0h							R/W-0h
15	14	13	12	11	10	9	8
LIMITHI							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITHI							
R/W-0h							

Table 20-66. ADCPPB2TRIPHI Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reserved
16	HSIGN	R/W	0h	High Limit Sign Bit. This is the sign bit (17th bit) to the LIMITHI bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITHI	R/W	0h	ADC Post Processing Block 2 Trip High Limit. This value sets the digital comparator trip high limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB2RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB2RESULT register. Reset type: SYSRSn

20.15.2.52 ADCPPB2TRIPLO Register (Offset = 4Eh) [Reset = 00000000h]

ADCPPB2TRIPLO is shown in [Figure 20-77](#) and described in [Table 20-67](#).

Return to the [Summary Table](#).

ADC PPB2 Trip Low/Trigger Time Stamp Register

Figure 20-77. ADCPPB2TRIPLO Register

31	30	29	28	27	26	25	24
REQSTAMP							
R-0h							
23	22	21	20	19	18	17	16
REQSTAMP				RESERVED			LSIGN
R-0h				R-0h			R/W-0h
15	14	13	12	11	10	9	8
LIMITLO							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITLO							
R/W-0h							

Table 20-67. ADCPPB2TRIPLO Register Field Descriptions

Bit	Field	Type	Reset	Description
31-20	REQSTAMP	R	0h	ADC Post Processing Block 2 Request Time Stamp. When a trigger sets the associated SOC flag in the ADCSOCFLG1 register the value of ADCCOUNTER.FREECOUNT is loaded into this bit field. Reset type: SYSRSn
19-17	RESERVED	R	0h	Reserved
16	LSIGN	R/W	0h	Low Limit Sign Bit. This is the sign bit (17th bit) to the LIMITLO bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITLO	R/W	0h	ADC Post Processing Block 2 Trip Low Limit. This value sets the digital comparator trip low limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB2RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB2RESULT register. Reset type: SYSRSn

20.15.2.53 ADCPPB3CONFIG Register (Offset = 50h) [Reset = 0000h]

ADCPPB3CONFIG is shown in [Figure 20-78](#) and described in [Table 20-68](#).

Return to the [Summary Table](#).

ADC PPB3 Config Register

Figure 20-78. ADCPPB3CONFIG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED		CBCEN	TWOSCOMPE N	CONFIG			
R-0h		R/W-0h	R/W-0h	R/W-0h			

Table 20-68. ADCPPB3CONFIG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-6	RESERVED	R	0h	Reserved
5	CBCEN	R/W	0h	ADC Post Processing Block Cycle By Cycle Enable. When set, this bit enables the post conversion hardware processing circuit to automatically clear the ADCEVTSTAT on a conversion if the event condition is no longer present. Reset type: SYSRSn
4	TWOSCOMPEN	R/W	0h	ADC Post Processing Block 3 Two's Complement Enable. When set this bit enables the post conversion hardware processing circuit that performs a two's complement on the output of the offset/reference subtraction unit before storing the result in the ADCPPB3RESULT register. 0 ADCPPB3RESULT = ADCRESULTx - ADCPPB3OFFREF 1 ADCPPB3RESULT = ADCPPB3OFFREF - ADCRESULTx Reset type: SYSRSn

Table 20-68. ADCPPB3CONFIG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-0	CONFIG	R/W	0h	ADC Post Processing Block 3 Configuration. This bit field defines which SOC/EOC/RESULT is associated with this post processing block. 0000 SOC0/EOC0/RESULT0 is associated with post processing block 3 0001 SOC1/EOC1/RESULT1 is associated with post processing block 3 0010 SOC2/EOC2/RESULT2 is associated with post processing block 3 0011 SOC3/EOC3/RESULT3 is associated with post processing block 3 0100 SOC4/EOC4/RESULT4 is associated with post processing block 3 0101 SOC5/EOC5/RESULT5 is associated with post processing block 3 0110 SOC6/EOC6/RESULT6 is associated with post processing block 3 0111 SOC7/EOC7/RESULT7 is associated with post processing block 3 1000 SOC8/EOC8/RESULT8 is associated with post processing block 3 1001 SOC9/EOC9/RESULT9 is associated with post processing block 3 1010 SOC10/EOC10/RESULT10 is associated with post processing block 3 1011 SOC11/EOC11/RESULT11 is associated with post processing block 3 1100 SOC12/EOC12/RESULT12 is associated with post processing block 3 1101 SOC13/EOC13/RESULT13 is associated with post processing block 3 1110 SOC14/EOC14/RESULT14 is associated with post processing block 3 1111 SOC15/EOC15/RESULT15 is associated with post processing block 3 Reset type: SYSRSn

20.15.2.54 ADCPPB3STAMP Register (Offset = 51h) [Reset = 0000h]

ADCPPB3STAMP is shown in [Figure 20-79](#) and described in [Table 20-69](#).

Return to the [Summary Table](#).

ADC PPB3 Sample Delay Time Stamp Register

Figure 20-79. ADCPPB3STAMP Register

15	14	13	12	11	10	9	8
RESERVED				DLYSTAMP			
R-0h				R-0h			
7	6	5	4	3	2	1	0
DLYSTAMP							
R-0h							

Table 20-69. ADCPPB3STAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-0	DLYSTAMP	R	0h	ADC Post Processing Block 3 Delay Time Stamp. When an SOC starts sampling the value contained in REQSTAMP is subtracted from the value in ADCCOUNTER.FREECOUNT and loaded into this bit field, thereby giving the number of system clock cycles delay between the SOC trigger and the actual start of the sample. Reset type: SYSRSn

20.15.2.55 ADCPPB3OFFCAL Register (Offset = 52h) [Reset = 0000h]

ADCPPB3OFFCAL is shown in [Figure 20-80](#) and described in [Table 20-70](#).

Return to the [Summary Table](#).

ADC PPB3 Offset Calibration Register

Figure 20-80. ADCPPB3OFFCAL Register

15	14	13	12	11	10	9	8
RESERVED						OFFCAL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
OFFCAL							
R/W-0h							

Table 20-70. ADCPPB3OFFCAL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	OFFCAL	R/W	0h	ADC Post Processing Block 3 Offset Correction. This bit field can be used to digitally remove any system level offset inherent in the ADCIN circuit. This 10-bit signed value is subtracted from the ADC output before being stored in the ADCRESULT register. 000h No change. The ADC output is stored directly into ADCRESULT. 001h ADC output - 1 is stored into ADCRESULT. 002h ADC output - 2 is stored into ADCRESULT. ... 200h ADC output + 512 is stored into ADCRESULT. ... 3FFh ADC output + 1 is stored into ADCRESULT. NOTE: In 16-bit mode, the subtraction will saturate at 0000h and FFFFh before being stored into the ADCRESULT register. In 12-bit mode, the subtraction will saturate at 0000h and 0FFFh before being stored into the ADCRESULT register. Note: in the case that multiple PPBs point to the same SOC, only the OFFCAL of the highest numbered PPB will be applied. Reset type: SYSRSn

20.15.2.56 ADCPPB3OFFREF Register (Offset = 53h) [Reset = 0000h]

ADCPPB3OFFREF is shown in [Figure 20-81](#) and described in [Table 20-71](#).

Return to the [Summary Table](#).

ADC PPB3 Offset Reference Register

Figure 20-81. ADCPPB3OFFREF Register

15	14	13	12	11	10	9	8
OFFREF							
R/W-0h							
7	6	5	4	3	2	1	0
OFFREF							
R/W-0h							

Table 20-71. ADCPPB3OFFREF Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFREF	R/W	0h	ADC Post Processing Block 3 Offset Correction. This bit field can be used to either calculate the feedback error or convert a unipolar signal to bipolar by subtracting a reference value. This 16-bit unsigned value is subtracted from the ADCRESULT register before being passed through an optional two's complement function and stored in the ADCPPB3RESULT register. This subtraction is not saturated. 0000h No change. The ADCRESULT value is passed on. 0001h ADCRESULT - 1 is passed on. 0002h ADCRESULT - 2 is passed on. ... 8000h ADCRESULT - 32,768 is passed on. ... FFFFh ADCRESULT - 65,535 is passed on. NOTE: In 12-bit mode the size of this register does not change from 16-bits. It is the user's responsibility to ensure that only a 12-bit value is written to this register when in 12-bit mode. Reset type: SYSRSn

20.15.2.57 ADCPPB3TRIPHI Register (Offset = 54h) [Reset = 00000000h]

ADCPPB3TRIPHI is shown in [Figure 20-82](#) and described in [Table 20-72](#).

Return to the [Summary Table](#).

ADC PPB3 Trip High Register

Figure 20-82. ADCPPB3TRIPHI Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							HSIGN
R-0h							R/W-0h
15	14	13	12	11	10	9	8
LIMITHI							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITHI							
R/W-0h							

Table 20-72. ADCPPB3TRIPHI Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reserved
16	HSIGN	R/W	0h	High Limit Sign Bit. This is the sign bit (17th bit) to the LIMITHI bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITHI	R/W	0h	ADC Post Processing Block 3 Trip High Limit. This value sets the digital comparator trip high limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB3RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB3RESULT register. Reset type: SYSRSn

20.15.2.58 ADCPPB3TRIPLO Register (Offset = 56h) [Reset = 00000000h]

ADCPPB3TRIPLO is shown in [Figure 20-83](#) and described in [Table 20-73](#).

Return to the [Summary Table](#).

ADC PPB3 Trip Low/Trigger Time Stamp Register

Figure 20-83. ADCPPB3TRIPLO Register

31	30	29	28	27	26	25	24
REQSTAMP							
R-0h							
23	22	21	20	19	18	17	16
REQSTAMP				RESERVED		LSIGN	
R-0h				R-0h		R/W-0h	
15	14	13	12	11	10	9	8
LIMITLO							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITLO							
R/W-0h							

Table 20-73. ADCPPB3TRIPLO Register Field Descriptions

Bit	Field	Type	Reset	Description
31-20	REQSTAMP	R	0h	ADC Post Processing Block 3 Request Time Stamp. When a trigger sets the associated SOC flag in the ADCSOCFLG1 register the value of ADCCOUNTER.FREECOUNT is loaded into this bit field. Reset type: SYSRSn
19-17	RESERVED	R	0h	Reserved
16	LSIGN	R/W	0h	Low Limit Sign Bit. This is the sign bit (17th bit) to the LIMITLO bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITLO	R/W	0h	ADC Post Processing Block 3 Trip Low Limit. This value sets the digital comparator trip low limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB3RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB3RESULT register. Reset type: SYSRSn

20.15.2.59 ADCPPB4CONFIG Register (Offset = 58h) [Reset = 0000h]

ADCPPB4CONFIG is shown in [Figure 20-84](#) and described in [Table 20-74](#).

Return to the [Summary Table](#).

ADC PPB4 Config Register

Figure 20-84. ADCPPB4CONFIG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED		CBCEN	TWOSCOMPE N	CONFIG			
R-0h		R/W-0h	R/W-0h	R/W-0h			

Table 20-74. ADCPPB4CONFIG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-6	RESERVED	R	0h	Reserved
5	CBCEN	R/W	0h	ADC Post Processing Block Cycle By Cycle Enable. When set, this bit enables the post conversion hardware processing circuit to automatically clear the ADCEVTSTAT on a conversion if the event condition is no longer present. Reset type: SYSRSn
4	TWOSCOMPEN	R/W	0h	ADC Post Processing Block 4 Two's Complement Enable. When set this bit enables the post conversion hardware processing circuit that performs a two's complement on the output of the offset/reference subtraction unit before storing the result in the ADCPPB4RESULT register. 0 ADCPPB4RESULT = ADCRESULTx - ADCPPB4OFFREF 1 ADCPPB4RESULT = ADCPPB4OFFREF - ADCRESULTx Reset type: SYSRSn

Table 20-74. ADCPPB4CONFIG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-0	CONFIG	R/W	0h	ADC Post Processing Block 4 Configuration. This bit field defines which SOC/EOC/RESULT is associated with this post processing block. 0000 SOC0/EOC0/RESULT0 is associated with post processing block 4 0001 SOC1/EOC1/RESULT1 is associated with post processing block 4 0010 SOC2/EOC2/RESULT2 is associated with post processing block 4 0011 SOC3/EOC3/RESULT3 is associated with post processing block 4 0100 SOC4/EOC4/RESULT4 is associated with post processing block 4 0101 SOC5/EOC5/RESULT5 is associated with post processing block 4 0110 SOC6/EOC6/RESULT6 is associated with post processing block 4 0111 SOC7/EOC7/RESULT7 is associated with post processing block 4 1000 SOC8/EOC8/RESULT8 is associated with post processing block 4 1001 SOC9/EOC9/RESULT9 is associated with post processing block 4 1010 SOC10/EOC10/RESULT10 is associated with post processing block 4 1011 SOC11/EOC11/RESULT11 is associated with post processing block 4 1100 SOC12/EOC12/RESULT12 is associated with post processing block 4 1101 SOC13/EOC13/RESULT13 is associated with post processing block 4 1110 SOC14/EOC14/RESULT14 is associated with post processing block 4 1111 SOC15/EOC15/RESULT15 is associated with post processing block 4 Reset type: SYSRSn

20.15.2.60 ADCPPB4STAMP Register (Offset = 59h) [Reset = 0000h]

ADCPPB4STAMP is shown in [Figure 20-85](#) and described in [Table 20-75](#).

Return to the [Summary Table](#).

ADC PPB4 Sample Delay Time Stamp Register

Figure 20-85. ADCPPB4STAMP Register

15	14	13	12	11	10	9	8
RESERVED				DLYSTAMP			
R-0h				R-0h			
7	6	5	4	3	2	1	0
DLYSTAMP							
R-0h							

Table 20-75. ADCPPB4STAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-0	DLYSTAMP	R	0h	ADC Post Processing Block 4 Delay Time Stamp. When an SOC starts sampling the value contained in REQSTAMP is subtracted from the value in ADCCOUNTER.FREECOUNT and loaded into this bit field, thereby giving the number of system clock cycles delay between the SOC trigger and the actual start of the sample. Reset type: SYSRSn

20.15.2.61 ADCPPB4OFFCAL Register (Offset = 5Ah) [Reset = 0000h]

ADCPPB4OFFCAL is shown in [Figure 20-86](#) and described in [Table 20-76](#).

Return to the [Summary Table](#).

ADC PPB4 Offset Calibration Register

Figure 20-86. ADCPPB4OFFCAL Register

15	14	13	12	11	10	9	8
RESERVED						OFFCAL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
OFFCAL							
R/W-0h							

Table 20-76. ADCPPB4OFFCAL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	OFFCAL	R/W	0h	ADC Post Processing Block 4 Offset Correction. This bit field can be used to digitally remove any system level offset inherent in the ADCIN circuit. This 10-bit signed value is subtracted from the ADC output before being stored in the ADCRESULT register. 000h No change. The ADC output is stored directly into ADCRESULT. 001h ADC output - 1 is stored into ADCRESULT. 002h ADC output - 2 is stored into ADCRESULT. ... 200h ADC output + 512 is stored into ADCRESULT. ... 3FFh ADC output + 1 is stored into ADCRESULT. NOTE: In 16-bit mode, the subtraction will saturate at 0000h and FFFFh before being stored into the ADCRESULT register. In 12-bit mode, the subtraction will saturate at 0000h and 0FFFh before being stored into the ADCRESULT register. Note: in the case that multiple PPBs point to the same SOC, only the OFFCAL of the highest numbered PPB will be applied. Reset type: SYSRSn

20.15.2.62 ADCPPB4OFFREF Register (Offset = 5Bh) [Reset = 0000h]

ADCPPB4OFFREF is shown in [Figure 20-87](#) and described in [Table 20-77](#).

Return to the [Summary Table](#).

ADC PPB4 Offset Reference Register

Figure 20-87. ADCPPB4OFFREF Register

15	14	13	12	11	10	9	8
OFFREF							
R/W-0h							
7	6	5	4	3	2	1	0
OFFREF							
R/W-0h							

Table 20-77. ADCPPB4OFFREF Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFREF	R/W	0h	ADC Post Processing Block 4 Offset Correction. This bit field can be used to either calculate the feedback error or convert a unipolar signal to bipolar by subtracting a reference value. This 16-bit unsigned value is subtracted from the ADCRESULT register before being passed through an optional two's complement function and stored in the ADCPPB4RESULT register. This subtraction is not saturated. 0000h No change. The ADCRESULT value is passed on. 0001h ADCRESULT - 1 is passed on. 0002h ADCRESULT - 2 is passed on. ... 8000h ADCRESULT - 32,768 is passed on. ... FFFFh ADCRESULT - 65,535 is passed on. NOTE: In 12-bit mode the size of this register does not change from 16-bits. It is the user's responsibility to ensure that only a 12-bit value is written to this register when in 12-bit mode. Reset type: SYSRStn

20.15.2.63 ADCPPB4TRIPHI Register (Offset = 5Ch) [Reset = 00000000h]

ADCPPB4TRIPHI is shown in [Figure 20-88](#) and described in [Table 20-78](#).

Return to the [Summary Table](#).

ADC PPB4 Trip High Register

Figure 20-88. ADCPPB4TRIPHI Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							HSIGN
R-0h							R/W-0h
15	14	13	12	11	10	9	8
LIMITHI							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITHI							
R/W-0h							

Table 20-78. ADCPPB4TRIPHI Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reserved
16	HSIGN	R/W	0h	High Limit Sign Bit. This is the sign bit (17th bit) to the LIMITHI bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITHI	R/W	0h	ADC Post Processing Block 4 Trip High Limit. This value sets the digital comparator trip high limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB4RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB4RESULT register. Reset type: SYSRSn

20.15.2.64 ADCPPB4TRIPLO Register (Offset = 5Eh) [Reset = 00000000h]

ADCPPB4TRIPLO is shown in [Figure 20-89](#) and described in [Table 20-79](#).

Return to the [Summary Table](#).

ADC PPB4 Trip Low/Trigger Time Stamp Register

Figure 20-89. ADCPPB4TRIPLO Register

31	30	29	28	27	26	25	24
REQSTAMP							
R-0h							
23	22	21	20	19	18	17	16
REQSTAMP				RESERVED			LSIGN
R-0h				R-0h			R/W-0h
15	14	13	12	11	10	9	8
LIMITLO							
R/W-0h							
7	6	5	4	3	2	1	0
LIMITLO							
R/W-0h							

Table 20-79. ADCPPB4TRIPLO Register Field Descriptions

Bit	Field	Type	Reset	Description
31-20	REQSTAMP	R	0h	ADC Post Processing Block 4 Request Time Stamp. When a trigger sets the associated SOC flag in the ADCSOCFLG1 register the value of ADCCOUNTER.FREECOUNT is loaded into this bit field. Reset type: SYSRSn
19-17	RESERVED	R	0h	Reserved
16	LSIGN	R/W	0h	Low Limit Sign Bit. This is the sign bit (17th bit) to the LIMITLO bit field when in 16-bit ADC mode. Reset type: SYSRSn
15-0	LIMITLO	R/W	0h	ADC Post Processing Block 4 Trip Low Limit. This value sets the digital comparator trip low limit. In 16-bit mode all 17 bits will be compared against the 17 bits of the PPBRESULT bit field of the ADCPPB4RESULT register. In 12-bit mode bits 12:0 will be compared against bits 12:0 of the PPBRESULT bit field of the ADCPPB4RESULT register. Reset type: SYSRSn

20.15.2.65 ADCINTCYCLE Register (Offset = 6Fh) [Reset = 0000h]

ADCINTCYCLE is shown in [Figure 20-90](#) and described in [Table 20-80](#).

Return to the [Summary Table](#).

ADC Early Interrupt Generation Cycle

Figure 20-90. ADCINTCYCLE Register

15	14	13	12	11	10	9	8
DELAY							
R/W-0h							
7	6	5	4	3	2	1	0
DELAY							
R/W-0h							

Table 20-80. ADCINTCYCLE Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	DELAY	R/W	0h	ADC Early Interrupt Generation Cycle Delay: Defines the delay from the fall edge of ADCSOC in terms of system clock cycles, for the interrupt to be generated. Reset type: SYSRStn

20.15.2.66 ADCINLTRIM1 Register (Offset = 70h) [Reset = 00000000h]

ADCINLTRIM1 is shown in [Figure 20-91](#) and described in [Table 20-81](#).

Return to the [Summary Table](#).

ADC Linearity Trim 1 Register

Figure 20-91. ADCINLTRIM1 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM31TO0																															
R/W-0h																															

Table 20-81. ADCINLTRIM1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM31TO0	R/W	0h	ADC Linearity Trim Bits 31-0. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.2.67 ADCINLTRIM2 Register (Offset = 72h) [Reset = 00000000h]

ADCINLTRIM2 is shown in [Figure 20-92](#) and described in [Table 20-82](#).

Return to the [Summary Table](#).

ADC Linearity Trim 2 Register

Figure 20-92. ADCINLTRIM2 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM63TO32																															
R/W-0h																															

Table 20-82. ADCINLTRIM2 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM63TO32	R/W	0h	ADC Linearity Trim Bits 63-32. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.2.68 ADCINLTRIM3 Register (Offset = 74h) [Reset = 00000000h]

ADCINLTRIM3 is shown in [Figure 20-93](#) and described in [Table 20-83](#).

Return to the [Summary Table](#).

ADC Linearity Trim 3 Register

Figure 20-93. ADCINLTRIM3 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM95TO64																															
R/W-0h																															

Table 20-83. ADCINLTRIM3 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM95TO64	R/W	0h	ADC Linearity Trim Bits 95-64. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.2.69 ADCINLTRIM4 Register (Offset = 76h) [Reset = 00000000h]

ADCINLTRIM4 is shown in [Figure 20-94](#) and described in [Table 20-84](#).

Return to the [Summary Table](#).

ADC Linearity Trim 4 Register

Figure 20-94. ADCINLTRIM4 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM127TO96																															
R/W-0h																															

Table 20-84. ADCINLTRIM4 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM127TO96	R/W	0h	ADC Linearity Trim Bits 127-96. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.2.70 ADCINLTRIM5 Register (Offset = 78h) [Reset = 00000000h]

ADCINLTRIM5 is shown in [Figure 20-95](#) and described in [Table 20-85](#).

Return to the [Summary Table](#).

ADC Linearity Trim 5 Register

Figure 20-95. ADCINLTRIM5 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM159TO128																															
R/W-0h																															

Table 20-85. ADCINLTRIM5 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM159TO128	R/W	0h	ADC Linearity Trim Bits 159-128. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.2.71 ADCINLTRIM6 Register (Offset = 7Ah) [Reset = 00000000h]

ADCINLTRIM6 is shown in [Figure 20-96](#) and described in [Table 20-86](#).

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ADC Linearity Trim 6 Register

Figure 20-96. ADCINLTRIM6 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INLTRIM191TO160																															
R/W-0h																															

Table 20-86. ADCINLTRIM6 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	INLTRIM191TO160	R/W	0h	ADC Linearity Trim Bits 191-160. This register should not be modified unless specifically indicated by TI Errata or other documentation. Modifying the contents of this register could cause this module to operate outside of datasheet specifications. Reset type: SYSRSn

20.15.3 ADC_RESULT_REGS Registers

Table 20-87 lists the memory-mapped registers for the ADC_RESULT_REGS registers. All register offset addresses not listed in Table 20-87 should be considered as reserved locations and the register contents should not be modified.

Table 20-87. ADC_RESULT_REGS Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	ADCRESULT0	ADC Result 0 Register		Go
1h	ADCRESULT1	ADC Result 1 Register		Go
2h	ADCRESULT2	ADC Result 2 Register		Go
3h	ADCRESULT3	ADC Result 3 Register		Go
4h	ADCRESULT4	ADC Result 4 Register		Go
5h	ADCRESULT5	ADC Result 5 Register		Go
6h	ADCRESULT6	ADC Result 6 Register		Go
7h	ADCRESULT7	ADC Result 7 Register		Go
8h	ADCRESULT8	ADC Result 8 Register		Go
9h	ADCRESULT9	ADC Result 9 Register		Go
Ah	ADCRESULT10	ADC Result 10 Register		Go
Bh	ADCRESULT11	ADC Result 11 Register		Go
Ch	ADCRESULT12	ADC Result 12 Register		Go
Dh	ADCRESULT13	ADC Result 13 Register		Go
Eh	ADCRESULT14	ADC Result 14 Register		Go
Fh	ADCRESULT15	ADC Result 15 Register		Go
10h	ADCPPB1RESULT	ADC Post Processing Block 1 Result Register		Go
12h	ADCPPB2RESULT	ADC Post Processing Block 2 Result Register		Go
14h	ADCPPB3RESULT	ADC Post Processing Block 3 Result Register		Go
16h	ADCPPB4RESULT	ADC Post Processing Block 4 Result Register		Go

Complex bit access types are encoded to fit into small table cells. Table 20-88 shows the codes that are used for access types in this section.

Table 20-88. ADC_RESULT_REGS Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Reset or Default Value		
-n		Value after reset or the default value
Register Array Variables		
i,j,k,l,m,n		When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula.
y		When this variable is used in a register name, an offset, or an address it refers to the value of a register array.

20.15.3.1 ADCRESULT0 Register (Offset = 0h) [Reset = 0000h]

ADCRESULT0 is shown in [Figure 20-97](#) and described in [Table 20-89](#).

Return to the [Summary Table](#).

ADC Result 0 Register

Figure 20-97. ADCRESULT0 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-89. ADCRESULT0 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 0 16-bit ADC result. After the ADC completes a conversion of SOC0, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.2 ADCRESULT1 Register (Offset = 1h) [Reset = 0000h]

ADCRESULT1 is shown in [Figure 20-98](#) and described in [Table 20-90](#).

Return to the [Summary Table](#).

ADC Result 1 Register

Figure 20-98. ADCRESULT1 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-90. ADCRESULT1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 1 16-bit ADC result. After the ADC completes a conversion of SOC1, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.3 ADCRESULT2 Register (Offset = 2h) [Reset = 0000h]

ADCRESULT2 is shown in [Figure 20-99](#) and described in [Table 20-91](#).

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ADC Result 2 Register

Figure 20-99. ADCRESULT2 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-91. ADCRESULT2 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 2 16-bit ADC result. After the ADC completes a conversion of SOC2, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.4 ADCRESULT3 Register (Offset = 3h) [Reset = 0000h]

ADCRESULT3 is shown in [Figure 20-100](#) and described in [Table 20-92](#).

Return to the [Summary Table](#).

ADC Result 3 Register

Figure 20-100. ADCRESULT3 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-92. ADCRESULT3 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 3 16-bit ADC result. After the ADC completes a conversion of SOC3, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.5 ADCRESULT4 Register (Offset = 4h) [Reset = 0000h]

ADCRESULT4 is shown in [Figure 20-101](#) and described in [Table 20-93](#).

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ADC Result 4 Register

Figure 20-101. ADCRESULT4 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-93. ADCRESULT4 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 4 16-bit ADC result. After the ADC completes a conversion of SOC4, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.6 ADCRESULT5 Register (Offset = 5h) [Reset = 0000h]

ADCRESULT5 is shown in [Figure 20-102](#) and described in [Table 20-94](#).

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ADC Result 5 Register

Figure 20-102. ADCRESULT5 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-94. ADCRESULT5 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 5 16-bit ADC result. After the ADC completes a conversion of SOC5, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.7 ADCRESULT6 Register (Offset = 6h) [Reset = 0000h]

ADCRESULT6 is shown in [Figure 20-103](#) and described in [Table 20-95](#).

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ADC Result 6 Register

Figure 20-103. ADCRESULT6 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-95. ADCRESULT6 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 6 16-bit ADC result. After the ADC completes a conversion of SOC6, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.8 ADCRESULT7 Register (Offset = 7h) [Reset = 0000h]

ADCRESULT7 is shown in [Figure 20-104](#) and described in [Table 20-96](#).

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ADC Result 7 Register

Figure 20-104. ADCRESULT7 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-96. ADCRESULT7 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 7 16-bit ADC result. After the ADC completes a conversion of SOC7, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.9 ADCRESULT8 Register (Offset = 8h) [Reset = 0000h]

ADCRESULT8 is shown in [Figure 20-105](#) and described in [Table 20-97](#).

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ADC Result 8 Register

Figure 20-105. ADCRESULT8 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-97. ADCRESULT8 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 8 16-bit ADC result. After the ADC completes a conversion of SOC8, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.10 ADCRESULT9 Register (Offset = 9h) [Reset = 0000h]

ADCRESULT9 is shown in [Figure 20-106](#) and described in [Table 20-98](#).

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ADC Result 9 Register

Figure 20-106. ADCRESULT9 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-98. ADCRESULT9 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 9 16-bit ADC result. After the ADC completes a conversion of SOC9, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.11 ADCRESULT10 Register (Offset = Ah) [Reset = 0000h]

ADCRESULT10 is shown in [Figure 20-107](#) and described in [Table 20-99](#).

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ADC Result 10 Register

Figure 20-107. ADCRESULT10 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-99. ADCRESULT10 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 10 16-bit ADC result. After the ADC completes a conversion of SOC10, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.12 ADCRESULT11 Register (Offset = Bh) [Reset = 0000h]

ADCRESULT11 is shown in [Figure 20-108](#) and described in [Table 20-100](#).

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ADC Result 11 Register

Figure 20-108. ADCRESULT11 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-100. ADCRESULT11 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 11 16-bit ADC result. After the ADC completes a conversion of SOC11, the digital result is placed in this bit field. Reset type: SYSRStn

20.15.3.13 ADCRESULT12 Register (Offset = Ch) [Reset = 0000h]

ADCRESULT12 is shown in [Figure 20-109](#) and described in [Table 20-101](#).

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ADC Result 12 Register

Figure 20-109. ADCRESULT12 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-101. ADCRESULT12 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 12 16-bit ADC result. After the ADC completes a conversion of SOC12, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.14 ADCRESULT13 Register (Offset = Dh) [Reset = 0000h]

ADCRESULT13 is shown in [Figure 20-110](#) and described in [Table 20-102](#).

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ADC Result 13 Register

Figure 20-110. ADCRESULT13 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-102. ADCRESULT13 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 13 16-bit ADC result. After the ADC completes a conversion of SOC13, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.15 ADCRESULT14 Register (Offset = Eh) [Reset = 0000h]

ADCRESULT14 is shown in [Figure 20-111](#) and described in [Table 20-103](#).

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ADC Result 14 Register

Figure 20-111. ADCRESULT14 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-103. ADCRESULT14 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 14 16-bit ADC result. After the ADC completes a conversion of SOC14, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.16 ADCRESULT15 Register (Offset = Fh) [Reset = 0000h]

ADCRESULT15 is shown in [Figure 20-112](#) and described in [Table 20-104](#).

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ADC Result 15 Register

Figure 20-112. ADCRESULT15 Register

15	14	13	12	11	10	9	8
RESULT							
R-0h							
7	6	5	4	3	2	1	0
RESULT							
R-0h							

Table 20-104. ADCRESULT15 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	RESULT	R	0h	ADC Result 15 16-bit ADC result. After the ADC completes a conversion of SOC15, the digital result is placed in this bit field. Reset type: SYSRSn

20.15.3.17 ADCPPB1RESULT Register (Offset = 10h) [Reset = 00000000h]

ADCPPB1RESULT is shown in [Figure 20-113](#) and described in [Table 20-105](#).

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ADC Post Processing Block 1 Result Register

Figure 20-113. ADCPPB1RESULT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SIGN																PPBRESULT															
R-0h																R-0h															

Table 20-105. ADCPPB1RESULT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	SIGN	R	0h	Sign Extended Bits. These bits reflect the same value as bit 16. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the SIGN bits extend down to bit 12, and all reflect the same value as bit 12. Reset type: SYSRSn
15-0	PPBRESULT	R	0h	ADC Post Processing Block Result 1 The result of the offset/reference subtraction post conversion processing is stored in this register. This result is available 1 SYSCLK cycle after the associated ADCRESULT is available. If ADCINTFLG is polled to determine when to read the PPBRESULT, it may be necessary to add a NOP instruction to ensure that the updated post conversion processing result has posted to the register. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the PPBRESULT bits are limited to bits 12:0. Reset type: SYSRSn

20.15.3.18 ADCPPB2RESULT Register (Offset = 12h) [Reset = 00000000h]

ADCPPB2RESULT is shown in [Figure 20-114](#) and described in [Table 20-106](#).

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ADC Post Processing Block 2 Result Register

Figure 20-114. ADCPPB2RESULT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SIGN																PPBRESULT															
R-0h																R-0h															

Table 20-106. ADCPPB2RESULT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	SIGN	R	0h	Sign Extended Bits. These bits reflect the same value as bit 16. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the SIGN bits extend down to bit 12, and all reflect the same value as bit 12. Reset type: SYSRSn
15-0	PPBRESULT	R	0h	ADC Post Processing Block Result 2 The result of the offset/reference subtraction post conversion processing is stored in this register. This result is available 1 SYSCLK cycle after the associated ADCRESULT is available. If ADCINTFLG is polled to determine when to read the PPBRESULT, it may be necessary to add a NOP instruction to ensure that the updated post conversion processing result has posted to the register. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the PPBRESULT bits are limited to bits 12:0. Reset type: SYSRSn

20.15.3.19 ADCPPB3RESULT Register (Offset = 14h) [Reset = 00000000h]

ADCPPB3RESULT is shown in [Figure 20-115](#) and described in [Table 20-107](#).

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ADC Post Processing Block 3 Result Register

Figure 20-115. ADCPPB3RESULT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SIGN																PPBRESULT															
R-0h																R-0h															

Table 20-107. ADCPPB3RESULT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	SIGN	R	0h	Sign Extended Bits. These bits reflect the same value as bit 16. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the SIGN bits extend down to bit 12, and all reflect the same value as bit 12. Reset type: SYSRSn
15-0	PPBRESULT	R	0h	ADC Post Processing Block Result 3 The result of the offset/reference subtraction post conversion processing is stored in this register. This result is available 1 SYSCLK cycle after the associated ADCRESULT is available. If ADCINTFLG is polled to determine when to read the PPBRESULT, it may be necessary to add a NOP instruction to ensure that the updated post conversion processing result has posted to the register. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the PPBRESULT bits are limited to bits 12:0. Reset type: SYSRSn

20.15.3.20 ADCPPB4RESULT Register (Offset = 16h) [Reset = 00000000h]

ADCPPB4RESULT is shown in [Figure 20-116](#) and described in [Table 20-108](#).

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ADC Post Processing Block 4 Result Register

Figure 20-116. ADCPPB4RESULT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SIGN																PPBRESULT															
R-0h																R-0h															

Table 20-108. ADCPPB4RESULT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	SIGN	R	0h	Sign Extended Bits. These bits reflect the same value as bit 16. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the SIGN bits extend down to bit 12, and all reflect the same value as bit 12. Reset type: SYSRSn
15-0	PPBRESULT	R	0h	ADC Post Processing Block Result 4 The result of the offset/reference subtraction post conversion processing is stored in this register. This result is available 1 SYSCLK cycle after the associated ADCRESULT is available. If ADCINTFLG is polled to determine when to read the PPBRESULT, it may be necessary to add a NOP instruction to ensure that the updated post conversion processing result has posted to the register. NOTE: If the conversion associated with this Post Processing Block is a 12-bit conversion, the PPBRESULT bits are limited to bits 12:0. Reset type: SYSRSn

20.15.4 ADC Registers to Driverlib Functions

Table 20-109. ADC Registers to Driverlib Functions

File	Driverlib Function
ADCCTL1	
adc.h	ADC_setInterruptPulseMode
adc.h	ADC_enableConverter
adc.h	ADC_disableConverter
adc.h	ADC_isBusy
ADCCTL2	
adc.c	ADC_setMode
adc.c	ADC_setNLTrim
adc.c	ADC_setOffsetTrim
adc.h	ADC_setPrescaler
ADCBURSTCTL	
adc.h	ADC_setBurstModeConfig
adc.h	ADC_enableBurstMode
adc.h	ADC_disableBurstMode
ADCINTFLG	
adc.h	ADC_getInterruptStatus
adc.h	ADC_clearInterruptStatus
ADCINTFLGCLR	
adc.h	ADC_clearInterruptStatus

Table 20-109. ADC Registers to Driverlib Functions (continued)

File	Driverlib Function
ADCINTOVF	
adc.h	ADC_getInterruptOverflowStatus
adc.h	ADC_clearInterruptOverflowStatus
ADCINTOVFCLR	
adc.h	ADC_clearInterruptOverflowStatus
ADCINTSEL1N2	
adc.h	ADC_enableInterrupt
adc.h	ADC_disableInterrupt
adc.h	ADC_setInterruptSource
adc.h	ADC_enableContinuousMode
adc.h	ADC_disableContinuousMode
ADCINTSEL3N4	
-	See INTSEL1N2
ADCSOCPRCTL	
adc.h	ADC_setSOCPriority
ADCINTSOCSEL1	
adc.h	ADC_setInterruptSOCTrigger
ADCINTSOCSEL2	
-	See INTSOCSEL1
ADCSOCFLG1	
-	
ADCSOCFRC1	
adc.h	ADC_forceSOC
adc.h	ADC_forceMultipleSOC
ADCSOCOVF1	
-	
ADCSOCOVFCLR1	
-	
ADCSOC0CTL	
adc.h	ADC_setupSOC
ADCSOC1CTL	
-	See SOC0CTL
ADCSOC2CTL	
-	See SOC0CTL
ADCSOC3CTL	
-	See SOC0CTL
ADCSOC4CTL	
-	See SOC0CTL
ADCSOC5CTL	
-	See SOC0CTL
ADCSOC6CTL	
-	See SOC0CTL
ADCSOC7CTL	
-	See SOC0CTL
ADCSOC8CTL	

Table 20-109. ADC Registers to Driverlib Functions (continued)

File	Driverlib Function
-	See SOC0CTL
ADCSOC9CTL	
-	See SOC0CTL
ADCSOC10CTL	
-	See SOC0CTL
ADCSOC11CTL	
-	See SOC0CTL
ADCSOC12CTL	
-	See SOC0CTL
ADCSOC13CTL	
-	See SOC0CTL
ADCSOC14CTL	
-	See SOC0CTL
ADCSOC15CTL	
-	See SOC0CTL
ADCEVTSTAT	
adc.h	ADC_getPPBEventStatus
ADCEVTCLR	
adc.h	ADC_clearPPBEventStatus
ADCEVTSEL	
adc.h	ADC_enablePPBEvent
adc.h	ADC_disablePPBEvent
ADCEVTINTSEL	
adc.h	ADC_enablePPBEventInterrupt
adc.h	ADC_disablePPBEventInterrupt
ADCOSDETECT	
adc.h	ADC_configOSDetectMode
ADCCOUNTER	
-	
ADCREV	
-	
ADCOFFTRIM	
adc.c	ADC_setOffsetTrim
ADCPPB1CONFIG	
adc.h	ADC_setupPPB
adc.h	ADC_enablePPBEventCBCClear
adc.h	ADC_disablePPBEventCBCClear
adc.h	ADC_enablePPBTwosComplement
adc.h	ADC_disablePPBTwosComplement
ADCPPB1STAMP	
adc.h	ADC_getPPBDelayTimeStamp
ADCPPB1OFFCAL	
adc.h	ADC_setPPBCalibrationOffset
ADCPPB1OFFREF	
adc.h	ADC_setPPBReferenceOffset

Table 20-109. ADC Registers to Driverlib Functions (continued)

File	Driverlib Function
ADCPPB1TRIPHI	
adc.c	ADC_setPPBTripLimits
ADCPPB1TRIPLO	
adc.c	ADC_setPPBTripLimits
ADCPPB2CONFIG	
-	See PPB1CONFIG
ADCPPB2STAMP	
-	See PPB1STAMP
ADCPPB2OFFCAL	
-	See PPB1OFFCAL
ADCPPB2OFFREF	
-	See PPB1OFFREF
ADCPPB2TRIPHI	
-	See PPB1TRIPHI
ADCPPB2TRIPLO	
-	See PPB1TRIPLO
ADCPPB3CONFIG	
-	See PPB1CONFIG
ADCPPB3STAMP	
-	See PPB1STAMP
ADCPPB3OFFCAL	
-	See PPB1OFFCAL
ADCPPB3OFFREF	
-	See PPB1OFFREF
ADCPPB3TRIPHI	
-	See PPB1TRIPHI
ADCPPB3TRIPLO	
-	See PPB1TRIPLO
ADCPPB4CONFIG	
-	See PPB1CONFIG
ADCPPB4STAMP	
-	See PPB1STAMP
ADCPPB4OFFCAL	
-	See PPB1OFFCAL
ADCPPB4OFFREF	
-	See PPB1OFFREF
ADCPPB4TRIPHI	
-	See PPB1TRIPHI
ADCPPB4TRIPLO	
-	See PPB1TRIPLO
ADCINTCYCLE	
adc.h	ADC_setInterruptCycleOffset
ADCINLTRIM1	
adc.c	ADC_setINLTrim
ADCINLTRIM2	

Table 20-109. ADC Registers to Driverlib Functions (continued)

File	Driverlib Function
adc.c	ADC_setINLTrim
ADCINLTRIM3	
-	
ADCINLTRIM4	
adc.c	ADC_setINLTrim
ADCINLTRIM5	
adc.c	ADC_setINLTrim
ADCINLTRIM6	
-	
ADCRESULT0	
adc.h	ADC_readResult
ADCRESULT1	
-	See RESULT0
ADCRESULT2	
-	See RESULT0
ADCRESULT3	
-	See RESULT0
ADCRESULT4	
-	See RESULT0
ADCRESULT5	
-	See RESULT0
ADCRESULT6	
-	See RESULT0
ADCRESULT7	
-	See RESULT0
ADCRESULT8	
-	See RESULT0
ADCRESULT9	
-	See RESULT0
ADCRESULT10	
-	See RESULT0
ADCRESULT11	
-	See RESULT0
ADCRESULT12	
-	See RESULT0
ADCRESULT13	
-	See RESULT0
ADCRESULT14	
-	See RESULT0
ADCRESULT15	
-	See RESULT0
ADCPPB1RESULT	
adc.h	ADC_readPPBResult
ADCPPB2RESULT	
-	See PPB1RESULT

Table 20-109. ADC Registers to Driverlib Functions (continued)

File	Driverlib Function
ADCPPB3RESULT	
-	See PPB1RESULT
ADCPPB4RESULT	
-	See PPB1RESULT